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**How to classify domain entities into
top-level ontology concepts using language
models: A study across multiple labels,
resources, domains, and languages**

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*"I have no special talents.
I am only passionately curious."*
— ALBERT EINSTEIN

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ABSTRACT

Classifying domain entities into their respective top-level ontology concepts is a complex problem that typically demands manual analysis and deep expertise in the domain of interest and ontology engineering. An efficient approach to classifying domain entities can enhance data integration and interoperability, which is crucial for structured knowledge representation and modeling. Motivated by this challenge, we propose an automated approach to classify domain entities into top-level ontology concepts using Natural Language Definitions (NLDs) of these domain entities. We hypothesize that NLDs encapsulate semantic information that is critical for associating domain entities with specific top-level ontology concepts. Also, we proposed two classification pipelines trained using fine-tuning and classic Machine Learning (ML) approaches and leveraging state-of-the-art language models to deal with the textual input. To evaluate our proposal, we constructed multi-label, multi-resource, and multi-language datasets by extracting data from BabelNet and domain ontologies available in the OBO Foundry repository. These datasets encompass mappings to DOLCE-Lite-Plus (DLP) and Basic Formal Ontology (BFO) top-level ontologies, ensuring a diverse and representative collection of domain entities and their corresponding top-level classifications, including different textual representations of domain entities, such as NLDs, definienda, definientia, and example sentences. In our experiments, we investigated the effectiveness of our approach across different textual representation techniques, languages, knowledge resources, knowledge domains, and domain ontologies. The results of our experiments highlight the value of leveraging Natural Language Definitions (NLDs) and embedding-based pipelines in classifying domain entities into top-level ontology concepts. Across multiple scenarios, NLDs consistently outperformed other textual representation approaches, achieving macro F1-scores exceeding 90% in most of the cases. Also, pipelines using the Mistral7B language model and a KNN classifier delivered a macro F1-score of 86.6% in cross-resource evaluations. Even in the challenging cross-domain experiments, where initial performance dropped below 40%, strategic inclusion of domain-specific data in the training data led to significant improvements, showcasing the differences between general and domains-specific NLDs in this classification task. Also, the findings underscore the potential of automated tools that can significantly support ontology engineers during the ontology development process. However, some limitations remain, such as the reliance on high-quality NLDs and challenges in handling underrepresented languages and knowledge domains.

Keywords: Top-level ontology classification. Natural language definition. Language model. Ontology learning.

Como classificar entidades de domínio em conceitos de ontologias de topo usando modelos de linguagem: Um estudo com múltiplos rótulos, recursos, domínios, e idiomas

RESUMO

Classificar entidades de domínio em seus respectivos conceitos de ontologias de topo é um problema complexo que normalmente exige análise manual e profundo conhecimento especializado no domínio de interesse e em engenharia de ontologias. Uma abordagem eficiente para classificar entidades de domínio pode melhorar a integração e a interoperabilidade de dados, aspectos cruciais para a representação e modelagem estruturada do conhecimento. Motivados por esse desafio, propomos uma abordagem automatizada para classificar entidades de domínio em conceitos de ontologias de topo, utilizando Definições em Linguagem Natural (NLDs) dessas entidades de domínio. Nossa hipótese é que as NLDs encapsulam informações semânticas essenciais para associar entidades de domínio a conceitos específicos de ontologias de topo. Além disso, propomos dois pipelines de classificação treinados utilizando técnicas de ajuste fino e abordagens clássicas de Aprendizado de Máquina (ML), aproveitando modelos de linguagem de última geração para lidar com os dados textuais. Para avaliar nossa proposta, construímos conjuntos de dados multi-rótulo, multi-recurso e multilíngues, extraídos do BabelNet e de ontologias de domínio disponíveis no repositório OBO Foundry. Esses conjuntos de dados abrangem mapeamentos para as ontologias de topo DOLCE-Lite-Plus (DLP) e Basic Formal Ontology (BFO), garantindo uma coleção diversa e representativa de entidades de domínio e suas respectivas classificações em ontologias de topo, incluindo diferentes representações textuais das entidades de domínio, como NLDs, definienda, definientia e sentenças de exemplo. Em nossos experimentos, investigamos a eficácia de nossa abordagem em diferentes técnicas de representação textual, idiomas, recursos de conhecimento, domínios de conhecimento e ontologias de domínio. Os resultados de nossos experimentos destacam o valor de aproveitar Definições em Linguagem Natural (NLDs) e pipelines baseados em embeddings na classificação de entidades de domínio em conceitos de ontologias de topo. Em diversos cenários, as NLDs superaram consistentemente outras abordagens de representação textual, alcançando macro F1-scores superiores a 90% na maioria dos casos. Além disso, pipelines utilizando o modelo de linguagem Mistral7B e um classificador KNN obtiveram um macro F1-score de 86,6% em avaliações entre diferentes fontes de da-

dos. Mesmo em experimentos entre diferentes domínios, onde o desempenho inicial caiu abaixo de 40%, a inclusão estratégica de dados específicos do domínio no treinamento resultou em melhorias significativas, evidenciando as diferenças entre NLDs gerais e específicas de domínio nessa tarefa de classificação. Os achados ressaltam o potencial do desenvolvimento de ferramentas automatizadas que possam apoiar os engenheiros de ontologias no processo de desenvolvimento de ontologias. No entanto, algumas limitações permanecem, como a dependência de NLDs de alta qualidade e os desafios no tratamento de idiomas e domínios de conhecimento sub-representados.

Palavras-chave: Classificação em ontologia de topo. Definição em linguagem natural. Modelo de linguagem. Aprendizagem de ontologia.

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LIST OF ABBREVIATIONS AND ACRONYMS

AI	Artificial Intelligence
BFO	Basic Formal Ontology
DOLCE	Descriptive Ontology for Linguistic and Cognitive Engineering
DLP	DOLCE-Lite-Plus
KNN	K-Nearest Neighbors
LM	Language Model
LLM	Large Language Model
NLD	Natural Language Definition
NLP	Natural Language Processing

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1 INTRODUCTION

In recent years, two significant areas in Artificial Intelligence (AI) have collided again. On the one hand, we have ontologies, which are a branch of philosophy and intersect areas such as metaphysics, epistemology, and philosophy of language, as ontologies consider how knowledge, language, and perception relate to the nature of reality (Studer; Benjamins; Fensel, 1998; Guarino, 1998). On the other hand, we have the advances in Natural Language Processing (NLP) for text representation and classification with Large Language Models (LLMs) (Devlin et al., 2018; Radford et al., 2018; Touvron et al., 2023; Jiang et al., 2024). This convergence has ushered in a new renaissance in ontology learning from text, in which the goal is to generate ontologies in an automatic or semi-automatic way, using text as input and state-of-the-art NLP techniques. While there are several approaches to tackle this challenge (He et al., 2022; He et al., 2024; Giglou; D’Souza; Auer, 2023), the field is still rife with open questions and opportunities for exploration.

Among various uses of ontologies, knowledge representation and data management are two usages that stand out because ontologies provide a structured and standardized way to represent and organize complex information, allowing for improved data integration and interoperability and enabling information retrieval and knowledge sharing across different systems or resources (Cicconeto et al., 2022; Kulvatunyou et al., 2022; Qu et al., 2024). Also, ontologies enhance semantic clarity by precisely defining domain entities and their relationships, reducing ambiguity, and ensuring a common understanding of complex knowledge domains. In this context, existing ontology engineering methodologies, such as NeOn (Suárez-Figueroa; Gómez-Pérez; Fernández-López, 2011), SABiO (Falbo, 2014) and SABiOx (Aguilar; Souza, 2024), recommend using a more general kind of ontology, called top-level ontology, to ensure these benefits when developing domain ontologies.

A top-level ontology defines a foundational and high-level structure for categorizing and organizing knowledge across various domains (Borgo et al., 2022; Guizzardi et al., 2022; Otte; Beverley; Ruttenberg, 2022). Also, a top-level ontology serves as a broad and abstract framework for representing fundamental concepts and relationships that are universally applicable and not tied to any specific domain. Thus, top-level ontologies are the starting point for organizing and categorizing domain-specific knowledge represented in domain ontologies and act as a common framework for the semantic integration of do-

main ontologies or data assets from distinct sources. However, identifying which top-level ontology concept a domain entity specializes in is a laborious and time-consuming task. This task is usually performed manually and requires expertise in the target domain and ontology engineering. Although ontology learning approaches often focus on tasks like entity recognition, extraction, and relationship extraction, which are crucial for building ontologies from textual sources (He et al., 2022; He et al., 2024; Giglou; D’Souza; Auer, 2023), classifying domain entities into top-level ontology concepts is particularly challenging, involving not only entity identification and extraction but also understanding the theoretical foundation and implications of choosing one top-level concept over another.

In this work, we address the problem of classifying domain entities into top-level ontology concepts using NLDs that represent domain entities textually. NLDs are textual descriptions written in natural language that articulate a specific domain entity’s meaning, characteristics, and context. From that, our central hypothesis is that the NLDs represent semantic information that allows domain entities to be related to top-level ontology concepts. In this context, we leverage two crucial ideas. Firstly, since top-level ontologies also contain a taxonomy of concepts (concepts related through hierarchical relationships, such as "is-a" and "subclass-of"), we used the notion of semantic similarity in taxonomies, which says that more specific entities grouped in the same more general entity share common features or attributes making them more similar to each other (Resnik, 1999; Sousa; Silva; Pesquita, 2024), i.e., we assume that NLDs of these entities include those common features (Robinson, 1950; Seppälä; Ruttenberg; Smith, 2016). Secondly, since our hypothesis uses a textual representation of domain entities from NLDs, we leverage the Distributional Hypothesis (Harris, 1954) (the basis of state-of-the-art language models), which suggests that words that occur in similar contexts are likely to have similar meanings, i.e., they are closer in distributional space, represented by the embedding vector. Therefore, since NLDs provide the intended meaning of a domain entity in a particular domain (Seppälä et al., 2016) and possibly similar domain entities have similar NLDs, we can use these NLDs to find the top-level ontology concepts of domain entities.

In order to validate our hypothesis, we proposed the extraction of multi-label, multi-language, multi-domain, and multi-resource datasets from BabelNet (Navigli et al., 2021) and domain ontologies available in the OBO Foundry repository (Jackson et al., 2021). These datasets encompass mappings to DOLCE-Lite-Plus (DLP) and Basic Formal Ontology (BFO) top-level ontologies, ensuring a diverse and representative collection of domain entities and corresponding top-level classifications. Also, these datasets

include different textual representations of domain entities, such as NLDs, definienda (terms), definientia (the explanatory part of the NLDs), and example sentences. Specifically, the DOLCE-Lite-Plus dataset from BabelNet covers 118 top-level concepts and contains data in 291 languages from various resources, such as WordNet (Miller, 1995), Wikipedia, and their variations. On the other hand, the BFO dataset from OBO Foundry ontologies covers 33 top-level concepts and includes data from 12 different knowledge domains, encompassing 110 domain ontologies. Also, we proposed two supervised classification pipelines for classifying domain entities into top-level concepts using text as input. The first pipeline utilizes a pre-trained language model that we fine-tuned to our target task. The second involves using a pre-trained language model to generate the embedding representation of the input text. From this embedding, we used classical machine learning classifiers (e.g., K-Nearest Neighbors, Decision Tree, and Support Vector Machine) to classify it into top-level ontology concepts.

In our experiments, we conducted five study cases, each addressing a distinct aspect of classifying domain entities into top-level ontology concepts. The first case focused on evaluating various textual representations, including NLDs, definiendum, and example sentences, and demonstrated the superior performance of NLDs. The second case investigated the approach’s effectiveness across datasets in multiple languages, revealing consistent results with high macro F1 scores. The third case explored cross-resource generalization, where NLDs from one knowledge resource were used for training and another for testing. The fourth case assessed cross-domain performance, testing the pipelines’ ability to generalize across diverse domains. The final case examined the impact of domain-specific NLDs in a cross-ontology scenario, evaluating how embeddings of the NLDs perform when trained and tested on distinct ontologies. The results of our experiments underscore the significant value of leveraging NLDs and embedding-based pipelines in classifying domain entities into top-level ontology concepts. Across the first study case, NLDs consistently outperformed other textual representations, achieving macro F1-scores exceeding 90%. In Study Case 2, NLDs proved robust across languages, with macro F1-scores surpassing 90% for English and Spanish datasets and slightly lower but still strong results for smaller datasets like Japanese and Russian. Study Case 3 demonstrated that embedding-based pipelines like Mistral7B + KNN maintained competitive macro F1-scores of 86.6% in cross-resource scenarios, outperforming fine-tuning approaches. While Study Case 4 highlighted the difficulty of generalizing across domains, with average macro F1-scores often below 40%, incorporating even small amounts of domain-

specific training data significantly improved results to above 60%. Finally, study case 5 revealed strong cross-ontology performance for specific ontologies such as DOID (Disease Ontology) and NCRO (Non-Coding RNA Ontology), which achieved macro F1-scores near 99%, while other ontologies demonstrated varying degrees of F1, underscoring the characteristics and limitations of domain-specific NLDs agist general NLDs of the other study cases.

In the remainder of this chapter, Section 1.1 introduces in detail our work’s context and motivation. Section 1.2 presents our main hypothesis and the objectives of this work. Section 1.3 describes the research questions we seek to answer in this proposal. Section 1.4 shows the intended contributions of our work. Section 1.5 describes the text structure of the thesis.

1.1 Context and Motivation

Ontology engineering is a crucial discipline in knowledge management and AI that involves the systematic development of ontologies, which are structured artifacts representing knowledge within a particular domain. These ontologies serve as a common vocabulary, enabling different systems to understand and process information in the same way. In addition, ontology engineering leverages principles from computer science, linguistics, and philosophy to ensure that the ontologies accurately reflect the knowledge representation of the target domain. The ontology engineering process includes creating, maintaining, and applying ontologies to facilitate data interoperability, knowledge sharing, and reasoning. In this context, ontology development methodologies were proposed in order to systematize and streamline the creation and evolution of ontologies. These methodologies offer guidelines and best practices to ensure the development of robust, scalable, and reusable ontologies. Common methodologies include METHONTOLOGY (Fernández-López; Gómez-Pérez; Juristo, 1997), the NeOn methodology (Suárez-Figueroa; Gómez-Pérez; Fernández-López, 2011), LOT (Poveda-Villalón et al., 2022), each with distinct phases such as specification, conceptualization, formalization, implementation, and maintenance. These phases help ontology engineers to capture domain knowledge systematically, convert this knowledge into formal representations, and implement these representations in ontology languages like OWL (Ontology Web Language).

At the core of ontology development is the use of top-level ontologies, which provide a standardized and generalized framework forming the backbone of more specific

ontologies. In this context, top-level ontologies define common concepts and relationships across various domains, such as objects, events, qualities, partonomic relationships, etc. These concepts and relationships serve as a reference point and ensure consistency and interoperability across different domains. For example, top-level ontologies help standardize the classification of physical objects in manufacturing, biological entities in healthcare, and geological objects and exploration activities in the petroleum and geology domain. Prominent examples of top-level ontologies include the Basic Formal Ontology (BFO) (Otte; Beverley; Ruttenberg, 2022), the Descriptive Ontology for Linguistic and Cognitive Engineering (DOLCE) (Borgo et al., 2022), and the Unified Foundational Ontology (UFO) (Guizzardi et al., 2022). However, identifying which top-level ontology concept a domain entity specializes in is laborious and time-consuming. This task is usually performed manually by an ontology engineer and requires high expertise in the target domain and ontology engineering. For instance, accurately categorizing new geological terms within a petroleum ontology demands geology knowledge and proficiency in ontology engineering, highlighting the complexity and skill needed in this meticulous process.

Reducing the manual effort required during ontology development while maintaining the quality of the produced ontology is a challenge addressed by several ontology learning approaches (He et al., 2022; He et al., 2024; Giglou; D’Souza; Auer, 2023). The main goal of these approaches is to automatically or semi-automatically construct ontologies from various data sources. This involves extracting knowledge from text, databases, or other structured or unstructured data and organizing it into a formal representation of concepts, relationships, and instances within a specific domain. The techniques involved in ontology learning approaches include Natural Language Processing (NLP) in order to extract domain entities and relationships from text, Data Mining for discovering patterns in structured data, and Machine Learning to automatically generate taxonomies or hierarchies and recognize and infer complex relationships. Despite the advancements in ontology learning, a significant gap remains in approaches that incorporate top-level ontologies into the resulting extracted ontologies. Specifically, there is a need for methods that provide automatic classification of domain entities into top-level ontology concepts or that serve as decision support tools for ontology engineers to verify and refine these classifications.

In our view, current approaches for ontology learning can not handle the task of classifying domain entities into top-level ontology concepts primarily because they do not

have this goal. Also, these approaches can not understand the implications of selecting one top-level concept over another since choosing a particular top-level concept can have significant consequences for the overall structure and interpretation of the ontology, influencing how domain knowledge is represented and utilized. For example, consider the domain entity "rock". In the DOLCE top-level ontology, "rock" could be classified under the concept "Amount of Matter", which emphasizes its composition as a specific quantity of material, focusing on its mass, chemical composition, and physical properties. Alternatively, it might be classified under "Physical Object", highlighting its status as a discrete, identifiable entity with a defined shape and boundary. These complexities highlight the need for novel methods to address these challenges and accurately and consistently use top-level ontologies in ontology learning.

In this thesis, we address the problem of classifying domain entities into top-level ontology concepts by exploring the potential of the combination of textual representation of domain entities, Language Models (LM), and Machine Learning (ML) classifiers. The main goal is to propose an approach that uses textual information about domain entities – generally present during the specification phase of ontology development methodologies – as input for a classification pipeline. This textual information includes terms, example sentences, and NLDs. In this context, we explore two different classification pipelines:

- Investigating Language Models as both encoders and classifiers, where the NLDs of domain entities are used as input, encoded with a state-of-the-art Language Model, and then utilized for classification into target top-level ontology concepts.
- Exploring the use of Language Models as encoders and traditional Machine Learning approaches as classifiers, where the input textual information of a domain entity is encoded using a state-of-the-art Language Model to generate a vectorized embedding. Then, we can classify these embeddings into target top-level ontology concepts using traditional Machine Learning algorithms, such as Decision Trees, Support Vector Machines, and K-Nearest Neighbors classifiers.

1.2 Hypothesis and Objectives

Our main hypothesis is that NLDs encapsulate crucial semantic information for associating domain entities with specific top-level ontology concepts. This hypothesis is based on the principles of ontology-based semantic similarity and the distributional hy-

pothesis of textual embedding representation. Ontology-based semantic similarity leverages the taxonomic structure of ontologies. This structure groups ontology entities into categories based on shared characteristics, similar to biological taxonomies. For instance, ontology entities within a category A_1 are more similar to each other than to those in a different category B_1 . Also, this taxonomic structure organizes entities hierarchically, i.e., if category A_1 belongs to a more abstract category A , all entities in A_1 also belong to A . As seen in OWL¹(Ontology Web Language) ontologies, ontology engineers typically construct these taxonomies through "is-a" or subclass-of relationships. On the other hand, the distributional hypothesis in linguistics suggests that words occurring in similar contexts tend to have related meanings (Harris, 1954). The concepts of semantic relatedness in textual embedding and semantic similarity in ontologies are often considered interchangeable. However, many authors view semantic similarity as a specific type of semantic relatedness, with the former focusing solely on taxonomic relationships and the latter encompassing any kind of relationship. Based on these notions, our hypothesis assumes that similar ontology entities have similar NLDs, which are closer to each other in the distributional space of textual embedding from Language Models (LMs) and can be classified using Machine Learning algorithms into top-level ontology concepts. From that, our main objectives in this work are:

- Identify methods to textually represent domain entities early in the ontology development process;
- Extract multi-source, multi-label, and multi-language datasets for benchmark validation;
- Propose two text classification pipelines that output the top-level concepts of domain entities in a multi-label scenario;
- Propose a training methodology to deal with the characteristics of the problem of classifying domain entities into top-level ontology concepts;
- Evaluate the effectiveness of our hypothesis by comparing its performance with other textual representation methods of domain entities;
- Evaluate the effectiveness of our hypothesis using different Language Models;
- Evaluate the effectiveness of our hypothesis across different languages.

¹<https://www.w3.org/TR/owl-ref/>

1.3 Research Questions

Based on our motivation and hypothesis, we define research questions encapsulating the main topic: classifying domain entities into top-level ontology concepts. These questions are designed to guide our investigation and ensure we thoroughly explore the relevant aspects of this classification process.

Research Question 1: How can NLDs of domain entities be leveraged to classify them into top-level ontology concepts?

Ontology engineering involves classifying domain entities into top-level ontology concepts, a laborious and time-consuming task requiring significant expertise, and current ontology learning approaches do not sufficiently address this challenge. Based on these factors, our research aims to investigate an approach that utilizes NLDs to automate this classification process using Language Models and Machine Learning classifiers in order to reduce the manual effort required to perform this task.

Research Question 2: What are the most effective Language Models and Machine Learning algorithms for encoding and classifying textual representations of domain entities into top-level ontology concepts?

This research question aims to explore and compare different Language Models and Machine Learning algorithms to determine their suitability for this classification task. Language Models can encode textual information into vectorized embeddings, which can be classified using various Machine Learning algorithms. Based on that, we aim to identify the most effective combinations for accurately classifying domain entities into top-level ontology concepts.

Research Question 3: How can the proposed classification approach be used in multi-source, multi-label, and multi-language scenarios?

Ontology development often involves data from multiple kinds of sources and languages, and domain entities may belong to multiple top-level ontology concepts. Our research aims to explore the proposed classification approach to handle these complexities effectively. We will investigate methods to create comprehensive datasets, develop training methodologies, and evaluate the performance of our approach across different languages and sources in a multi-label classification scenario. From that, we aim to ensure the robustness and applicability of our approach in diverse contexts, enhancing its usability for ontology engineers and the broader knowledge engineering community.

1.4 Intended Contributions

This thesis proposal aims to address several key challenges in ontology engineering by leveraging advanced Natural Language Processing (NLP) techniques and Machine Learning algorithms. The intended contributions of this work are as follows:

Automated classification of domain entities into top-level ontology concepts:

The thesis proposes a novel approach for automatically classifying domain entities into top-level ontology concepts using NLDs. This approach reduces the manual effort required from ontology engineers and aims to improve the accuracy and efficiency of ontology development processes.

Utilization of NLDs: By leveraging NLDs, the research demonstrates that this kind of textual representation encapsulates crucial semantic information, enabling more accurate classification of domain entities. In this context, we use the principles of ontology-based semantic similarity and the distributional hypothesis of textual embedding representation to support this hypothesis.

Multi-label, Multi-language, and Multi-resource Datasets: The thesis introduces a method for extracting and aligning multi-label, multi-language, and multi-resource datasets from OntoWordNet, BabelNet, and DOLCE-Lite-Plus, and from OBO Foundry domain ontologies and Basic Formal Ontology (BFO). From that, we enable the classification approach to be validated across diverse languages and resources, enhancing the robustness and applicability of the proposed method across the entire structure of the top-level ontology, from the most generic to the most specific concepts.

Supervised Classification Pipelines: Two independent and alternative supervised classification pipelines are proposed. The first utilizes a pre-trained language model fine-tuned for the target task. In contrast, the second uses a pre-trained Language Model to generate embeddings that are then classified using classical Machine Learning algorithms. We designed these pipelines to efficiently handle the complexities of classifying domain entities into top-level ontology concepts.

Evaluation of Textual Representation Approaches: The thesis evaluates various textual representation approaches for domain entities, including terms, example sentences, and NLDs. The results suggest that using NLDs is the most effective method for representing domain entities and classifying them into top-level ontology concepts.

Training Methodology: A comprehensive training methodology is proposed to deal with the characteristics of the classification problem. Also, we propose a data aug-

mentation approach to increase the number of instances in the extracted dataset by changing the term defined in the Natural Language Definition (NLD).

1.5 Thesis Proposal Structure

Chapter 2 delves into the theoretical background underpinning this work, starting with a discussion on ontologies, top-level ontologies, and ontology development methodologies, highlighting their importance in knowledge modeling. Also, we present the evolution of text representation methods in Natural Language Processing and cover multi-label classification techniques, setting the stage for the proposed classification pipelines in this work.

Chapter 3 reviews existing literature on text classification and ontology learning, identifying gaps and opportunities for new contributions. Also, we revise the current trends in ontology learning using Language Models in order to align domain ontologies or domain ontologies and top-level ontologies.

Chapter 4 details the methodology for classifying domain entities into top-level ontology concepts using NLDs and Large Language Models, describing the extraction of the datasets from existing ontologies developed based on Dolce-Lite-Plus (DLP) and the Basic Formal Ontology (BFO), top-level ontologies. Also, we provide a deep discussion about how domain entities can be represented textually and the strengths and weaknesses of different approaches, such as using definiendum, definiens, example sentences, and NLDs. In addition, we outline two main classification pipelines employing Language Models as classifiers and generating embeddings for classical machine learning classifiers, respectively. Finally, we discuss how we can use different training methodologies and data augmentation, considering the NLDs of domain entities.

Chapter 5 presents the experimental setup and the results of five study cases conducted to evaluate the proposed approach, examining different textual representation methods, multiple languages, various knowledge resources, knowledge domains, and domain ontologies. Also, we conclude by discussing the findings and highlighting the strengths and limitations of the proposed methods.

Chapter 6 summarizes the key findings, reaffirming the hypothesis that NLDs can effectively be used to classify domain entities into top-level ontology concepts using Large Language Models. Also, we discuss the broader implications and future directions for the research and emphasize the significance of developing automated tools to assist

ontology engineers and the contributions to ontology engineering.

2 THEORETICAL BACKGROUND

This chapter presents the role of ontologies in knowledge modeling and computer science, contrasting its philosophical and computational perspectives. In this context, we examine top-level ontologies and underscore their importance for consistency and interoperability in knowledge modeling. Additionally, we describe the linguistic and cognitive aspects of definitions in ontologies. After that, we explore the evolution of Natural Language Processing (NLP), detailing advancements from basic text representation methods like one-hot encoding to sophisticated transformer-based language models. We also discuss how semantic similarity works in ontologies, emphasizing how ontology engineers structurally organize domain entities in ontologies. In addition, we also review various machine learning algorithms for multi-label classification, focusing on Decision Trees, K-Nearest Neighbors (KNN), and Artificial Neural Networks (ANNs).

2.1 Ontologies

In this section, we delve into various aspects of ontologies in the context of knowledge modeling and computer science while also touching on philosophical perspectives. Firstly, we introduce the concept of ontology, highlighting its varied meanings in philosophy and computer science. Also, we discuss the classification of ontologies based on generality levels, ranging from top-level ontologies to domain ontologies. Secondly, we focus on top-level ontologies, explaining their significance in ensuring consistency and interoperability in knowledge modeling. Finally, we address the role of definitions in ontologies, focusing on their linguistic and cognitive aspects.

2.1.1 What is an Ontology?

Ontologies are powerful tools to support knowledge modeling and computer science. The term "ontology" originates from philosophy and has different meanings and uses in various communities. According to Guarino, Oberle and Staab (2009), the most divergent senses of ontology come between Philosophy and Computer Science, wherein the first ontology meaning refers to the study of the nature and structure of things per se, independently of any further considerations, and even independently of their actual

existence. On the other hand, in Computer Science, an ontology is a special kind of information object or computation artifact that is used to model the structure of a system formally. In computer science, there are two main different perspectives about ontologies: conceptualism and realism.

In the Conceptualism perspective, Studer, Benjamins and Fensel (1998) merged the definitions of ontology proposed by Gruber (1993) and Borst and Borst (1997) into "*an ontology is a formal and explicit specification of a shared conceptualization*", where: A *conceptualization* refers to an abstract model of some phenomenon in the world by having identified the relevant concepts of that phenomenon; *explicit* means that the type of concepts used, and the constraints on their use are explicitly defined. For example, in medical domains, the concepts are diseases and symptoms, the relations between them are causal and a constraint is that a disease cannot cause itself; *formal* refers to the fact that the ontology should be machine-readable, which excludes natural language; *shared* reflects the notion that an ontology captures consensual knowledge, that is, it is not private to some individual, but accepted by a group.

In the Realism perspective, Smith et al. (2006) proposed one of the most popular definitions of ontology, that says: *An ontology is a representational artifact, comprising a taxonomy as proper part, whose representations are intended to designate some combination of universals, defined classes, and certain relations between them.* In this definition, a *representational artifact* refers to a structured entity that embodies information in a way that is accessible and interpretable by both humans and machines. A *taxonomy* is a hierarchical structure that organizes concepts or entities into groups and subgroups, often based on their similarities or relationships. An *universal* is a general or broad category that applies to many entities that share common properties that give them identity within the ontology, serving as a fundamental building block. A *defined class* is a more specific category within the ontology, characterized by a set of attributes or relations that distinguish their instances from other class instances. A *relation* is a connection or association between two or more concepts or entities within the ontology.

Besides these philosophical views of ontologies, ontologies in computer science are often implemented using formal languages like the Web Ontology Language (OWL), enabling machines to process and understand structured information. From that, ontologies bridge human conceptual understanding and machine readability, thus facilitating more efficient and accurate data processing, analysis, and decision-making in complex systems. Therefore, ontologies are prevalent in various fields, including AI, semantic

web, information retrieval, and bioinformatics, because they play a critical role in enabling semantic interoperability between different systems. In addition, there are several ways to classify ontologies in terms of the domain covered, the formal language used, or the level of generality of the entities defined in the ontology structure (Guarino, 1998; Prestes et al., 2013). In this last case, a consensus in the literature proposes four levels of generality in which an ontology can be classified:

- **Level 1 - Top-level ontologies:** the most general type of ontology, containing concepts like space, time, matter, object, event, and action, independent of a particular problem or domain.
- **Level 2 - Core ontologies:** are viewed as mid-level, bridging top-level and domain ontologies. They reuse concepts from top-level ontologies and specify new ones that can be used across various domains and tasks.
- **Level 3 - Domain and task ontologies:** are more specific types of ontology, detailing vocabulary related to a generic domain (like medicine) or a generic task/activity (like diagnosing), respectively.
- **Level 4 - Application ontologies:** are even more specific, combining domain and task concepts tailored to a particular application, often representing roles played by domain entities in specific activities.

In this work, we use the term "domain entity", referring to any entity described in Levels 2, 3, and 4 of the types of ontology described above. Also, the term "top-level ontology concept" refers to any entity described in a top-level ontology. From that, we aim to propose an approach to automatically classify these domain entities into concepts defined in top-level ontologies (Level 1) using their NLDs, Natural Language Processing (NLP) approaches, and Machine Learning (ML) algorithms.

2.1.2 What is a Top-Level Ontology?

Top-level ontologies, often referred to as upper-level ontologies, foundational ontologies, or general ontologies, serve as abstract frameworks that provide the most general concepts (e.g., space, time, matter, object, event, action, etc.) and relationships (e.g., quality-of, part-of, cause-effect, etc.) applicable across various domains (Otte; Beverley; Ruttenberg, 2022; Borgo et al., 2022; Guizzardi et al., 2022). These ontologies establish a universal scaffold, ensuring consistency and interoperability in knowledge representa-

tion. In this sense, by defining a standardized set of categories and principles, top-level ontologies facilitate the integration and alignment of domain ontologies from diverse domains, promoting a common understanding across different fields of study. This universal approach helps to build a shared knowledge artifact that can be extended and specialized for specific applications (Guarino, 1998).

The philosophical foundations of top-level ontologies are deeply rooted in the philosophy discipline of metaphysics, particularly the study of being and existence. For instance, the Basic Formal Ontology (BFO) (Otte; Beverley; Ruttenberg, 2022) exemplifies a realist approach by categorizing entities based on their existence in reality, independent of human cognition. BFO focuses on entities that can be objectively observed and classified, such as biological processes, physical objects, and spatial-temporal regions, grounding the ontology in a metaphysical framework of existence and reality. In another example, the Descriptive Ontology for Linguistic and Cognitive Engineering (DOLCE) (Borgo et al., 2022) is more concerned with the cognitive and linguistic aspects of how humans conceptualize and categorize the world. DOLCE incorporates distinctions that are more closely aligned with human perception and linguistic structures, making it suitable for applications involving human cognition and natural language processing. Also, DOLCE strongly emphasizes the descriptive aspects of entities, considering how they are represented in human thought and language. From that, DOLCE is more attuned to human conceptualizations than the more reality-focused categorization of BFO.

In addition to the complexities of the ontological commitments of different top-level ontologies, they can also present different taxonomic structures. In Figure 2.1, we present the taxonomy of the OWL¹ (Ontology Web Language) version of BFO and DOLCE-Lite-Plus (DLP) top-level ontologies. This figure shows that while the BFO is divided into "continuant" and "occurrent", the DLP concentrates on the nature of the existence of their concepts, divided into "abstract" and "spatio-temporal" particulars. Going deeper into these taxonomies, we see that their top concepts are increasingly becoming ontologically different. Although the ontological structure of both taxonomies are different, we can note certain characteristics in common, such as the use of different concepts to represent entities that persist through time (continuants and endurants) and entities that present different temporal parts over time (occurrents and perdurants), and the notion of qualities, which describe attributes or characteristics that entities possess

So far, we have been relatively shallow in describing the complexities of top-level

¹<https://www.w3.org/TR/owl-ref/>

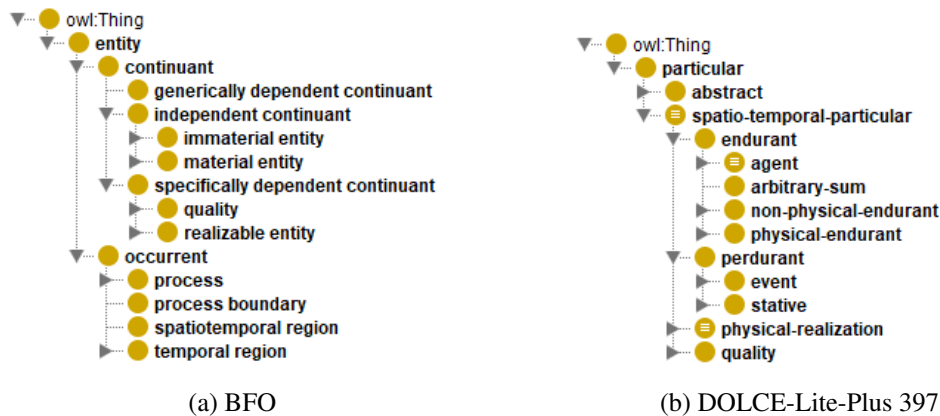


Figure 2.1 – Taxonomic structure of BFO and DOLCE-Lite-Plus top-level ontologies. (Source: author)

ontologies, which ontology engineers must know (or even master) when developing a domain ontology based on the ontological commitments of the top-level ontology concepts. Along the same line, we started to ask ourselves: how can we reduce the effort required from the ontology engineer to carry out this task? The answer seems to be quite simple: Machine Learning. However, what should we use? How can we design a classification pipeline to perform this task? We will discuss these questions during the course of this proposal.

2.1.3 Definitions in Ontologies

According to Robinson (1950), a statement that clarifies the meaning of a term or entity conceptualizes a definition. In his work, Robinson posits that definitions intend to elucidate the essence of what is being defined, thereby distinguishing it from other entities. He emphasizes that a good definition must be clear, precise, and concise, avoiding ambiguity or circular reasoning. Definitions are not just seen as linguistic tools; they are fundamental to logical analysis and philosophical inquiry. Also, definitions play an instrumental role in advancing understanding, facilitating communication, and providing a foundation for further exploration across various fields of knowledge. From this perspective, definitions are crucial elements in the structure of knowledge, aiding in the articulation and differentiation of entities essential to intellectual discourse.

Other works, such as Seppälä, Ruttenberg and Smith (2016) and Seppälä et al. (2016), extend the contributions on definitions in ontologies. According to them, the definition of the term "definition" can vary in nature, depending on their context of use and target audience. However, definitions are essential tools for communication and un-

derstanding, specifically designed to align with the cognitive and linguistic requirements of their intended context. From a linguistic perspective, definitions aim to align with a certain pre-existing lexical use, conveying the semantic value of a term by delimiting its intension and extension, adjusting the overall lexical competence of users by enhancing their inferential competences, particularly semantic inferential competence. On the other hand, from a cognitive perspective, definitions in ontologies bring about a reconfiguration of the receiver's existing body of knowledge regarding the intended referent of the defined term, working to augment and reconfigure the knowledge and beliefs of the user to align them more closely with those of relevant expert communities. Overall, definitions help in disambiguating terms and ensuring consistency in their use by providing clear and unambiguous intended meaning (often using Aristotelian principles) to distinguish a term from neighboring terms.

In addition to the differences between the above perspectives about definitions, they can also differ based on their original resources. Table 2.1 describes the relationships between the several criteria about definitions from different kinds of definition resources, in this case, examining lexicographic, lexico-semantic, terminological resources, like WordNet (Miller, 1995), versus ontologies, standards, normative terminologies, such as IOF (Kulvatunyou et al., 2022) or the ontologies that follow the guidelines of OBO Foundry (Jackson et al., 2021; Smith et al., 2007). Also, in our work, we used this subdivision to split our discussion about NLDs and our experiments and results. In this context, as described in Seppälä et al. (2016), there are four criterion to compare the definition resources:

- **Lexical Use:** Refers to how words are employed within resources. Lexicographic resources tend to describe, focusing on typical usage and meaning in language. Ontologies, on the other hand, prescribe, setting standards for usage to ensure clarity and prevent ambiguity.
- **Logical Form of Definitions:** This criterion relates to the structure and reasoning underlying the definitions. Lexicographic resources often employ prototypical forms, describing general characteristics or typical examples. Ontologies use necessary and sufficient conditions, establishing precise and formal criteria for term application.
- **Utility:** This aspect considers the purpose of the resource. Lexicographic resources primarily aim to convey meaning, offering insights into word usage and interpretation. Ontologies are more focused on disambiguating meaning, providing clear and

Table 2.1 – Relationships between the types of resources where definitions can be extracted.
(Source: Seppälä et al. (2016))

Criteria	Type of resource	
	Lexicographic, lexico-semantic, terminological	Ontologies, standards, normative terminologies
Lexical use	Describe	Prescribe
Logical form of definitions	(Proto)typical	Necessary & sufficient conditions
Utility	Conveying meaning	Disambiguating Meaning
Cognitive effect	Knowledge/belief augmenting	Knowledge/belief reconfiguring

unambiguous definitions to aid in precise communication and understanding.

- **Cognitive Effect:** This refers to the impact on the user's knowledge or beliefs. Lexicographic resources tend to augment knowledge or beliefs by adding new information or perspectives. Ontologies, conversely, reconfigure knowledge or beliefs, often changing or refining the user's understanding of a term or concept.

Following these notions of definitions in ontologies from Robinson (1950) and Seppälä, Ruttenberg and Smith (2016) works, in this thesis, we will talk about only NLDs, which are textual descriptions written in natural language that explain the meaning, characteristics, or context of a domain entity within an ontology. The NLDs have a canonical form that derives from the Aristotelian form of the definition "X is a Y that Z". For example, "An amount of rock is a solid consolidated earth material that is constituted by an aggregate of particles made of mineral matter or material of biological origin". - example extracted from GeoCore (Garcia et al., 2020). In this example, "X" is the *Definiendum*, "Y that Z" is the *Definiens*, and "is a" is the *Copula*. The *Definiendum* is the defined term, the subject of the NLD, i.e., what the NLD is about. For example, "An amount of rock". The *Definiens* is the part that expresses the content of the NLD, i.e., the explanatory part of the NLD that specifies what the *Definiendum* is. For example, "solid consolidated earth material that is constituted by an aggregate of particles made of mineral matter or material of biological origin". The *Copula* is the linking part that expresses an equivalence or subsumption between the *Definiendum* and the *Definiens*, i.e., the *Copula* connects the defined term to its explanation or description, establishing a relationship of identity or classification between them.

The use of NLDs is the main part of the hypothesis of this work, which says that "NLDs encapsulate crucial semantic information for associating domain entities with specific top-level ontology concepts". At first view, using the Aristotelian form of definitions, the association between a domain entity and its top-level ontology concept can

be "easily" achieved if the "Y" part of the *Definiens* is the label of the top-level ontology concept. However, in ontology development methodologies such as METHONTOLOGY (Fernández-López; Gómez-Pérez; Juristo, 1997), NeOn (Suárez-Figueroa; Gómez-Pérez; Fernandez-Lopez, 2015), and LOT (Poveda-Villalón et al., 2022), NLDs are provided early than the conceptualization stage. At this point, there is no established relation between the domain entities being defined and any top-level ontology. Consequently, the NLDs are constructed in isolation without reusing information contained in a top-level ontology.

In this work, we focused on two kinds of NLDs: general NLDs (from lexicographic, lexico-semantic, and terminological resources) and domain-specific NLDs (from domain ontologies, standards, and normative terminologies). In our view, this distinction allowed us to analyze the differences in the structure, intent, and semantic precision of NLDs based on their origins. For instance, general NLDs from glossaries prioritize user comprehension and linguistic flexibility. On the other hand, domain-specific NLDs from domain ontologies prioritize logical clarity and formal reasoning, often requiring domain expertise to interpret or apply them effectively. In this context, we extracted NLDs from resources such as WordNet, Wikipedia, BabelNet, and domain ontologies contained in the OBO Foundry repository in order to validate our hypothesis. Also, we considered as valid NLDs those that follow the whole structure of the Aristotelian form of definitions containing the *Definiendum*, the *Definiens*, and the *Copula* as the input sentences of our approach.

2.1.4 Methodologies for Developing Ontologies

Ontology development methodologies have evolved significantly to address the complex challenges of knowledge representation, sharing, and reuse in diverse domains. In this context, methodologies including METHONTOLOGY (Fernández-López; Gómez-Pérez; Juristo, 1997), NeOn (Suárez-Figueroa; Gómez-Pérez; Fernández-López, 2011; Suárez-Figueroa; Gómez-Pérez; Fernandez-Lopez, 2015), LOT (Poveda-Villalón et al., 2022), SABiO (Falbo, 2014), and SABiOx (Aguilar; Souza, 2024) offer distinct approaches such as goal definition, scalability, and flexibility, while contributing uniquely to the field of ontology engineering, filling gaps and improving practices identified during the ontology development process. METHONTOLOGY, for example, emphasizes defining clear goals and domain-specific applications through systematic specification and conceptualization phases. This structured approach ensures that the ontology meets its intended

objectives, whether for knowledge sharing, interoperability, or reasoning. The NeOn methodology builds on this by focusing on the creation of ontology networks, enabling distributed teams to collaboratively construct ontologies through the reuse and reengineering of both ontological and non-ontological resources. LOT further extends this by targeting lightweight ontologies for industrial applications, particularly in alignment with Semantic Web principles, addressing challenges related to publishing and interoperating ontologies in linked data ecosystems. SABiO and SABiOx, meanwhile, emphasize the development of reference and operational ontologies, catering to rigorous domain-specific needs by incorporating foundational ontologies and ontological analysis into the development process.

The selection of methodologies is guided by criteria such as domain relevance, scalability, and flexibility. METHONTOLOGY provides a robust foundation for building ontologies from scratch but is less suited for the dynamic, collaborative needs addressed by NeOn. NeOn introduces nine scenarios for modular ontology development, making it well-suited for large-scale, distributed projects where reuse and alignment are essential. LOT, on the other hand, offers a lightweight and streamlined approach, making it particularly relevant for projects with time constraints or limited resources. SABiOx extends the SABiO framework by integrating agile principles, enabling iterative development cycles that support continuous refinement and evolution of ontologies, which is particularly beneficial for novice ontology engineers.

Although each ontology development methodology was proposed with different goals, there are several development stages common across these methodologies, including (in order) specification, knowledge acquisition, conceptualization, formalization, implementation, evaluation, and maintenance. METHONTOLOGY provides detailed guidelines for the specification and conceptualization stages, laying the groundwork for structured ontology creation. NeOn enriches this process by integrating modularity and reuse scenarios, enabling developers to iteratively refine and extend ontologies within a collaborative environment. SABiO and SABiOx offer a distinct phase named ontological analysis, to ensure that domain concepts are rigorously defined and grounded in top-level ontologies. LOT simplifies the development process by focusing on requirement elicitation, implementation, and publication, ensuring compatibility with Semantic Web best practices and agile workflows. These methodologies also propose diverse evaluation techniques to ensure ontology quality. METHONTOLOGY and SABiO stress the importance of validation through competency questions, while NeOn incorporates metrics such as

consistency and alignment across ontology networks. LOT emphasizes the evaluation of interoperability and adherence to linked data standards, addressing challenges in publishing and consuming ontologies in web-based ecosystems. SABiOx integrates iterative testing and refinement cycles, leveraging both user feedback and formal validation methods to ensure the ontology’s fitness for its intended purpose. However, challenges persist in ontology development methodologies, mainly in making the stages less time-consuming and improving the efficiency of the ontology development process.

Some of the above ontology development stages can be (semi-)automated using ontology learning approaches (as we present in Section 3.2). However, the task of associating domain entities with top-level ontology concepts, as recommended in NeOn, SABiO, and SABiOx methodologies, remains a significant bottleneck in ontology development due to its reliance on manual effort and domain expertise. In this context, we propose automating this process through the use of NLDs, state-of-the-art Language Models, and Machine Learning classifiers. Our approach is specifically designed to be integrated into the ontology development workflow immediately after the knowledge acquisition stage, where informal definitions of domain entities have been gathered, and before the conceptualization stage, where these entities are formalized into structured ontological concepts. By automating the classification of domain entities into top-level ontology concepts at this intermediate stage, we aim to improve efficiency and fill a gap in one of the most labor-intensive aspects of ontology engineering.

2.1.5 Semantic Similarity and Relatedness in Ontologies

A typical ontology is structured to represent knowledge in a domain by organizing concepts and their relationships formally and hierarchically, as described by Gruber (1993) and Studer, Benjamins and Fensel (1998). Therefore, the foundation of any ontology lies in its taxonomic structure, which organizes classes (concepts) and instances (individuals) hierarchically through subclass and instance-of relationships. Typically represented as a directed acyclic graph, this hierarchy facilitates the classification of classes and instances based on their degree of generality or specificity. For example, in a biological ontology, "Animal" might be a superclass with subclasses such as "Mammal", "Bird", and "Reptile". A key property of this hierarchical organization is inheritance, where subclasses inherit the properties and restrictions of their superclasses, ensuring a structured and logically consistent organization of domain knowledge (Guarino; Oberle; Staab,

2009). In addition to taxonomic relationships, ontologies also define non-hierarchical relationships, such as "hasParent", "isPartOf", or "teaches", which capture domain-specific relationships and attributes of classes and individuals, enriching the taxonomy with detailed context (Smith, 2004).

From the structure of ontologies, ontology-based semantic similarity measures the degree of similarity, likeness, or resemblance between two ontology entities (Chandrasekaran; Mago, 2021; Kulmanov et al., 2021; Sousa; Silva; Pesquita, 2024). In this context, the hierarchical structure of an ontology allows for the calculation of similarity based on the position of entities within the taxonomy, i.e., entities that share a common ancestor (Lin, 1998; Resnik, 1999), especially one that is closer in the hierarchy, are considered more similar. For example, in a biological ontology, "Lion" and "Tiger" would be deemed more similar than "Lion" and "Fish" due to their closer common ancestor, "Feline". Other approaches rely on measuring the path length between entities based on only taxonomic relationships (Rada et al., 1989), where closer entities are considered more similar. In addition to taxonomic distance and common ancestors, properties are also employed to measure semantic similarity, where two entities that share more properties in common are more similar (Tversky, 1977). In contrast, semantic relatedness evaluates how much two ontology entities are associated, encompassing both hierarchical and non-hierarchical relationships (Formica; Taglino, 2023). For example, while "Doctor" and "Hospital" are not similar in a taxonomic sense, they are strongly related due to their functional association.

At first view, the differences between ontology-based semantic similarity and relatedness measures might seem to rely solely on the type of relationships used—hierarchical (taxonomic) for similarity and a mix of hierarchical and non-hierarchical for relatedness. However, in our view, this distinction is not merely practical but reflects deeper cognitive perspectives that influence how these measures are applied and the meaning of their results. While similarity often assumes a structured, deductive framework grounded in shared properties or proximity within a taxonomy, relatedness incorporates a more associative reasoning process that accounts for broader contextual relationships. For instance, when categorizing domain entities into top-level ontology concepts, the nature of similarity aligns well with the hierarchical organization of ontologies. Conversely, the relatedness notion may be more suitable for tasks that require associative reasoning, such as linking entities in a knowledge graph or identifying contextual relationships (Khadir; Aliane; Guessoum, 2021).

This structured perspective on similarity is particularly relevant when considering NLDs, which inherently embed features of generic concepts within specific ones (as described in Section 2.1.3). NLDs create a semantic bridge between hierarchical levels, reflecting the structured organization of ontologies. For instance, "A dog is a common four-legged animal, especially kept by people as a pet or to hunt or guard things" not only describes the specific entity "Dog" but also includes attributes of broader entities, such as "Animal" and "Pet." This layered semantic inheritance aligns with the taxonomic nature of similarity, reinforcing its role in ontology-based classification. Consequently, this thesis employs the similarity notion intrinsic to hierarchical ontology structures to evaluate the capacity of language models to encode NLDs and classify them into general entities contained in top-level ontologies.

2.2 Vectorized Text Representation

In this section, we deeply describe the evolution and advancements in the field of Natural Language Processing (NLP), mainly focusing on text representation and embedding learning methods. Firstly, we discuss the Distributional Hypothesis, illustrating its importance in linguistics and computational linguistics, which influenced the development of word embedding and state-of-the-art language models. After this, we review the initial stages of computational linguistics, highlighting the use of one-hot encoding and Bag of Words (BOW) for representing textual data. Following, we introduce the significant advancement in NLP with Word2Vec, detailing the CBOW and Skip-Gram architectures. Finally, we present the state-of-the-art text representation approaches, explaining the contextual word embedding approaches, like ELMo, and their advancements with transformer-based models, like BERT and GPT, and cutting-edge technologies, such as T5, LLAMA, Gemini, and Mixtral language models.

2.2.1 Distributional Hypothesis and Semantic Relatedness

The Distributional Hypothesis, proposed by Harris (1954), presents a foundational idea in linguistics that suggests words occurring in similar contexts likely have similar meanings. In his work, Harris proposed a method for inferring the semantics of words by analyzing their distributional patterns across texts. Also, his work emphasized the

importance of context in understanding linguistic meaning by discovering the semantic attributes of words through a systematic examination of their usage in various linguistic environments. In their experiments, Harris employed a framework for empirical language analysis that relies on observable, quantitative data, shifting away from more introspective or purely theoretical methods. As a result, the framework influenced how linguists and computational linguists could study language. His hypothesis also significantly impacted the development of technologies like word embedding (Mikolov et al., 2013b; Mikolov et al., 2013a; Pennington; Socher; Manning, 2014) and language models (Radford et al., 2018; Devlin et al., 2018; Touvron et al., 2023; Jiang et al., 2024). These models represent text as vectors in a high-dimensional space, where the proximity between these vectors indicates their semantic relatedness.

Although the Distributional Hypothesis states that "words occurring in similar contexts likely have similar meanings", the interpretation of "similar" should not be conflated with purely "semantic similarity" (as described in Section 2.1.5). In this context, the term "similar" is better understood as "related", because the hypothesis is strongly influenced by word co-occurrence patterns, which capture not only similarity but also broader relatedness. Word co-occurrence reflects both hierarchical relationships, such as those between generic and specific concepts, and non-hierarchical relationships, such as those between contextually linked but taxonomically unrelated entities. For instance, the co-occurrence of "doctor" and "hospital" indicates relatedness through functional association, not semantic similarity, while the co-occurrence of "lion" and "tiger" reflects a closer taxonomic similarity. Therefore, since the basis of state-of-the-art language models is constructed based on the Distributional Hypothesis, this distinction highlights that the distributional nature of language models reflects patterns of semantic relatedness in the training data, not only similarity.

2.2.2 Early Methods and One-Hot Encoding

In the nascent stages of computational linguistics, methods for representing textual data were rudimentary, primarily focusing on direct, discrete representations of words. The main proposals aimed to validate the distributional hypothesis introduced by Harris (1954), which posits that words appearing in similar contexts are likely to share similar meanings. A straightforward example of these initial methods is one-hot encoding, where every word in a given vocabulary is assigned a unique vector.

One-hot encoding is a method for representing text where each word in the vocabulary is assigned a unique vector. The vector has the same length as the total vocabulary size, with a value of "1" at the index corresponding to the word and "0" at all other positions. For example, in a vocabulary of 10,000 words, each word is represented by a 10,000-dimensional vector, making the representation sparse and high-dimensional. While this approach provides a simple way to encode text for computational purposes, it has several limitations. One major drawback is the inefficiency caused by the high dimensionality and sparsity of the textual data, which makes it computationally expensive, particularly for large corpora. Additionally, one-hot encoding does not capture any semantic or syntactic relationships between words. All words are treated as equally distant from one another in the vector space, failing to encode contextual or meaningful relationships between terms, as highlighted in Mikolov et al. (2013b).

The Bag-of-Words (BoW) (Zhang; Jin; Zhou, 2010) model extends one-hot encoding by representing text as a collection of word frequencies rather than individual word vectors. In this model, a text document is converted into a vector where each dimension corresponds to a word in the vocabulary, and the value at each position indicates the word's frequency in the document. Although this approach addresses some sparsity issues in representing entire documents, it still suffers from critical limitations. Like one-hot encoding, BoW does not capture the order or relationships between words, treating a document as a "bag" where context is entirely lost. This lack of contextual representation makes BoW insufficient for tasks that require an understanding of word meanings or sequences. Both one-hot encoding and BoW have served as stepping stones for more advanced text representation methods, which aim to address their inability to capture semantic and syntactic information.

2.2.3 Word2Vec, GloVe and Embedding Learning

Previous approaches to text representation, such as one-hot encoding and Bag-of-Words (BoW), have significant disadvantages that underscore the need for dense embedding vectors, such as high-dimensional vectors where each word is treated as an independent entity. Additionally, the dimensionality of one-hot vectors grows linearly with the size of the vocabulary and the dataset, leading to inefficiency and scalability issues for large datasets. In this context, Word2Vec (Mikolov et al., 2013a; Mikolov et al., 2013b) and GloVe (Pennington; Socher; Manning, 2014) represent two foundational methods for

generating dense, semantically rich word embeddings.

Word2Vec (Mikolov et al., 2013a; Mikolov et al., 2013b) employs a predictive modeling approach, offering two key architectures: Continuous Bag of Words (CBOW) and Skip-Gram. The CBOW model predicts a target word based on its surrounding context, while the Skip-Gram model does the reverse, predicting context words from a given target word. This bidirectional prediction mechanism allows Word2Vec to capture nuanced semantic and syntactic relationships within the embedding space, demonstrated through vector arithmetic capabilities, such as the well-known analogy "king - man + woman = queen". These embeddings reflect the model's ability to encode linguistic patterns beyond simple co-occurrence statistics, significantly enriching word representations in a dense vector.

In contrast, GloVe (Pennington; Socher; Manning, 2014) employs a count-based approach, leveraging a global word-word co-occurrence matrix constructed from a corpus. This matrix stores how frequently pairs of words appear together in a given context window (range of words surrounding a target word that is considered to capture its contextual relationship). GloVe uses matrix factorization to reduce this high-dimensional data into dense vectors, ensuring that words with similar co-occurrence patterns are close in the embedding space. Unlike Word2Vec, which focuses on local context, GloVe emphasizes global statistical patterns, allowing it to capture both fine-grained semantic and syntactic relationships across the corpus.

Word2Vec and GloVe, while pivotal in advancing dense word embeddings, both methods generate static embeddings, meaning each word is assigned a single vector regardless of its context in different sentences. From that, these language models can not capture polysemy or context-specific meanings, such as "bank" referring to a financial institution versus the side of a river. Additionally, these models rely on pre-trained embeddings with a fixed size, which makes them less adaptable to new data. Also, both models rely on co-occurrence patterns, which can lead to embeddings that reflect biases inherent in the training data. Therefore, this kind of language model is limited to low-resource datasets or specialized domains where such corpora are unavailable. However, these limitations have driven the development of contextual embeddings, which aim to address these issues by dynamically adjusting word representations based on context.

2.2.4 Sentence Embeddings and Language Models

The evolution of natural language processing (NLP) from word embeddings to sentence embeddings and language models marks a significant advancement in understanding and processing human language. Early approaches to sentence embeddings were primarily extensions of word embedding techniques, where sentence representations were derived from the embeddings of individual words they contained. Methods like simple averaging of word vectors were common, but these approaches lacked the ability to capture the nuanced semantics and syntax of sentences (Le; Mikolov, 2014).

With the intent to better capture the meaning of entire sentences, models such as the Skip-Thought vectors were developed (Kiros et al., 2015). Based on the principles of the Skip-Gram model in Word2Vec Mikolov et al. (2013a), Mikolov et al. (2013b), Skip-Thought aimed to encode a sentence in such a way that it could predict the sentences around it. Although this model marked progress, it still struggled with capturing the full context and complexity of sentence structures.

The introduction of contextual word embeddings was a major leap forward. One of the first notable models in this field was ELMo (Embeddings from Language Model), introduced by Peters et al. (2018). This model uses a bidirectional LSTM (Long Short-Term Memory) trained on a large text corpus. From that, it provides word embeddings that are functions of the entire input sentence, as opposed to previous methods that generate a fixed embedding for each word. This means that the word "play" would have different embeddings based on its usage in different contexts ("to play a game" vs. "a play in theater"), capturing the nuances and polysemy of words effectively.

The next significant advancement came with the development of transformer-based models, such as BERT (Bidirectional Encoder Representations from Transformers) (Devlin et al., 2018) and GPT (Generative Pre-trained Transformer) (Radford et al., 2018). These models revolutionized NLP by using attention mechanisms to understand the full context of a word in relation to all other words in a sentence or even across multiple sentences. In this context, BERT, in particular, marked a paradigm shift by pre-training on a large corpus of text and then fine-tuning for specific tasks, achieving unprecedented performance across a range of NLP benchmarks. Its bidirectional nature meant it could effectively understand the context from both the left and the right of each word in a sentence, providing a more thorough understanding of language. On the other hand, GPT and its successors, GPT-2, GPT-3, and GPT-4, further expanded on this concept, using

massive amounts of data and computational power to generate highly context-sensitive embeddings and even generating coherent extended text.

DistilBERT (Sanh et al., 2020), RoBERTa (Liu et al., 2019), and ALBERT (Lan et al., 2019) are advancements over BERT focusing on efficiency and performance. DistilBERT reduces the model size by 40% while retaining 97% of BERT’s capabilities through knowledge distillation, making it faster and lighter. RoBERTa optimizes BERT’s pretraining by removing the Next Sentence Prediction objective, extending training, and dynamically changing the masking pattern, significantly outperforming BERT on major benchmarks. ALBERT introduces parameter-reduction techniques and a Sentence Order Prediction loss, achieving better results with fewer parameters and focusing on inter-sentence coherence.

T5 (Text-to-Text Transfer Transformer) (Raffel et al., 2023) and LLAMA (Touvron et al., 2023) represent significant advancements in language model architecture. T5 adopts a unified text-to-text framework, simplifying the diverse landscape of NLP tasks by treating them all as text-to-text problems, which allows for systematic study and comparison of different approaches. This model leverages transfer learning with a new clean crawled corpus, achieving promising results in numerous NLP tasks. On the other hand, LLAMA represents a collection of foundation language models focusing on scalability and accessibility. Ranging from 7 billion to 65 billion parameters and trained on trillions of tokens using publicly available datasets, LLAMA models demonstrate that it is possible to train state-of-the-art models without proprietary data, with some versions outperforming much larger counterparts like GPT-3.

Gemini (Team et al., 2023) and Mixtral (Eliseev; Mazur, 2023) represent cutting-edge multimodal and efficient inference language models. Gemini, developed by Google, is a family of multimodal models with three sizes, each tailored for different computational and application needs. These models are trained across various data types, including image, audio, video, and text. They have shown unique capabilities across different benchmarks, advancing the state-of-the-art in many of them. On the other hand, Mixtral focuses on fast inference with offloading techniques, making it a practical solution for using large language models on consumer hardware.

The field of language models continues to evolve rapidly, with ongoing research exploring even more sophisticated models and techniques for capturing the nuances of human language. The relation between these models and the distributional hypothesis lies in how the models learn: by predicting words based on their contexts (for example, the

words before and/or after a given word in a sentence). This predictive process inherently relies on the assumption that words with similar contexts have similar meanings as the models adjust their internal parameters to reduce prediction errors based on the contexts they observe. Consequently, language models not only learn to predict words accurately but also implicitly learn rich, context-sensitive embeddings for words and phrases that reflect their meanings and relationships, embodying the principles of the distributional hypothesis. From that, our goal in this work is to explore language models in two different architectures: (1) generating embedding of the informal textual definitions and using classical machine learning to classify them; (2) fine-tuning a language model to the specific task of classifying domain entities into top-level ontology concepts using their informal textual definitions as input.

2.3 Multi-label Classification

In this section, we provide an overview of various machine learning algorithms and their application in multi-label classification, focusing on how they handle the complexity of predicting multiple labels simultaneously. Firstly, we explain how decision trees are used in multi-label classification, presenting the advantages and disadvantages. Secondly, we describe the K-Nearest Neighbor (KNN), which can be used for classification and regression tasks and multi-class and multi-label classification problems. After that, we discuss Artificial Neural Networks (ANNs) in multi-label classification, also presenting their evolution, including Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM) networks, attention mechanisms, and transformers. Finally, we explain why we do not review specific approaches in this thesis proposal, such as Support Vector Machines (SVMs), Convolution Neural Networks (CNNs), among others.

2.3.1 Decision Tree

Decision trees (Safavian; Landgrebe, 1991) are widely used in machine learning, particularly in multi-label classification tasks where an instance may be associated with multiple labels simultaneously. Their straightforward structure, where each node represents a feature and each branch a decision, makes them easily interpretable and popular among machine learning researchers (Madjarov et al., 2012).

One significant advantage of decision trees in multi-label classification is their inherent simplicity (Gasparetto et al., 2022). They can be easily visualized and understood, even by those with limited statistical knowledge, making them useful for decision support systems (Zhang; Zhou, 2014). Furthermore, decision trees are versatile and capable of handling numerical and categorical data without extensive preprocessing, such as normalization or dummy variable creation (Gibaja; Ventura, 2015).

However, decision trees have limitations, such as they are prone to overfitting, especially when dealing with complex multi-label datasets. Overfitting occurs when the model captures noise in the training data, leading to poor generalization to new data (Gasparetto et al., 2022). Another disadvantage is their instability; small changes in the data can lead to significantly different tree structures. This problem is often mitigated by using Random Forests, which build multiple trees and average their predictions for improved stability and accuracy (Ho, 1995; Fernández-Delgado et al., 2014).

2.3.2 K-Nearest Neighbor (KNN)

The k-Nearest Neighbors (KNN) algorithm is a widely recognized machine learning method for classification and regression tasks (Tarekegn; Giacobini; Michalak, 2021; Bogatinovski et al., 2022). This algorithm classifies a new instance based on the majority vote of its k nearest neighbors, i.e., it operates on the principle that similar training examples are likely to have the same label. Also, its effectiveness is primarily determined by the choice of k (the number of neighbors considered) and the metric used for measuring the distance between instances (e.g., Euclidean, Manhattan). In addition, it is a lazy learning algorithm, as it does not build a model to classify a new instance but instead makes decisions based on the entire training data for every prediction (Spyromitros-Xioufis et al., 2011).

One of the primary advantages of KNN is its simplicity and ease of implementation. As a non-parametric method, KNN makes no assumptions about the underlying data distribution, making it versatile across different applications. Also, KNN is adaptable because we can use it for multi-class and multi-label classification, regression, and search problems. However, the effectiveness of the KNN is significantly impacted when the decision boundary is irregular.

Despite its simplicity and flexibility, KNN suffers from several drawbacks. The main disadvantage is that KNN is computationally expensive, particularly for large datasets,

as it requires calculating the distance of each query instance to all training samples. The performance heavily depends on the choice of k and the distance metric. A small value of k means that noise will have a higher influence on the result, while a large value makes it computationally expensive (Chen; Guestrin, 2016). Also, KNN is sensitive to the scale of the data and irrelevant features, necessitating appropriate preprocessing steps like feature scaling and dimensionality reduction.

For multi-label tasks, managing the label space's complexity and dependencies between labels becomes an additional challenge (Zhang; Zhou, 2007). As the number of labels increases, the complexity of the model increases, leading to potential overfitting and increased computational demand. Additionally, the need to understand and model the relationships between labels to improve prediction accuracy adds to the algorithm's complexity and computational requirements. Therefore, while KNN and its variants offer a simple and often effective method for classification, they require careful tuning and consideration of the computational and methodological implications, especially in complex tasks like multi-label classification.

2.3.3 Artificial Neural Network

Artificial Neural Networks (ANNs) have significantly impacted multi-label classification, leveraging their capacity for modeling complex patterns and relationships (Tarekegn; Giacobini; Michalak, 2021; Bogatinovski et al., 2022). They adapt to multi-label classification through architecture modifications, loss function adjustments, and incorporating label correlation strategies, demonstrating profound adaptability and effectiveness. In this context, a common approach involves modifying the network's output layer to correspond to multiple labels and employing loss functions capable of handling multiple predictions simultaneously. For instance, a network might feature an output node for each potential label, using sigmoid activations to predict the presence or absence of each label independently. Loss functions like binary cross-entropy are then applied across all output nodes, effectively training the network to make accurate and independent predictions for each label.

Advancing on ANNs, Recurrent Neural Networks (RNNs) (Rumelhart; Hinton; Williams, 1986; Sherstinsky, 2020) and Long Short-Term Memory (LSTM) (Hochreiter; Schmidhuber, 1997; Yu et al., 2019) networks are foundational in processing sequential data, each with unique mechanisms for handling temporal dependencies. RNNs in-

troduced a revolutionary looped architecture, enabling a form of memory and dynamic temporal behavior crucial for tasks such as language understanding and speech recognition. However, they suffer from vanishing and exploding gradients, hindering their ability to learn from long sequences. LSTMs evolved to address these shortcomings with a gating system, allowing better preservation and regulation of information over extended sequences and significantly mitigating the vanishing gradient problem.

Attention mechanisms (Vaswani et al., 2017) further advanced the field by enabling models to dynamically focus on different parts of the input sequence, enhancing the context sensitivity of the outputs. This innovation was particularly transformative in sequence-to-sequence tasks, such as machine translation, where the relevance of different input parts can vary significantly. Attention mechanisms have led to more nuanced and effective models by allowing a selective focus on the most pertinent information at each step of the sequence.

Transformers (Wolf et al., 2020) emerged as a groundbreaking architecture, leveraging the power of attention mechanisms to the fullest. By dispensing recurrence and convolution entirely, transformers process data in parallel, drastically improving efficiency and scalability. This architecture, powered by self-attention and multi-head attention techniques, has set new performance benchmarks across a wide array of tasks, especially in natural language processing. Transformers' ability to handle long-range dependencies and complex patterns in data has made them the dominant model in many cutting-edge applications.

Despite the advances brought by RNNs, LSTMs, attention mechanisms, and transformers, challenges remain, including computational demands, large data requirements, and interpretability issues. Each development has pushed the boundaries further, leading to more powerful and efficient models. The ongoing evolution in this space reflects a commitment to overcoming these challenges and improving our understanding and modeling of sequential data, with the ultimate goal of creating more robust, efficient, and interpretable systems.

2.3.4 Final Considerations

Our study aimed to assess the performance of two distinct classification architectures in classifying domain entities into top-level ontology concepts using NLDs as input. Specifically, we examined the impact of fine-tuning language models for our research

problem needs and using embeddings from language models with traditional machine learning techniques. Consequently, our review of multi-label classification intentionally excluded specific methodologies, such as Convolutional Neural Networks (CNNs), Support Vector Machines (SVMs), and specific multi-label approaches like algorithm adaptation, problem transformation, and ensemble learning. In this context, we deliberately omitted CNNs because transformer-based methods have shown superior performance across various NLP benchmarks despite their applicability in text and multi-label classification. Similarly, SVMs were not included in our review. While they were initially designed for multi-class problems, numerous strategies have been proposed to adapt SVMs for multi-label contexts (Madjarov et al., 2012; Tarekegn; Giacobini; Michalak, 2021; Bogatinovski et al., 2022). Therefore, we chose to concentrate on methods inherently suited for multi-label and text classification from embeddings, as these were more aligned with our research objectives. This focus allowed us to provide a more targeted and relevant analysis of the applicability of techniques for classifying domain entities into top-level ontology concepts.

3 RELATED WORKS

In this chapter, we revise the two main areas related to the proposed approach in this work: text classification and ontology learning. We intend to show that text classification can be employed in many different tasks and that there is a gap in ontology learning in not considering the top-level concepts of domain entities in their classification pipeline.

3.1 Text Classification

In this section, we split the text classification into two classification architectures: dense text vectorization with classical machine learning and fine-tuning pre-trained language models. The first architecture involves converting text into fixed-length numerical vectors using models that capture semantic meanings. From that, current approaches use classical machine-learning techniques for various text classification tasks. The second architecture employs fine-tuning a pre-trained language model on a task-specific dataset, where transformer-based language models are highlighted.

3.1.1 Dense Text Vectorization with Classical Machine Learning

Dense text vectorization is a technique that converts text into fixed-length numerical vectors using models that capture semantic meanings. These vectors enable machines to understand textual nuances, relationships, and context more effectively. From that, classical machine-learning techniques, like Support Vector Machines (SVMs), Decision Trees (DTs), K-Nearest Neighbors (KNNs), etc., are employed to solve complex tasks, such as sentiment analysis, spam detection, and many other text classification tasks.

Classical text representation approaches like Term Frequency-Inverse Document Frequency (TF-IDF) are still used for text representation and classification. TF-IDF transforms text into numerical vectors by emphasizing significant words within a document while reducing the influence of common terms. Shah et al. (2020) demonstrated the effectiveness of TF-IDF in a text classification task on the BBC News dataset, comparing Logistic Regression, Random Forest, and K-Nearest Neighbors (KNN). Logistic Regression achieved the highest accuracy of 97%, indicating its robustness for structured text datasets. However, such methods have limitations in capturing contextual relationships

between words (Minaee et al., 2021). Park, Hong and Kim (2020) enhanced classical text representation approaches by integrating cosine similarity to measure the closeness between documents and categories, improving the performance of classifiers such as Multinomial Naïve Bayes (MNB) and Support Vector Machines (SVM). The study underscored the limitations of classical techniques, particularly in datasets requiring richer semantically understating of the domain context.

Extending the exploration of contextual embeddings, Kale et al. (2023) investigated their application in malware classification tasks. Their work uses text embedding approaches, such as Word2Vec, BERT, and ELMo, to process opcode sequences as features to train classification models like SVM, Random Forest, and CNN. The study highlighted the strengths and limitations of these representations, with static embeddings like Word2Vec providing general semantic insights but needing more adaptability to specific contexts. On the other hand, contextual embeddings, particularly BERT, demonstrated better performance by learning patterns within opcode sequences. From this adaptability, higher classification accuracies were achieved with BERT-based combinations, though the computational demands of such models presented scalability challenges.

Recent surveys have reviewed advancements in text classification, providing insights into using different types of vectorized representations of text, machine learning classifiers, and practical challenges. Minaee et al. (2021) analyzed over 150 deep learning models, highlighting architectures like RNNs, CNNs, and transformers such as BERT, which excel in capturing context and handling large-scale datasets. However, high computational costs and limited interpretability remain key challenges. Gasparetto et al. (2022) complemented this perspective by focusing on linguistic challenges in applying text classification to non-English languages, using Italian as a case study. Their study emphasized the scarcity of task-specific datasets and the need for preprocessing adaptations, underscoring the importance of developing efficient, multilingual models that balance computational demands with linguistic diversity.

In this thesis, we employed dense text vectorization combined with classical machine learning techniques, such as K-Nearest Neighbors (KNN), in a pipeline to classify domain entities into top-level ontology concepts based on textual representations of these entities. Based on the presented works, dense text vectorization from transformer-based techniques, like BERT, provided a means to transform the text into fixed-length numerical vectors, capturing semantic relationships and enabling machines to process and classify textual nuances effectively. While classical text representation techniques like TF-IDF

and Word2Vec have proven effective for certain tasks, their limitations in capturing contextual semantics underscored the relevance of dense vectorization for this research. As a result, this work addresses key challenges in aligning domain-specific textual data with top-level ontology concepts by integrating dense vector representations with traditional machine-learning methods.

3.1.2 Fine-Tuning a Pre-Trained Language Model

Dense text vectorization with classical machine-learning approaches was the most popular approach to dealing with text classification before popularizing transformer-based language models trained with vast amounts of textual data. Although the older approach is less computationally intensive, the newer approach broke barriers of contextual text understanding in complex and specific Natural Language Processing (NLP) tasks. In this context, fine-tuning these transformer-based language models for specific applications involves adapting the general knowledge trained from a vast and general dataset captured during pre-training to perform effectively on a target task with a smaller dataset.

Before the popularization of language models such as BERT (Devlin et al., 2018) and GPT (Radford et al., 2018), the journey of fine-tuning began with Howard and Ruder (2018) work, which introduced Universal Language Model Fine-Tuning (ULMFiT). Their work proposes using a three-stage process to fine-tune a pre-trained language model for text classification tasks: general-domain pre-training, task-specific language model fine-tuning, and classifier fine-tuning. From these steps, ULMFiT achieved state-of-the-art results across multiple datasets and reduced error rates significantly compared to prior methods. Furthermore, ULMFiT achieves high performance with minimal labeled data, showcasing its effectiveness in low-resource settings and laying the groundwork for subsequent research in fine-tuning methodologies.

Building on the principles of ULMFiT, Sun et al. (2020) systematically explored fine-tuning strategies for BERT. They also proposed a three-step approach: further pre-training on task-specific or domain-specific data, optional multi-task fine-tuning when related tasks are available, and targeted task-specific fine-tuning. Their work highlighted the significance of carefully selecting which layers to fine-tune and adjusting learning rates across different layers. As a result, their experiments show that incorporating domain-specific pre-training improves performance while applying different learning rates to BERT layers, which addresses issues such as catastrophic forgetting. Their experiment evaluated

the proposed methods across several English and Chinese datasets, achieving state-of-the-art results and providing a practical framework for leveraging BERT in domain-specific and resource-limited scenarios.

Expanding on the theme of domain adaptation, Gu et al. (2021) and Tinn et al. (2023) focused on the biomedical field, a domain with abundant specialized text but distinct linguistic characteristics. Gu et al. (2021) emphasized the benefits of pre-training language models like BERT from scratch using biomedical-specific corpora rather than relying on mixed-domain pre-training. Their approach demonstrates better performance across diverse biomedical tasks, leveraging a biomedical benchmark, BLURB, which includes applications such as named entity recognition, relation extraction, and question answering. Also, their work highlights the utility of domain-specific vocabularies and simplified labeling schemes. Another study proposed by Tinn et al. (2023) addressed the challenges of fine-tuning language models in low-resource biomedical settings. In their work, they explored techniques such as layer freezing, layerwise decay, and reinitializing upper layers to improve the stability of fine-tuning processes. Combined with task-specific pre-training adjustments, these techniques achieved new state-of-the-art results in benchmarks across biomedical datasets.

Addressing another challenge in fine-tuning pre-trained language models, particularly for long-text document classification, Kong, Wang and Zhang (2022) proposed a hierarchical BERT model with an adaptive fine-tuning strategy. Standard pre-trained language models like BERT have limitations on input length, i.e., they can only process long documents with truncation, potentially omitting critical information. To overcome this, they introduced a hierarchical structure combining local and global encoders to capture sentence- and document-level information. From this adaptive fine-tuning strategy, BERT layers are dynamically chosen to fine-tune based on the input data, optimizing computational resources while maintaining performance. Empirical results demonstrate improved performance across multiple datasets, highlighting its capacity to process short and long documents while maintaining contextual integrity.

In this thesis, we employed fine-tuning pre-trained language models, such as BERT and GPT, within another classification pipeline to classify domain-specific textual data with top-level ontology concepts. Based on the works described in this section, fine-tuning a pre-trained language model offers distinct advantages over static embeddings as a classifier input. When we fine-tune a pre-trained language model, we adapt extensive pre-trained knowledge to our specific task using fewer labeled data. Also, fine-tuning

integrates the classification task into the model training process, enabling the model to adapt its internal representations to the nuances of the input text and the target task. In addition, fine-tuning optimizes the entire classification model, including its embeddings, to maximize performance on our specific classification objective.

3.2 Ontology Learning

Ontology learning is the semi-automated or automated process of creating ontologies by extracting and organizing knowledge from unstructured or semi-structured data sources, such as text or databases (Wong; Liu; Bennamoun, 2012; Khadir; Aliane; Guessoum, 2021). This process involves identifying key terms, grouping them into concepts, establishing relationships (taxonomic and non-taxonomic), and defining rules or axioms that formalize the knowledge. The main objective of ontology learning is to reduce the manual effort required for ontology development and leverages techniques from Natural Language Processing (NLP), Machine Learning (ML), and data mining to discover patterns, infer structures, and ensure scalability. Based on this perspective, this work aligns with the ontology learning field by addressing the automation of a critical and complex task: classifying domain entities into top-level ontology concepts.

In this section, we revise the most important contributions and the recent advancements in ontology learning. Firstly, we present famous ontology learning systems that use natural language processing approaches and ruled-based methods to extract entities and their relationships from unstructured text. After that, we revise the cutting-edge approaches that use the recent advances in language models for application in ontology learning.

3.2.1 Classical Ontology Learning Systems

In this section, we revised only ontology approaches that do not use language models, which we call classical ontology learning systems. Classical ontology learning systems represent the foundational phase in developing automated approaches to ontology construction. These systems primarily focused on extracting knowledge from unstructured text using linguistic patterns, statistical methods, and rule-based frameworks (Wong; Liu; Bennamoun, 2012; Khadir; Aliane; Guessoum, 2021). These systems aimed

to streamline manual ontological creation efforts by automating tasks such as term recognition, relationship extraction, and hierarchical concept structuring. Early systems, such as Text2Onto (Maedche; Staab, 2000; Cimiano; Völker, 2005), OntoLearn (Navigli; Velardi, 2004; Velardi et al., 2005), and others, relied heavily on predefined rules, lexical resources like WordNet (Miller, 1995) and Wikipedia, or domain-specific dictionaries to interpret and organize textual data into coherent ontological structures. While these systems were instrumental in establishing the field of ontology learning, they faced challenges that we revised below.

Text-to-Onto represents one of the earliest efforts in the field of ontology learning, addressing the challenge of extracting and structuring knowledge from unstructured text (Maedche; Staab, 2000). The system utilizes Natural Language Processing (NLP) techniques and lexical databases, such as WordNet (Miller, 1995), to preprocess and identify conceptual structures within the textual input, emphasizing the role of linguistic constraints such as syntax, morphology, and semantics for concept and relation identification. Also, the system uses lexical databases for semantic enrichment and validation. In addition, the Text-to-Onto system allows user intervention to validate and refine the generated ontologies in order to ensure alignment with domain expectations. However, while pioneering, Text-to-Onto faces challenges in scalability and domain independence due to its reliance on handcrafted constraints and predefined corpora.

Building on early advancements like Text-to-Onto, OntoLearn introduces a methodology for automatically constructing domain ontologies from web sources and document repositories (Navigli; Velardi, 2004; Velardi et al., 2005). The system extracts domain-relevant terminology and semantically interprets terms to build taxonomies. A key innovation of this system is the use of Structural Semantic Interconnections (SSI), a word sense disambiguation algorithm that maps extracted terms to WordNet (Miller, 1995) concepts. This approach enables the creation of domain-specific ontologies that extend or enhance general-purpose ontologies such as WordNet. Also, the OntoLearn system integrates its outputs into broader frameworks, supporting applications in domains like tourism and finance. However, the main disadvantage of the OntoLearn system is the dependency on the quality of external resources and the need for human intervention during validation phases.

Following the Text-to-Onto system, Text2Onto introduced several innovations in order to enhance ontology learning processes (Cimiano; Völker, 2005). In this context, the framework incorporates a Probabilistic Ontology Model (POM) that assigns confi-

dence levels to the learned concepts, improving transparency and traceability of the learning process. From the POM approach, the system allows users to focus on the most confident predictions while user interactions remain central to refining the results. Additionally, Text2Onto introduces data-driven change discovery, enabling selective updates to the ontology when the underlying corpus changes, thus improving efficiency in ontology maintenance. Also, the Text2Onto system relies on external resources, such as WordNet, for semantic validation and integration into various ontology development languages (e.g., OWL¹, RDFS², etc.) to ensure broader applicability. However, the need for user validation and challenges in managing inconsistencies in learned concepts persist. Recent developments in the field have seen efforts to adapt Text2Onto for other languages like French (Hajji; Qbadou; Mansouri, 2020) and to integrate it into broader ontology learning pipelines (Ramayanti et al., 2020).

Expanding on earlier efforts, Elnagar, Yoon and Thomas (2022) proposed a framework that addresses the limitations of domain-specific and manually intensive ontology generation systems by proposing a domain-independent approach. This framework integrates Knowledge Graphs (KGs) to dynamically represent knowledge and combines them with existing ontologies to improve quality and structure. A key contribution of this framework is the refinement of the KGs through correction and alignment with reference ontologies, such as the middle-level ontology YAGO (Suchanek et al., 2024) and the top-level ontology DOLCE (Gangemi et al., 2002; Borgo et al., 2021), ensuring correctness, completeness, and consistency. As the other systems previously described, this framework focuses on unstructured text as the input source and automates ontology construction while reducing the need for manual intervention. Despite these advancements, the framework acknowledges limitations in scalability and the absence of hierarchical representations (e.g., OWL format). The emphasis on combining the dynamism of KGs with the structural rigor of middle- and top-level ontologies reflects an evolution in addressing the challenges identified in earlier systems.

Table 3.1 presented an overview comparison of classical ontology learning systems, focusing on their approaches, findings, use of external resources, challenges, and integration with top-level ontologies. However, only the most recent approach, Elnagar, Yoon and Thomas (2022), incorporates a top-level ontology in the system design. In our view, using top-level ontologies in an ontology learning system offers clear advantages: it clarifies domain-specific terms by mapping them to top-level concepts, improves in-

¹<https://www.w3.org/OWL/>

²<https://www.w3.org/TR/rdf12-schema/>

teroperability across systems, enables reusability of ontologies, reduces manual effort in alignment, and enhances the scalability of the ontology development process. In this context, this work proposes an approach to classify domain entities into top-level ontology concepts using NLDs of these entities as input.

3.2.2 Ontology Learning with Language Models

Ontology learning from text has evolved significantly with advancements in Natural Language Processing (NLP). Traditional approaches like Text2Onto focused on extracting concepts and relations using handcrafted rules and statistical measures (Maedche; Staab, 2000). Recent efforts have shifted to embedding-based models (Schmidt; Santos; Vieira, 2016; Schmidt et al., 2016; Schmidt et al., 2019; Schmidt; Trojahn, 2019; Schmidt, 2020) or approaches based on language models (He et al., 2022; Chen et al., 2023; Julien; Valentino; Freitas, 2022; Sousa et al., 2023). These approaches perform alignment, matching, classification, and refinements between domain ontologies or between domain ontologies and top-level ontologies, considering the ontology structures or relying only on text representation of the domain ontology entities as input.

Daniela Schmidt’s research provides valuable contributions to the field of ontology alignment and integration (Schmidt; Santos; Vieira, 2016; Schmidt et al., 2016; Schmidt et al., 2019; Schmidt, 2020), which closely aligns with the goals of this work. All her proposed works fit to align top-level ontologies (called foundational ontologies), such as SUMO, DOLCE, and BFO, with domain-specific ontologies, addressing key challenges, such as semantic interoperability, semantic coherence, and consistency. In Schmidt et al. (2016), the authors introduce a method to automatically validate equivalence and subsumption relations in alignments generated by existing ontology matching tools. The approach uses WordNet for synset disambiguation and semantic relations identification, such as synonymy and hypernymy, to validate correspondences between domain concepts and top-level ontologies. Experiments on the OAEI Conference dataset reveal that while matchers generate equivalences, they often fail to identify subsumptions or address ambiguities caused by insufficient terminological richness or missing annotations in domain ontologies. In another work, Schmidt et al. (2019) focus on manually aligning Conference Ontology (Zamazal; Svátek, 2017) with SUMO, where the author also employed WordNet as a bridging resource. Their work found two main challenges in manually aligning ontologies that are also challenges in automated approaches. The first is the difficulty in

interpreting the semantics of domain ontology concepts, which often need more terminological detail (e.g., clear NLDs). The second one regards the structural limitations of domain ontologies, which often represent relationships or constraints as independent concepts, leading to ambiguities.

More situated with automated alignment between domain ontologies and top-level ontologies, in Schmidt et al. (2018), the author proposed an approach involving two steps: first, selecting the most appropriate WordNet synset for domain concepts via Word Sense Disambiguation (WSD) using Lesk-based overlap measures and word embeddings similarity evaluation; second, mapping the domain concepts to top-level ontology classes through pre-established alignments between WordNet and top-level ontologies, such as in OntoWordNet and SUMO-WordNet. The method incorporates the notion of context, derived from terminological annotations and hierarchical relationships, to disambiguate concepts and compute alignments. The evaluation, conducted across ontologies from the Semantic Sensor Network Ontology (SSN), Core Ontology for Robotic and Automation (CORA), and OAEI Conference datasets and using word embedding from GloVe (Pennington; Socher; Manning, 2014) and GoogleNews, reveals higher performance in aligning general domain concepts compared to more specialized ones. While the approach improves traditional ontology matching tools in recall and F-measure, the author identified challenges regarding insufficient terminological richness in domain ontologies, lack of domain-specific embeddings, and limitations in using general-purpose word embeddings. Another critical point is that, despite the significant contributions of Schmidt's works, her proposals precede the advent and widespread adoption of Large Language Models, which have revolutionized ontology engineering by enabling more robust, context-aware, and scalable solutions for tasks like ontology learning.

Following the proposal of the BERT language model (Devlin et al., 2018), He et al. (2022) proposes a system that leverages pre-trained BERT embeddings for ontology alignment, providing an innovative approach to mapping concepts across ontologies by combining contextual word semantics with ontology structure and logic. BERTMap addresses ontology learning challenges by focusing on predicting equivalence mappings using fine-tuned BERT models trained on synonym and non-synonym label pairs extracted from existing ontologies. Additionally, BERTMap employs mapping refinement techniques, including extension based on semantic locality principles and logic-based repair. The system outperforms traditional methods ontology alignment methods like LogMap (Jiménez-Ruiz; Grau, 2011) and AML (Faria et al., 2013) in some tasks, highlighting

its effectiveness in utilizing contextual embeddings from pre-trained language models to improve alignment quality.

Another work that also uses the BERT language model was proposed in Chen et al. (2023). In their work, the authors proposed the BERTSubs framework to address ontology learning by predicting subsumption relationships within and across ontologies. BERTSubs focuses on predicting both intra-ontology and inter-ontology subsumptions, including named class subsumptions and existential restrictions. Also, BERTSubs uses customized templates, such as Isolated Class (IC), Path Context (PC), and Breadth-first Context (BC), to capture contextual and textual information from ontological structures to improve embedding quality and prediction accuracy. In addition, BERTSubs framework leverages hierarchical information and textual annotations from ontologies like FoodOn (Dooley et al., 2018), GO (Consortium, 2019), NCIT (Sioutos et al., 2007), and DOID (Schriml et al., 2019) to demonstrate improved performance over traditional geometric and ontology embedding methods like OWL2Vec* (Chen et al., 2021) and TransE (Bordes et al., 2013).

Language models such as BERT and their variations have been explored to automate the classification of domain-specific concepts into top-level ontology concepts, particularly DOLCE-Lite-Plus top-level concepts. In Jullien, Valentino and Freitas (2022), the authors investigate the capacity of transformer-based language models, such as BERT and RoBERTa, to encode abstract categories from top-level ontologies. Their analysis, using example sentences from the Brown Corpus (Francis, 1965) to represent domain entities, reveals that these models do, in fact, encode essential aspects of top-level ontology information during pre-training. In their experiments, they evaluate the ability of language models to classify domain entities into six Dolce-Lite-Plus categories: Socially-Constructed-Person, Cognitive-Event, Geographical-Role, Biological-Object, Non-Agentive-Functional-Object, and Information-Object. As a result, the experiment shows that while transformers incidentally encode top-level ontology information during pre-training, fine-tuning improves their performance, achieving up to 90% accuracy on certain top-level ontology concepts.

In another work, Sousa et al. (2023) proposes a methodology for predicting top-level ontology concepts of domain entities using Machine Learning models trained on datasets derived from OntoWordNet. In their work, they proposed datasets addressing issues of multi-inheritance and imbalance in the distribution of top-level ontology concepts. The experimental evaluation compares the performance of various classification models,

including the classification pipeline proposed in (Lopes et al., 2022) (one of our publications derived from this thesis work), using features like *rdfs:comments* (a.k.a NLDs) and entity labels. The findings reveal that leveraging *rdfs:comments* improves model performance, with BERT outperforming other Machine Learning classifiers across datasets. However, challenges remain, such as misclassification between closely related concepts like *Perdurants* and *Situations*, and limited generalization due to sparse comments in real-world ontologies.

With the advancements in Large Language Models (LLMs) and conversational systems like LLAMA (Touvron et al., 2023), GPT-3.5, and GPT-4 (OpenAI et al., 2023), approaches utilizing techniques such as in-context learning, zero-shot learning, and fine-tuning archives the state-of-the-art results in several ontology learning approaches. In Giglou, D’Souza and Auer (2023), the authors investigate the application of LLMs to three key ontology learning tasks: term typing, taxonomy discovery, and non-taxonomic relation extraction. The study evaluates these models on domain-specific ontologies, including WordNet, GeoNames, UMLS, and schema.org, to assess their effectiveness in ontology construction. As a result, their findings indicate that foundational LLMs, in their pre-trained state, lack the reasoning capabilities and domain-specific expertise required for ontology construction. However, with proper fine-tuning, these models show potential as effective tools to support and streamline the knowledge acquisition process for ontology development.

In order to fill a gap in systems and frameworks for ontology learning using language models and deep learning, He et al. (2024) proposed the DeepOnto system. This system is designed to integrate deep-learning methods into ontology alignment, completion, and reasoning tasks. DeepOnto supports multiple ontology engineering processes by leveraging pre-trained language models and other machine learning methods. Also, DeepOnto incorporates the BERTMap (He et al., 2022) for equivalence matching, BERTSubs (Chen et al., 2023) for subsumption prediction, and modules for pruning, verbalization, and ontology projection. These tools aim to address limitations of traditional ontology engineering systems (as presented in Section 3.2.1), such as reliance on lexical similarity and rule-based methods. In addition, DeepOnto demonstrates its utility across tasks like the alignment of biomedical ontologies and ontology curation for healthcare applications, providing a flexible platform for combining linguistic and structural data.

Despite their advancements, current approaches to ontology learning using language models face several limitations that impact their applicability in classifying do-

main entities into top-level ontology concepts. Many approaches, such as those proposed by Daniela’s works (Schmidt; Santos; Vieira, 2016; Schmidt et al., 2016; Schmidt et al., 2019; Schmidt; Trojahn, 2019; Schmidt, 2020), BERTMap (He et al., 2022), and BERTSubs (Chen et al., 2023), focus narrowly on ontology alignment or subsumption prediction, in which have different goals rather than classification. Additionally, most approaches are restricted to multiclass classification and lack support for multi-label classification, which is essential for accurately representing complex concepts that fall under multiple top-level ontology categories or the entire branch of the top-level ontology in which a domain entity subsumes, ranging from specific to more general top-level concepts. Also, these approaches need a deeper analysis of the types of text used as input for the proposed classifier, such as the differences between using terms, annotations, example sentences, NLDs, etc., as input of language models. Based on that, this work aims to address these gaps by proposing an approach that combines Machine Learning pipelines with Large Language models, supporting multilingual and multi-label classification tasks, and leveraging a deep analysis in diverse inputs such as terms, example sentences, and NLDs. Also, we investigate the differences between general and domain-specific NLDs and how these differences affect the results of our proposed classification pipelines using LLMs. Table 3.2 presents an overall analysis of the works revised in this section and our proposed approach.

Table 3.1 – Comparison of ontology learning systems.

System	Proposed Approach	Main Findings	Uses External Resources?	Challenges	Uses Top-Level Ontologies?
Text-to-Onto (2000)	Extract conceptual structures from unstructured text using linguistic constraints and user interaction.	Demonstrated feasibility of automating ontology construction with user validation.	WordNet for semantic enrichment.	Scalability limited by reliance on pre-defined rules and handcrafted constraints.	No
OntoLearn (2004)	Builds domain ontologies by extracting and semantically interpreting terms aligned with general-purpose ontologies.	Introduced Structural Semantic Interconnections (SSI) for word sense disambiguation and hierarchical arrangement of terms.	WordNet for mapping and semantic validation.	Dependent on high-quality input data and human oversight for validation.	No
Text2Onto (2005)	Uses a Probabilistic Ontology Model (POM) with confidence scores and data-driven change discovery for ontology updates.	Improved efficiency by selectively updating affected ontology parts, ensuring transparency and traceability.	WordNet for validation and semantic alignment.	Managing inconsistencies and reliance on user validation.	No
Automatic Ontology Generation Framework (2020)	Integrates Knowledge Graphs (KGs) with ontologies to generate domain-independent ontological structures from unstructured text.	Combines KGs' dynamic features with ontology quality, introducing KG refinement for consistency and correctness.	Reference ontologies and external corpora.	Scalability, absence of hierarchical representation (e.g., OWL), and quality assurance.	Yes, DOLCE

Table 3.2 – Comparison of approaches in ontology learning using language models.

Approach	Ontology Task	Learning	Output Classification	Input to Classifier	Foundational Ontology Usage	Language Model Integration	Multilingual Support
Schmidt et al. (2016-2020)	Alignment		Aligned ontology	WordNet synsets, hierarchical context	DOLCE-Lite-Plus, SUMO, BFO	No	No
BERTMap (He et al., 2022)	Alignment		Aligned ontology	Ontology labels, annotations	Not mentioned	Yes	No
BERTSubs (Chen et al., 2023))	Subsumption	Prediction	Intra/inter ontology subsumptions	Textual annotations, hierarchical context	Not mentioned	Yes	No
Jullien, Valentino and Freitas (2022)	Classification		Multiclass	Domain entity labels, example sentence from Brown corpus	DOLCE-Lite-Plus	Yes	No
Sousa et al. (2023)	Classification		Multiclass	<i>rdfs:comments</i> , labels	DOLCE-Lite-Plus	Yes	No
LLMs4OL (Giglou; D’Souza; Auer, 2023)	Term Typing, Taxonomy Discovery, Non-Taxonomic relation extraction		Taxonomy, typed terms	Ontology terms, textual definitions	Not mentioned	Yes	Yes
DeepOnto (He et al., 2024)	Alignment, Completion, Reasoning		Aligned ontology	Ontology labels, annotations	Not mentioned	BERTMap, BERTSubs	No
This Ph.D. work	Classification		Multiclass, Multilabel	Natural Language Definitions (NLDs)	DOLCE-Lite-Plus, BFO, other top-level ontologies	Yes	Yes

4 PROPOSED APPROACH

This chapter presents a methodology to classify domain entities into top-level ontology concepts using datasets aligned with DOLCE-Lite-Plus (DLP) and Basic Formal Ontology (BFO). We align domain entities between OntoWordNet and BabelNet for DLP to build a multi-language, multi-resource dataset. For BFO, we select and process domain ontologies from the OBO Foundry repository, ensuring their alignment with BFO’s structure. Also, we convert the datasets from multi-class to multi-label formats to capture hierarchical relationships and address issues like multiple inheritance and disjoint concepts. In addition, we delve into the textual representation of domain entities, supporting our hypothesis that NLDs capture semantic information, enabling effective classification into top-level ontology concepts. We propose two pipelines from this: fine-tuning language models for direct classification and using pre-trained embeddings with classical Machine Learning algorithms. Finally, we propose training methodologies exploring data augmentation for the task of classifying domain entities into top-level ontology concepts.

4.1 DOLCE-Lite-Plus Dataset Extraction

This section outlines the methodology for extracting multi-language and multi-resource datasets to evaluate our hypothesis using the DOLCE-Lite-Plus top-level ontology. First, we introduce the OntoWordNet and BabelNet resources, explaining how we used their structures to establish a match between them. Next, we detail the algorithm employed to align domain entities from OntoWordNet and BabelNet, resulting in a dataset where BabelNet is mapped to the DOLCE-Lite-Plus top-level ontology. Finally, we discuss the challenges encountered in this process, including partial coverage, discrepancies caused by special characters, and the imbalanced nature of top-level ontology concepts.

4.1.1 OntoWordNet Ontology and BabelNet Semantic Network

The starting point for extracting the datasets used in this work began with the ontology from the OntoWordNet project. This project establishes a mapping between WordNet—a structured lexical database of English words organized into synonym sets (synsets)—and the top-level ontology concepts of DOLCE-Lite-Plus (DLP). This cate-

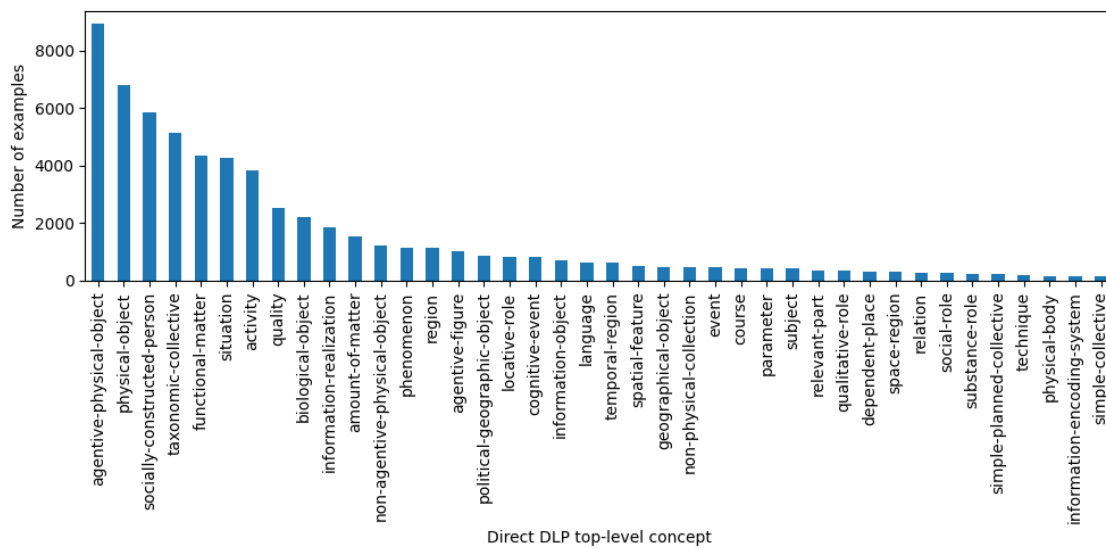


Figure 4.1 – Top 40 most populated direct DLP top-level concepts of WordNet synsets in the OntoWordNet.

gorization was pivotal to transforming WordNet from merely a lexical database into an ontology by clearly separating concept-synsets, relation-synsets, meta-property-synsets, and individual-synsets. The output of this alignment comprises a total of 65,973 domain entities. Also, OntoWordNet aligns its domain entities with 120 top-level concepts from the 244 of the whole structure of DOLCE-Lite-Plus. An important clarification is that the terms "synsets" and "domain entities" will be used interchangeably throughout this section for dataset description purposes. In this context, the term "synset" refers to a set of synonyms in WordNet representing a single lexicalized domain entity, while "domain entity" refers to an entity described within a domain ontology that belongs to our extracted datasets

Figure 4.1 provides a detailed visualization of the number of WordNet synsets categorized under the DOLCE-Lite-Plus (DLP) top-level concepts in the OntoWordNet ontology. The figure focuses exclusively on direct top-level ontology concepts from DLP where WordNet synsets subsumes. The most populated concepts are defined based on the sheer count of synsets mapped to them, which serves as the metric for this analysis. Also, this figure highlights the imbalanced nature of the OntoWordNet ontology, illustrating how certain top-level DLP concepts have a high concentration of associated WordNet synsets, while others are significantly less populated. In this context, this disparity is crucial for understanding the distribution and representation of concepts within the top-level ontology, offering insights into potential areas of richness or sparsity of domain entities.

WordNet also divided the NLDs of its synsets into two parts for technical purposes: *definiendum* and *definiens*. The *definiendum* part refers to each term within a synset, indicating the subject of the *definiens*. Conversely, the *definiens* provides the explanation or the descriptive content of the synset’s meaning, encapsulating the synset’s essence. In WordNet, each synset contains only one *definiens* to maintain the synset clarity and precision. Therefore, the domain entities in the OntoWordNet ontology also have their NLDs divided into *definiendum* and *definiens* parts. This organization facilitates mapping the OntoWordNet domain entities to a more expansive and interconnected semantic network. One such network is BabelNet, which also integrates the comprehensive lexical coverage of WordNet extended with the extensive encyclopedic knowledge from Wikipedia in multiple languages. From that, it is possible to align the OntoWordNet domain entities with the domain entities in BabelNet because both use English WordNet *definienda* and *definientia*, as presented in the next section.

4.1.2 Alignment Between OntoWordNet and BabelNet

As previously described, we leverage the inclusion of WordNet synsets in both the OntoWordNet ontology and the BabelNet semantic network to propose an algorithm for aligning the domain entities between them. In this context, BabelNet encompasses more synsets than WordNet, from adding Wikipedia and synsets for more recent linguistic concepts not present in WordNet, such as streamer and influencer. This broadness of BabelNet provides an opportunity for the enhancement and expansion of OntoWordNet domain entities to multi-language and multi-resource datasets. From that, we take the advantage that both the structure of WordNet inside OntoWordNet and BabelNet share almost the same *definienda* and *definientia* to propose the Algorithm 1 to match the domain entities between them.

Algorithm 1 inputs the OntoWordNet ontology (*OWN*) and the BabelNet semantic network (*BN*) and outputs the alignment (*align*) between them. Initially, *align* is an empty list that will store the matched synsets from BabelNet and their corresponding DOLCE-Lite-Plus (DLP) top-level concepts. The algorithm starts by retrieving all domain entities from *OWN*, and for each domain entity, it extracts the *definienda* (terms representing the entity), the *definiens* (definition of the entity), and the *dlp_cls* (the DLP top-level concept the entity belongs to). It then initializes a dictionary, *rank*, to keep track of the similarity scores for each synset. For each term in the *definienda*, the algorithm

retrieves all associated BabelNet synsets, focusing on WordNet synsets in English to reduce the number of comparisons. For each synset, if it is not already present in *rank*, the algorithm initializes its entry with a tuple $(0, 0)$, representing the cumulative similarity score and the number of similarity evaluations performed. The algorithm calculates the similarity between the domain entity’s *definiens* and the synset’s *bn_definiens* (definition) using the Jaro-Winkler distance¹ and updates the similarity score and comparison count in *rank*. After processing all synsets, the algorithm normalizes the similarity scores by dividing them by the maximum comparison count to penalize synsets with fewer common terms. Finally, it selects the synset with the highest normalized similarity score as the best match for the domain entity and adds this synset, along with the associated DLP concept, to the *align* list. The algorithm continues this process for all domain entities and returns the final *align* list containing the best-matching synsets and their corresponding DLP concepts.

In addition, Algorithm 1 has an asymptotic complexity of $O(n^3)$. However, it offers a significant efficiency over a method that compares all domain entities of OntoWordNet with all entities in BabelNet, with a complexity of $O(n^2)$. This efficiency gain is particularly relevant considering the size of BabelNet, which contains more synsets than WordNet, the basis of OntoWordNet. Since OntoWordNet comprises 65,973 domain entities, a naive approach that compares these entities with only relevant BabelNet synsets would entail up to 4.3 billion comparisons. In contrast, the proposed algorithm exploits the fact that BabelNet synsets are indexed by synset term or ID, allowing for their retrieval in $O(1)$ time. Consequently, the algorithm iterates over the terms naming OntoWordNet entities and retrieves only the associated BabelNet synsets, significantly reducing the number of comparisons to less than 1 million.

The resulting alignment contains a total of 65,018 domain entities if we consider the main dataset that comprises domain entities represented by informal WordNet definitions in English (as presented in Table 4.1). Following the application of Algorithm 1, we transitioned from a single-language resource, relying solely on the definitions from WordNet, to multi-language and multi-resource datasets. This advancement enabled the extraction of datasets for predicting the top-level concept of a domain entity across 291 languages and from many kinds of resources, such as WordNet, Wikipedia, Wiktionary, Open Multilingual WordNet, WordNet 2020, OmegaWiki, etc. Also, this alignment made it possible to access other types of representation approaches for domain entities beyond

¹Jaro-Winkler distance fits our purposes in this work perfectly because it is fast and the compared sentences are almost identical if the compared entities are the same.

Algorithm 1 The algorithm used to align OntoWordNet and BabelNet

Input: OntoWordNet OWN ; BabelNet BN
Output: Alignment $align$ between OntoWordNet and BabelNet

```

1: Initialize  $align$  as an empty list.
2:  $domain\_entities \leftarrow DomainEntities(OWN)$ 
3: for  $domain\_entity$  in  $domain\_entities$  do
4:    $definienda \leftarrow Definienda(domain\_entity)$ 
5:    $definiens \leftarrow Definiens(domain\_entity)$ 
6:    $dlp\_cls \leftarrow Concept(domain\_entity)$ 
7:   Initialize  $rank$  as an empty dict.
8:   for  $definiendum$  in  $definienda$  do
9:      $synsets \leftarrow BabelNetSynsets(definiendum, 'WN', 'EN')$ 
10:    for  $synset$  in  $synsets$  do
11:      if  $synset \notin rank$  then
12:        Initialize  $rank[synset]$  as a tuple (0, 0)
13:      end if
14:       $bn\_definiens \leftarrow Definiens(synset)$ 
15:       $rank[synset][0] += JaroWinklerDistance(definiens, bn\_definiens)$ 
16:       $rank[synset][1] += 1$ 
17:    end for
18:  end for
19:   $max\_count \leftarrow \max(rank[synset][1] \text{ for } synset \text{ in } rank)$ 
20:  for  $synset$  in  $rank$  do
21:     $rank[synset][0] \leftarrow rank[synset][0] / max\_count$ 
22:  end for
23:   $best\_synset \leftarrow \arg \max(rank[synset][0] \text{ for } synset \text{ in } rank)$ 
24:  Add the tuple ( $best$ ,  $dlp\_cls$ ) to  $align$ 
25: end for
26: return  $align$ 

```

NLDs, as well as example sentences, Wikipedia source pages, images, etc. Integrating these diverse elements significantly enhances the breadth and depth of linguistic and semantic information within our datasets, thereby enriching the scope and utility of the datasets that can be extracted.

Table 4.1 – Number of domain entities of each BabelNet resource after matching

Domain	Resource	Number of Domain Entities
General	WN	65,018
General	WN2020	64,993
General	WIKI	33,547
General	WIKIDATA	23,874
General	WIKIDIS	11,264
General	WIKT	42,506
General	WIKTLB	24,119
General	OMWIKI	12,654

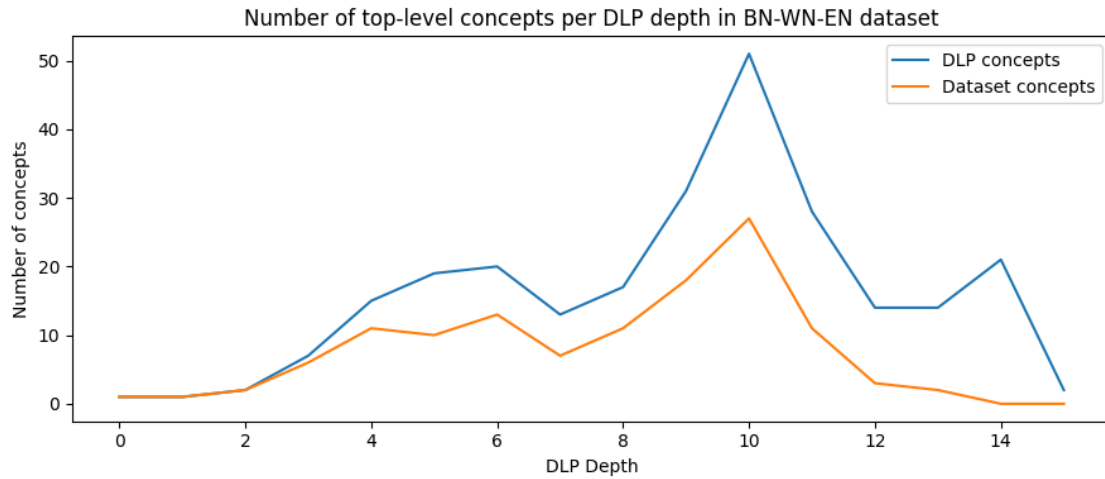


Figure 4.2 – The number of top-level concepts per level in DOLCE-Lite-Plus and from the alignment between OntoWordNet and Babel.

4.1.3 Coverage and Limitations

The OntoWordNet project mapped WordNet synsets to 120 from the 244 top-level concepts of DOLCE-Lite-Plus (DLP), which means that OntoWordNet covers less than half of the DLP top-level concepts. Figure 4.2 compares level by level the number of top-level concepts in the DLP and the number of top-level concepts in the OntoWordNet after we converted the dataset from multi-class to multi-label (described in Section 4.7). As presented, after level 2 of the DLP hierarchy, there is a discrepancy between the number of top-level concepts between DLP and OntoWordNet, which mostly follows the same growth pattern as DLP until level 12. Based on this, we can say that although OntoWordNet does not include some of the top-level concepts of DLP, OntoWordNet still represents all the most generic concepts in DLP.

As another limitation, OntoWordNet does not handle special characters in the definiendum of WordNet synsets, i.e., OntoWordNet removed the special characters before the alignment with DLP. On the other hand, BabelNet indexes its synsets with special characters such as underscores, percent signs, or non-ASCII characters. Based on that, these discrepancies can cause searches to fail or return incorrect data if not adequately handled. Considering that we have a BabelNet synset that was erroneously retrieved, the ranking strategy adopted in Algorithm 1 can handle this problem by assigning a lower similarity score between the two compared definiens because they are different. Also, the ranking approach further reduces the value of the erroneous retrieved synset because the compared synsets share fewer definiendum than the right synset. In the other case, if there are no results in the BabelNet search for all definienda of an OntoWordNet domain

entity, we do not include this domain entity in the alignment between OntoWordNet and BabelNet.

As a final limitation, top-level ontologies often exhibit an imbalanced nature, as we can see in Figure 4.1, due to the inherent complexity and diversity of the concepts they aim to represent. The imbalanced nature can be perceived as the difference in the number and depth of different branches in the top-level ontology hierarchy. Also, some top-level concepts can represent only a limited number of domain entities due to their restrictions, and others can represent a broader number of domain entities because they are not so restrictive. Also, this imbalance can be attributed to the differential emphasis placed on certain areas or categories over others, reflecting the subjective priorities of the ontology designers or the specific needs of the intended application domain. Consequently, datasets derived from these top-level ontologies tend to inherit this imbalance, manifesting in skewed distributions of instances across different categories. Based on that, the performance of classification models that use such datasets can be negatively affected, and approaches to handle data augmentation in this scenario tend to be complex.

4.2 Basic Formal Ontology (BFO) Dataset Extraction

The extraction of datasets aligned with the Basic Formal Ontology (BFO) aims to create a tailored dataset for evaluating the classification of domain entities into BFO top-level concepts considering different domains and ontologies. While the OBO Foundry already provides BFO-aligned ontologies, this work refines their selection and organization for task-specific evaluations. This section details the methodology used, including the manual curation of ontologies from the OBO Foundry repository, the extraction and organization of domain entities into a cohesive dataset, and an analysis of coverage and limitations.

4.2.1 Manual Selection Methodology

In addition to the datasets extracted for the DOLCE-Lite-Plus top-level ontology, as discussed in Section 4.1, we also extracted datasets for the Basic Formal Ontology (BFO) top-level ontology to ensure broader coverage and compatibility with different philosophical foundations (as presented in Section 2.1.2) and different types of NLDs (as

presented in Section 2.1.3). To do that, we used the OBO Foundry repository, a well-established source of interoperable and scientifically validated ontologies, as our source of BFO-aligned domain ontologies. We know that OBO Foundry contains many ontologies already aligned with BFO top-level ontology. However, our goal here is to extract datasets to validate our work hypothesis. In this context, the OBO Foundry repository contains 191 domain ontologies² grouped into 14 knowledge domains (e.g., health, agriculture, chemistry and biochemistry, etc.). Also, most of these ontologies were developed in the Ontology Web Language (OWL), containing domain entities with labels (a.k.a. *definiendum*) and comments (a.k.a. *definiens*). From this, we conducted a systematic and manual selection process, starting with excluding ontologies that do not have labels or comments from the pre-selected ones. In addition, we used two more methodological steps in order to decide whether to include a domain ontology in our dataset, ensuring the chosen domain ontologies align with the BFO structure:

1. **Presence of BFO top-Level concepts:**

we selected domain ontologies that explicitly referenced BFO top-level concepts. To do this, we analyzed each ontology for domain entities aligned with BFO's concepts, evaluating whether the ontologies reused BFO concepts directly or established mappings through equivalence relationships. For instance, since the ChEBI entity "water" subsumes (directly or indirectly) the BFO concept "Object" in our dataset, the domain entity "water" is classified as an "object". As another example, BFO top-level ontology and ChEBI domain ontology define the concept of "Role" (a realizable entity in the BFO), which are considered equivalents. Consequently, in our dataset, all subentities of the ChEBI "role" are treated as subentities of the BFO "Role" rather than as realizable entities. An important point here is that we did not modify or reinterpret the hierarchical organization of the ontologies. Instead, we used reasoners to extract these relationships between domain and top-level ontologies.

2. **Extension of another ontology aligned with BFO:** we also considered domain ontologies that do not directly reference BFO top-level concepts if they extend or rely on another ontology already aligned with BFO. To evaluate these cases, we identified dependencies by examining the ontology's import statements, modular references, and explicit equivalence or subclass mappings to its parent ontology. During the import process, we systematically verified the consistency of the im-

²Revised in 19/12/2024

ported modules with the BFO top-level ontology using reasoning tools (e.g., ELK reasoner). In this context, we used the reasoner to validate whether the imported ontology maintained coherent subclass and equivalence relationships consistent with BFO. For example, if the target ontology extended a parent ontology that defined "cell" as a subclass of BFO "Object", we verified that this classification propagated correctly through the target ontology's hierarchy without introducing conflicts or logical inconsistencies. For domain ontologies that rely on modular extensions, i.e., they extend only a part of the parent ontology, we analyzed the specific modules extended and imported the whole ontology structure to which these modules belong. In these cases, we used reasoning to validate whether the alignment propagated correctly within the scope of the imported ontology and did not conflict with other parts of the ontology. For instance, if a modular extension imported the "molecular entity" branch from a parent ontology aligned with BFO, we verified that all subclasses of "molecular entity" adhered to BFO's definition of "Object" and maintained consistency in their hierarchical relationships.

From the methodology used to manually select domain ontologies that align with the BFO top-level ontology, we selected 110 domain ontologies from the 194 available in the OBO Foundry repository. These selected ontologies belong to 12 distinct knowledge domains. However, we have to exclude two knowledge domains: Simulation and Phenotype. We excluded the Simulation domain because we did not select any ontology using our methodology. Also, we excluded the Phenotype domain due to a lack of consensus in the literature regarding the classification of "phenotype" within the BFO structure. Specifically, some studies classify phenotype as a Specifically Dependent Continuant, while others classify it as an Independent Continuant. This inconsistency in classification made it challenging to establish a clear alignment with BFO, leading to its exclusion from our selection process and dataset.

4.2.2 Extraction and Organization of BFO Dataset

After selecting the domain ontologies as described in the previous section, we proceeded to extract the definienda (labels) and definientia (comments) of each domain entity in these ontologies to build a dataset that encompasses NLDs. We input the set of selected domain ontologies into the extraction process, organized into folders based on their re-

Table 4.2 – Selected domain ontologies in our manual analysis of OBO Foundry ontologies.

Domain	Extracted Domain Ontologies	Number of Domain Entities
Agriculture	AGRO, PSO	1,071
Anatomy and Development	AISM, BSPO, CARO, CL, CLAO, CLO, CLYH, COLAO, CTENO, ECAO, FBBT, FBDV, HSAPDV, LEPAO, OARCS, PLANA, PO, PORO, UBERON, WBBT, WBLs, XAO, ZFA, ZFS	99,568
Biological Systems	GENO, GO, INO, NBO, OHPI	43,933
Chemistry and Biochemistry	CHEBI, CHEMINF, DIDEO, GNO, MOD, MOP, OMIT, PR, PROCO, RXNO, TXPO	739,133
Diet, Metabolomics, and Nutrition	CDNO, FIDEO, FOODON, ONE, ONS	30,139
Environment	ECOCORE, ECTO, ENVO, GEO, PCO, RBO	7,380
Health	ADO, SV, CHMO, CIDO, CTO, CVDO, DOID, DRON, GENEPIO, HSO, HTN, IDO, MAXO, MF, MFOEM, MFOMD, MONDO, MPIO, OAE, OGMS, OHD, OMRSE, ONTONEO, OOSTT, OVAE, SYMP, TRANS, VO	818,467
Information	CIO, CRO, DUO, IAO, LABO, OCCO, PDRO, PSDO, T4FS	3,377
Information Technology	MCRO, OBSCS, STATO, SWO	1,889
Investigations	ECO, ICO, MCO, NCRO, NGBO, OBI, OBIB, OPMI, ORNASEQ, PECO, SEPIO, SLSO	9,417
Microbiology	ICEO	4,100
Organisms	AFPO, BCO, GECKO, GSSO, HANCESTRO, OHMI, OPL, VBO	28,453

spective knowledge domains. Each ontology contained specific domain entities, and in most cases, entities from other ontologies were also included. To ensure that the extracted dataset was specific to the target ontology and its associated knowledge domain, we used the ontology's unique namespace as a filter. For example, in the ChEBI ontology, all domain entities have the namespace "CHEBI" in their Internationalized Resource Identifier (IRI), such as CHEBI_000001, CHEBI_000002, etc. Using this namespace, we extracted only those domain entities whose IRIs matched the namespace of the target ontology. This approach ensured that the resulting dataset contained only the entities specific to the target ontology and excluded unrelated entities imported from other ontologies.

Algorithm 2 describes a step-by-step process for extracting a dataset aligned with

Algorithm 2 Algorithm for extracting BFO-aligned datasets

Input: Ontology file *ontologyPath*, Namespace *namespace*
Output: Extracted dataset *d*

```

1: Initialize d as an empty dataset.
2: Load ontology from ontologyPath.
3: Identify domain entities entities using namespace.
4: for each entity in entities do
5:   Extract labels and comments from entity.
6:   if labels is null then
7:     Skip to next entity.
8:   end if
9:   Find the deepest BFO concept deepestBFOConcept that entity subsumes.
10:  if deepestBFOConcept is null then
11:    Skip to next entity.
12:  end if
13:  Add labels, comments, and deepestBFOConcept to d
14: end for
15: return Dataset d

```

Basic Formal Ontology (BFO) concepts from a given ontology file. The algorithm begins by initializing an empty dataset, *d*, which stores the extracted information and then loads the ontology from the specified file path (*ontologyPath*). Next, the algorithm identifies the domain entities within the ontology using the specified *namespace* of this ontology. For each domain entity, it extracts labels and comments, which provide textual descriptions of the entity and are essential for constructing their NLDs. If the domain entity does not have labels, the algorithm skips the current entity and moves to the next one. After that, the algorithm traverses the BFO class hierarchy and identifies the deepest BFO concept that the domain entity subsumes. We need this step to ensure that the domain entity subsumes at least one BFO concept. If no subsumption relationship with any BFO concept is found, the algorithm skips the current entity and moves to the next one. For entities successfully aligned with a BFO concept, the algorithm appends the extracted labels, comments, and the identified deepest BFO concept to the dataset *d*. After processing all domain entities, the algorithm returns the completed dataset, which contains the aligned domain entities along with their labels, comments, and hierarchical relationships to BFO concepts.

Table 4.2 summarizes the extracted domain ontologies from the OBO Foundry repository, grouping domain entities by ontology and predefined knowledge domains. Each domain represents a distinct area of knowledge, such as Agriculture, Health, or Information Technology, and lists the corresponding ontologies alongside the total number

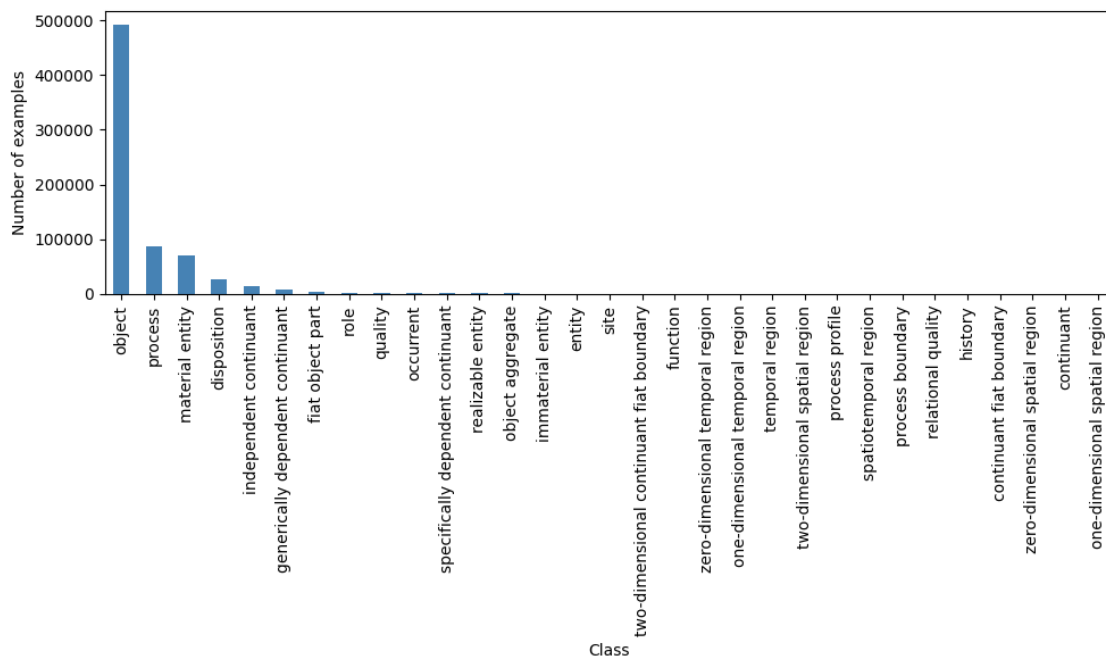


Figure 4.3 – Top 40 most populated direct DLP top-level concepts of WordNet synsets in the OntoWordNet (in linear scale).

of entities extracted. For instance, the Agriculture domain includes 1,071 entities from AGRO (Agronomy Ontology) and PSO (Plant Stress Ontology), while the Health domain includes 818,467 entities from multiple ontologies like DOID (Disease Ontology), DRON (Drug Ontology), and OGMS (Ontology for General Medical Science). These variations illustrate differences in the granularity and scope of ontologies across domains. Larger datasets in domains like Health and Chemistry reflect their broader range of ontological structures, while domains like Information Technology and Microbiology contain fewer entities, indicating narrower focus or fewer ontological resources.

Finally, the extracted datasets are structured by the predefined domains in the OBO Foundry and they are separated based on individual ontologies to maintain clear boundaries between the data sources. For example, entities extracted from the Gene Ontology (GO) and the Environment Ontology (ENVO) are stored in separate datasets to ensure no overlap or mixing of data. This structure allows us to analyze each ontology independently while also supporting cross-domain analyses where multiple datasets may be combined. Also, we can address updates or changes to individual ontologies without affecting others and ensure alignment with the OBO Foundry's classification. This approach also enables scalable and consistent analyses, providing flexibility for our experiments to perform domain-specific studies or integrate data across multiple domains.

4.2.3 Coverage and Limitations

The process of selecting, extracting, and organizing domain ontologies aligned with Basic Formal Ontology (BFO) top-level concepts, while systematic, is subject to several limitations. These limitations mainly arise from the inherent diversity of the OBO Foundry repository, as described below:

- **Limitations of manual selection:** we rigorously followed the selection methodology with careful attention to each step of the process to ensure consistency and alignment with the defined criteria. However, the manual selection methodology, while systematic, required significant time and effort and remained susceptible to human error. Decisions about including or excluding ontologies, particularly those with partial or ambiguous alignment to BFO, relied on subjective judgment. This subjectivity, inherent in manual evaluation, introduced variability into the dataset composition and may have excluded certain ontologies that could have contributed valuable domain entities to the dataset. For example, there are some alignments between NCBI Taxonomy and BFO in the OBO Foundry ontologies, articles, or presentations, but there is no official version. Also, since the OBO Foundry Repository is still growing, the manual selection and evaluation methodology may face scalability challenges. The inclusion of new ontologies or updates to existing ones requires re-evaluation to ensure alignment with BFO, which may not be feasible in the long term. Additionally, the increasing complexity of modular dependencies and extensions further complicates the extraction process.
- **Variability in ontology design practices:** the analyzed ontologies show considerable heterogeneity in design practices, including differences in naming conventions, annotation styles, and metadata completeness. For example, while most domain ontologies use Persistent Uniform Resource Locator (PURL) for namespaces, some adopt their own custom conventions, making automation of the extraction process more difficult. Similarly, some ontologies use underscores ("_") to separate words in labels, while others follow different formatting conventions, leading to inconsistencies during data parsing. Another notable issue is the lack of standardized annotations, as many ontologies do not consistently provide labels or comments for their entities. This lack of annotations was a primary reason for excluding several OBO Foundry ontologies from the dataset.
- **Imbalanced representation of BFO concepts:**

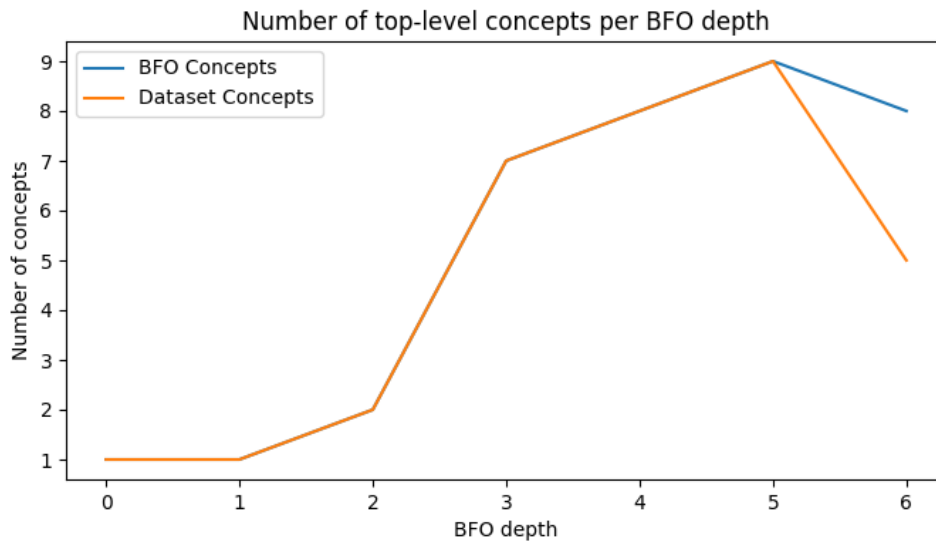


Figure 4.4 – The number of top-level concepts per level in Basic Formal Ontology (BFO) and of the domain ontologies from OBO Foundry repository.

As in the DLP dataset (described in Section 4.1), the distribution of domain entities across BFO top-level concepts is heavily imbalanced. Figure 4.3 illustrates that the "Object" concept is the most populated, while concepts such as "Relational Quality" and "Spatial Region" have significantly fewer entities. This imbalance arises primarily from the inherent focus of many domain ontologies on representing tangible, concrete entities with clear, observable characteristics, like in ontologies of the Chemistry and Biochemistry domains.

Figure 4.4 compares the coverage of top-level concepts in our extracted dataset against the BFO hierarchy based on their depth levels after we converted the datasets from multi-class to multi-label (described in Section 4.7). The blue line represents the distribution of BFO concepts, while the orange line represents the concepts from our dataset. By analyzing the lines, we observe that our dataset aligns closely with the BFO hierarchy at shallower depths but diverges at greater depths, where concepts such as "zero-dimensional continuant fiat boundary", "three-dimensional spatial region", and "one-dimensional continuant fiat boundary" are not represented. While a few top-level concepts remain uncovered in our BFO dataset, its overall coverage of BFO concepts is significantly more comprehensive than that of the DLP dataset.

4.3 From a Multi-Class to a Multi-Label Dataset

The transformation of a multi-class dataset into a multi-label dataset has significant implications for representing and analyzing hierarchical relationships between domain entities and top-level ontologies. Conventional approaches for classifying domain entities into top-level concepts, as presented in Section 3.2, adopt a multi-class scenario, where each entity is associated with a single label selected from a set of categories representing the top-level concepts. For example, the entity "rock" may be labeled as "Amount of Matter" in datasets such as DLP, ignoring its hierarchical associations with more general top-level concepts such as "Physical Endurant". This scenario inadequately captures the complexity of top-level hierarchies because it flattens this structure by restricting entities to a single label, thereby obscuring the inherent connections across the top-level ontology hierarchy depths. From that, the multi-class scenario loses important contextual information and misrepresents the relationships among domain entities and top-level ontology concepts. Also, in Figure 4.1 and Figure 4.2, we can see several top-level concepts that are more generic than others in the same branch of the top-level ontology with fewer number of domain entities, resulting in a wrongly under-representation of these more general top-level concepts. For instance, since the top-level concept of "Functional Matter" is a subclass of "Amount of Matter", and the top-level concept of "Agentive Physical Object" is a subclass of "Physical Object", "Amount of Matter" and "Physical Object" must encompass all domain entities inside "Amount of Matter" and "Agentive Physical Object", respectively.

From the ontology development perspective, the process of associating domain entities with top-level ontology concepts involves critical decisions about the level of specificity required when analyzing the hierarchy of the top-level ontology. When developing a domain ontology based on top-level concepts, ontology engineers are not always compelled to use the most specific top-level concept that a domain entity subsumes. Instead, they can choose more generic top-level concepts to guide their decisions, typically from shallower depths in the hierarchy, depending on the nature of the analyzed domain entity. Figure 4.5 and Figure 4.6 illustrate the distribution of domain entities across different hierarchical depths in DLP and BFO multi-class datasets, respectively. Figure 4.5 shows that a majority of domain entities of the DLP dataset are associated with depths between 6 and 8, while shallower depths and the deepest levels have significantly fewer entities. In contrast, Figure 4.6 exhibits a much simpler distribution, with most domain entities

of the BFO dataset concentrated at depth five and very few entities at shallower levels. These distributions are not accidental but instead reflect the inherent design philosophies and intended applications of each ontology. However, these datasets exhibit clear biases in their representation of domain entities, highlighting the need for a more balanced reflection of generic concepts in the top-level ontology's structure to ensure comprehensive and equitable classification across the entire hierarchy.

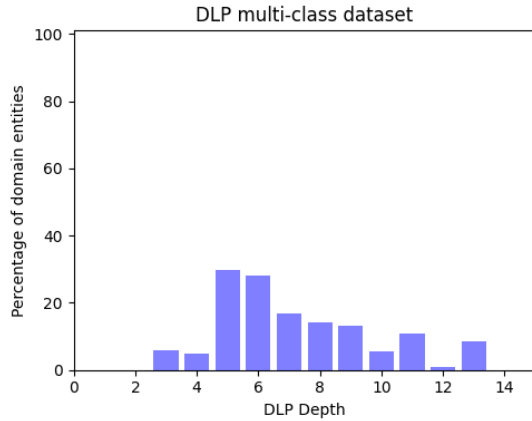


Figure 4.5 – Percentage of domain entities per depth in the DLP multi-class dataset.

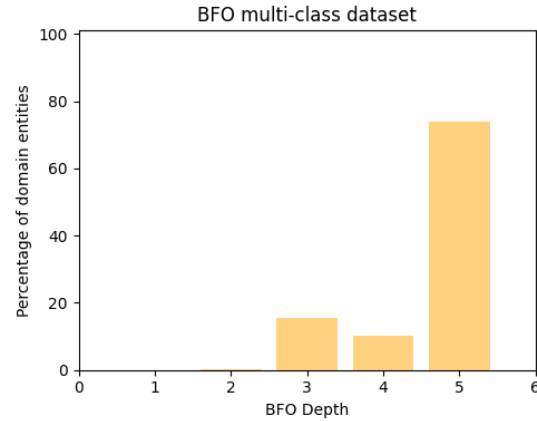


Figure 4.6 – Percentage of domain entities per depth in the BFO multi-class dataset.

The multi-class scenario becomes particularly limiting when a domain entity spans multiple distinct but equivalent branches in a top-level ontology by subsuming equivalent top-level concepts. For instance, the domain entity "Accord" (a written agreement between two states or sovereigns) can simultaneously belong to "Situation" and "Information Realization" due to its dual characteristics as both a formal event and a tangible informational artifact. In a multi-class dataset, this dual nature forces the entity to be arbitrarily assigned to a single concept, which inadequately represents its complexity. Conversely, a multi-label approach allows "Accord" to retain associations with both "Situation" and "Information Realization", preserving its full semantic meaning. This ensures that the dataset accurately captures the diverse characteristics of such entities, improving the performance and interpretability of classification models by reducing the distortion and redundancy observed in multi-class scenarios.

In order to solve all the above drawbacks of the multi-class scenario, we handled the problem of classifying domain entities into top-level ontology concepts as a multi-label classification problem, where each label (top-level concept) represents each depth of the top-level ontology in which the domain entity subsumes, from the most specific to the most generic one. In this context, we take advantage of the fact that top-level ontologies, such as DLP and BFO, are structured using subclass-of relations in order to

extract all the labels of a specific domain entity. The subclass-of is a taxonomic relation that is inherently transitive, meaning that if a concept A is a subclass of concept B, and B is a subclass of concept C, then A is also a subclass of C. This transitivity implies that domain entities classified under a specific concept also inherently belong to all the more general concepts in the hierarchy. For instance, if "rock" is a subclass of "Amount of Matter", and "Amount of Matter" is a subclass of "Physical Endurant", then "rock" is also an "Physical Endurant".

Unlike the multi-class approach, in a multi-label scenario, a domain entity can simultaneously be assigned to multiple labels. This raises an important question: why should classifying domain entities into top-level ontology concepts be handled as a multi-label problem? The answer lies in the distinct roles and structures of top-level and domain ontologies, as discussed in Section 3.2. Domain ontologies are dynamic, frequently evolving to incorporate new entities and relationships as knowledge in their respective domains expands. In contrast, top-level ontologies are intentionally designed to be stable frameworks, providing a consistent foundation for organizing knowledge across diverse domains. Their static structure is crucial for maintaining long-term consistency and ensuring interoperability across systems that depend on the top-level ontology. For this reason, we can take advantage of the stability in the top-level ontology structure and represent its entire concept hierarchy as labels in a classification pipeline. In this context, explicitly representing all levels of the top-level ontology hierarchy as labels in a multi-label classification approach offers several benefits during ontology development, such as allowing for capturing the full semantic richness of the ontology by associating each domain entity with every relevant abstraction level, from the most specific concept to the most general. This explicit representation supports decision-making by ontology engineers, enabling them to visualize how an entity fits into the top-level structure of the ontology and ensuring that its classification aligns with the intended goals of the domain entity in the developed ontology.

We used Algorithm 3 to convert the multi-class datasets, described in Section 4.1 and Section 4.2, into multi-label datasets by systematically associating each domain entity with all hierarchical levels from their respective top-level ontology (*TLO*). The starting point is the initialization of an empty dictionary D , where each unique domain entity is a key, and its associated labels are stored as a set to ensure deduplication and better performance for updates. For each domain entity e in the multi-class dataset E , the algorithm retrieves its assigned top-level concept c and all superclasses of c in the *TLO*, including

Algorithm 3 Algorithm for transforming a multi-class dataset into a multi-label dataset

Input: Multi-class dataset E , Top-Level Ontology TLO
Output: Multi-label dataset D

- 1: Initialize D as an empty dictionary, where keys are unique domain entities and values are sets of labels.
 - 2: **for** each domain entity e in E **do**
 - 3: Retrieve the class c of the entity e
 - 4: Retrieve all superclasses $labels$ of c in TLO , including itself
 - 5: **if** e exists in D **then**
 - 6: Update $D[e]$ with the union of $D[e]$ and $labels$.
 - 7: **else**
 - 8: Add e to D with $labels$ as its value.
 - 9: **end if**
 - 10: **end for**
 - 11: Convert D into a structured dataset with merged labels and deduplicated domain entities.
 - 12: **return** D
-

c itself. These superclasses are stored as a set of labels. If the domain entity e already exists in D , its set of labels is updated by taking the union of its current labels with the newly retrieved labels. If e does not already exist in D , the domain entity and its labels are added as a new entry. After processing all entities, D is converted into a structured dataset where each domain entity is uniquely represented along with its merged and deduplicated set of labels. The output of the algorithm is a multi-label dataset D that solves the problem of accurately reflecting the domain entity and the hierarchical structure of the top-level ontology that the domain entity subsumes and the problem of duplicated domain entities when a domain entity subsumes two different top-level concepts.

Figure 4.7 illustrates the implications of transforming a multi-class to a multi-label representation of domain entities categorized under a top-level ontology. On the left, the multi-class scenario depicts a simplified assignment, where each entity is given only one label (top-level concept). For example, "Tea (dinner)" is only labeled under "Functional Matter". On the right, the multi-label scenario provides a more accurate depiction of the hierarchy by assigning multiple labels to each domain entity, reflecting its placement in all top-level ontology levels. For example, "Tea (dinner)" receives labels for "Functional Matter", "Amount of Matter", and "Physical Endurant". From that, we underscored the key advantage of the multi-label representation in capturing the hierarchical nature of top-level ontologies while addressing the limitations of the multi-class format described before. For example, the top-level concept "Physical Endurant", which had no domain entity associations in the multi-class scenario, now has three domain entities properly

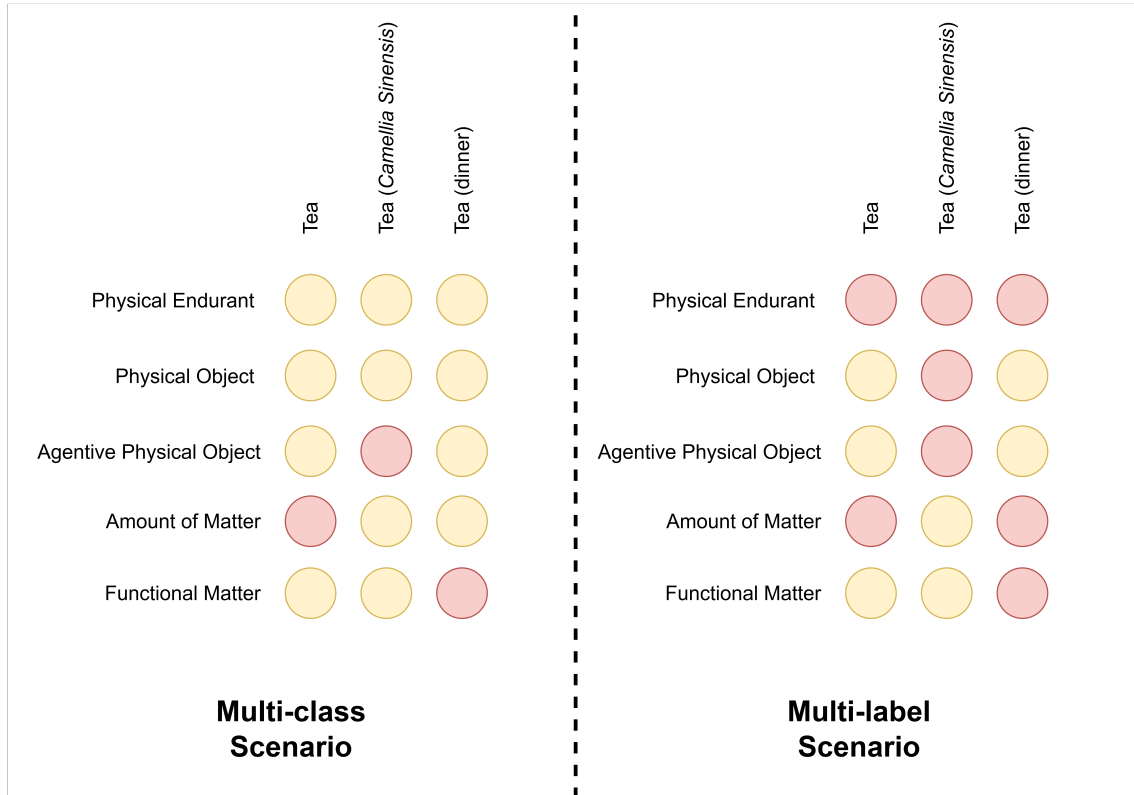


Figure 4.7 – Example of considering the DLP dataset in multi-class and multi-label scenarios.

associated in the multi-label scenario, illustrating how hierarchical relationships are accurately captured.

Figure 4.8 and Figure 4.9 show the representation of the complete taxonomy of DLP and BFO top-level ontologies in the extracted datasets after applying Algorithm 2. In comparison to Figure 4.5 and Figure 4.6, which represent the multi-class datasets, the multi-label datasets accurately reflect shallower depths in each respective top-level ontology, as well describes precisely domain entities that subsumes multiple top-level concepts. In this context, the resulting DLP dataset decreased from 65,018 domain entities in the original dataset to 61,483 domain entities in the multi-label dataset. This reduction occurred due to removing duplicate domain entities during the transformation process. For the BFO dataset, the algorithm preserved the number of domain entities, as no duplication existed in the extracted datasets.

The transformation from a multi-class to a multi-label dataset significantly enhances the ability to represent the hierarchical relationships inherent in top-level ontologies. By assigning multiple labels to each domain entity, this approach captures the full spectrum of generalization and specialization within the ontology structure, effectively preserving the taxonomic dependencies often lost in a multi-class setting. This refinement ensures that domain entities are not constrained to a single, overly specific classification

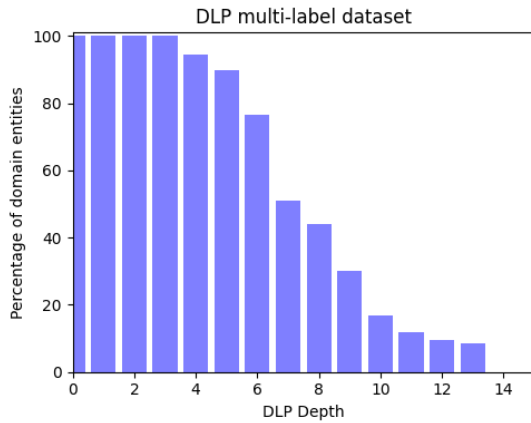


Figure 4.8 – Percentage of domain entities per depth in the DLP multi-label dataset.

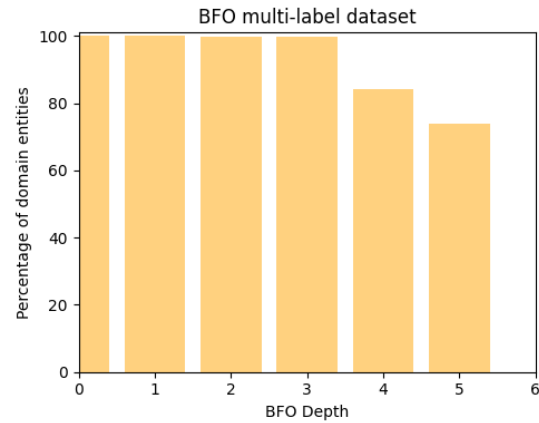


Figure 4.9 – Percentage of domain entities per depth in the BFO multi-label dataset.

but instead reflect all relevant levels of abstraction. For example, in the DLP dataset, transforming "Tea (dinner)" from a single "Functional Matter" label to a multi-label structure including "Functional Matter", "Amount of Matter", and "Physical Endurant" ensures a richer representation of the concept's ontological placement. This expanded labeling not only resolves the issue of under-representation in higher-level concepts but also improves the dataset's alignment with formal ontological structures. However, while this methodology improves hierarchical fidelity, it also introduces complexities, such as label redundancy and the need to manage potential conflicts between top-level concepts. Addressing these challenges requires careful curation and algorithmic adjustments to ensure semantic integrity while maximizing classification accuracy in classifying domain entities into top-level ontology concepts.

4.4 Textual Representation of Domain Entities

In the previous sections, we described our methodology for extracting datasets for the task of classifying domain entities into top-level ontology concepts. Also, we transform multi-class datasets into multi-label datasets to encompass the full range of generality within top-level ontologies. Through this process, we identify distinct textual representation approaches for describing and representing domain entities, each with specific structural characteristics and implications for the ontology development process. These approaches include definiendum (plural definienda), definiens (plural definientia), example sentences, and NLDs. Understanding these approaches provides a foundational basis for analyzing their utility in classifying domain entities into top-level ontology con-

cepts.

Table 4.3 describes examples of definiendum and definiens extracted from the DLP dataset described in Section 4.1 for material entities "gold", "iron", "silver", the periods of the day "evening", "night", and "day" and the colors "gold", "silver", and "green". Usually, NLDs are written using the Aristotelian format ("X is a Y that Z"), containing the definiendum "X", the copula "is a" and the definiens "Y that Z". As discussed before, the definiendum is the defined term, the copula is the relationship between the definiendum and the definiens, and the definiens is the explanatory part of the NLD. On the other hand, example sentences are pieces of text where the definiendum of the domain entity occurs in plain text. For example, an example sentence for the "gold" domain entity should be "The gold necklace gleamed brilliantly under the soft light of the chandelier". The examples in Table 4.3 highlight different examples of definienda and definientia, which we analyze in detail below.

Table 4.3 – Examples of definienda and definientia extracted from BabelNet.

Definiendum	Definiens
Gold	A soft yellow malleable ductile (trivalent and univalent) metallic element; occurs mainly as nuggets in rocks and alluvial deposits; does not react with most chemicals but is attacked by chlorine and aqua regia
Iron	A heavy ductile magnetic metallic element; is silver-white in pure form but readily rusts; used in construction and tools and armament; plays a role in the transport of oxygen by the blood
Silver	A soft white precious univalent metallic element having the highest electrical and thermal conductivity of any metal; occurs in argentite and in free form; used in coins and jewelry and tableware and photography
Day	The time after sunrise and before sunset while it is light outside
Night	The time after sunset and before sunrise while it is dark outside
Evening	The latter part of the day (the period of decreasing daylight from late afternoon until nightfall)
Gold	A color tone resembling the gold chemical element
Silver	A color tone resembling gray that is a representation of the color of polished silver
Green	The color between cyan and yellow on the visible spectrum

As we can see in Table 4.3, using the definiendum is the shortest way to represent a domain entity since the definiendum names a domain entity through a combination of a few words. We can find definienda in many formats from the resources we used to extract our datasets. For example, in WordNet, a definiendum is part of a synset, which also serves as groups of synonymous terms that share a common meaning. In Wikipedia, the

definiendum is employed as page titles or article headings, aligning it with the domain entity the article aims to describe. In OWL (Ontology Web Language) ontologies, the definiendum is represented through *rdfs:label*. The definiendum offers simplicity and brevity, making it an efficient way to represent domain entities. However, its reliance on a minimal textual format introduces significant limitations, particularly distinguishing entities with the same definiendum but different meanings. As demonstrated in Table 4.3, the definiendum "gold" may refer to a material entity in one context and color in another. This polysemy problem underscores the necessity of additional textual information, such as the definiens, example sentences, or NLDs, to resolve ambiguities and provide a more complete understanding of the domain entity. Despite these limitations, the definiendum remains an essential component of ontology design, serving as the initial identifier upon which more detailed textual representations are constructed.

The definiens offers a more detailed textual representation of a domain entity than the definiendum. From that, the definiens provides a descriptive explanation, conveying the semantic content and context necessary to differentiate one domain entity from others. Resources like WordNet, Wikipedia, and OWL ontologies utilize definiens in distinct ways. WordNet includes glosses as definiens, delivering concise descriptions of synset meanings. Wikipedia employs the introductory sections of articles to serve as definiens, where detailed descriptions of the entity align with its definiendum. In OWL ontologies, the definiens is represented using properties such as *rdfs:comment* or *IAO_0000115* (definition), allowing formal and explicit descriptions of domain entities. Also, the definiens addresses many limitations of the definiendum by providing context and specificity. From that the definiens resolves ambiguities introduced by polysemy, as it explicitly describes the unique characteristics of each sense of a domain entity. For example, the definiens distinguishes the material and color senses of "gold" by highlighting their distinct properties and contexts. However, the richness of the definiens can vary significantly across resources. In the DLP dataset, the definientia, while succinct, may omit critical details since we extracted them from WordNet and Wikipedia, whereas the BFO dataset definientia adhere to strict formalization and domain-specific vocabulary, which can make them less interpretable for non-experts. We will discuss more about this characteristic later in this section.

Example sentences provide contextual illustrations of how a domain entity is used or understood in practical scenarios. These sentences supplement the definiendum and definiens by demonstrating the entity in action, offering additional clarity, and supporting

disambiguation. In WordNet, example sentences accompany many synsets to show the usage of terms within their intended semantic contexts. These sentences illustrate the relationships between the definiendum and its broader linguistic environment. For instance, the synset containing "gold" as a material entity might include the example sentence "The necklace was made of pure gold", whereas the synset for "gold" as a color might include "The curtains were a rich gold color". Wikipedia incorporates example sentences within the narrative of its articles, using them to illustrate concepts, highlight use cases, or describe scenarios where the domain entity plays a role. These sentences appear organically throughout the article, often complementing the definiendum and definiens by situating them in broader contexts. For example, the Wikipedia article on "Gold" may describe its use in jewelry with a sentence like "Gold has been valued for its use in jewelry for centuries due to its malleability and resistance to tarnish". On the other hand, OWL ontologies rarely include explicit example sentences, as their focus lies in formal and logical representation rather than natural language use. Although we can use example sentences to disambiguate the meaning of a definiendum (as in WordNet), if an example sentence was not directly provided (as in Wikipedia), we depend on word sense disambiguation in order to extract the real meaning of the definiendum inside the example sentence. For example, in the example sentence "The gold necklace gleamed brilliantly under the soft light of the chandelier", it is confusing if the sentence is talking about the gold as the material that the necklace is composed of or about its color.

While definiendum, definiens, and example sentences each contribute with unique strengths to the textual representation of domain entities, they also exhibit limitations. In this context, this work proposes using NLDs to represent domain entities textually. In our view, these NLDs act as a bridge between linguistic descriptions and the structure of top-level ontologies by linking the definiendum and the definiens through the copula in a unique and unambiguous textual sample. This bridging role overcomes the limitations of relying solely on definiendum, definiens, or example sentences for textual representation. Also, based on this bridging role and their ability to encapsulate semantic information, we hypothesize that NLDs enable the systematic association of domain entities with top-level ontology concepts. This hypothesis was grounded in two main notions: the semantic similarity in taxonomies (described in Section 2.1.5) and the semantic relatedness in the Distributional Hypothesis of language models (described in Section 2.2.1).

The first notion behind our hypothesis is how domain entities are classified within an ontology through a taxonomy. A taxonomy is a classification system that organizes

concepts, domain entities, or information into categories hierarchically based on common characteristics or criteria. Also, taxonomies are the base structure for ontologies and top-level ontologies, facilitating knowledge organization. Based on that, domain entities grouped in the same top-level ontology concept mean that they share common features or attributes, i.e., they are more similar to each other than those of other domain entities under another top-level concept. For example, all domain entities grouped under the "Amount of Matter" top-level concept, such as "gold", "iron", and "silver", have a greater similarity score because they share common characteristics about physical substances or materials. These entities are inherently more similar to one another when contrasted with entities classified under a completely different concept, such as "evening", "night", and "day" under the "Temporal Region" top-level concept. In this context, NLDs play a critical role in classifying domain entities into top-level ontology concepts by encapsulating the distinguishing features of domain entities within structured textual patterns. These patterns enable the accurate association of entities with their respective top-level ontology concepts, taking advantage of how the Distributional Hypothesis works in current language models.

The second notion underlying the hypothesis in this thesis is the Distributional Hypothesis, which posits that words appearing in similar contexts tend to have similar meanings, as reflected in their embedding representations in language models. This hypothesis is particularly useful in identifying semantic similarities among domain entities through their NLDs. For instance, entities like "gold" and "silver", both described with attributes such as malleability and conductivity, are likely to be classified under a common top-level ontology concept like "Amount of Matter" due to their shared contextual features. Figure 4.10 illustrates a PCA visualization of the embedding representations generated by the BERT language model for the definiendum, definiens, and NLDs of the domain entities listed in Table 4.3. This figure reveals significant differences in how NLDs, definiens, and definiendum represent domain entities in relation to their semantic distribution and polysemy challenges. In the NLDs (c) and definiens (b) plots, domain entities are more clearly separated into distinct clusters, such as the grouping of "gold" (material), "iron", and "silver" (material) under material entities, "day", "night", and "evening" under temporal entities, and "gold" (color), "green", and "silver" (color) under quality entities. This clustering reflects the ability of these representations to capture richer contextual and semantic information, aligning with the Distributional Hypothesis and the taxonomy similarity. In contrast, the definiendum (a) plot shows a less distinct separation between

polysemous terms, particularly evident in the overlap between "gold" (color) and "gold" (material), highlighting its limitation in resolving ambiguity due to minimal textual context. These findings suggest that NLDs and definiens are better suited for tasks involving text classification. Also, it is important to note that while the Distributional Hypothesis helps to identify contextual relationships, it does not always align perfectly with taxonomic similarity. Language models rely on statistical patterns of word co-occurrence, which are influenced by the frequency and contexts present in the training data. For example, entities like "apple" and "green" often appear in similar contexts but belong to different top-level ontology concepts. "Apple" is classified as a material entity, while "green" falls under the quality concept. This distinction highlights a limitation of the Distributional Hypothesis, as language models may cluster these terms based on contextual co-occurrence rather than their underlying ontological differences.

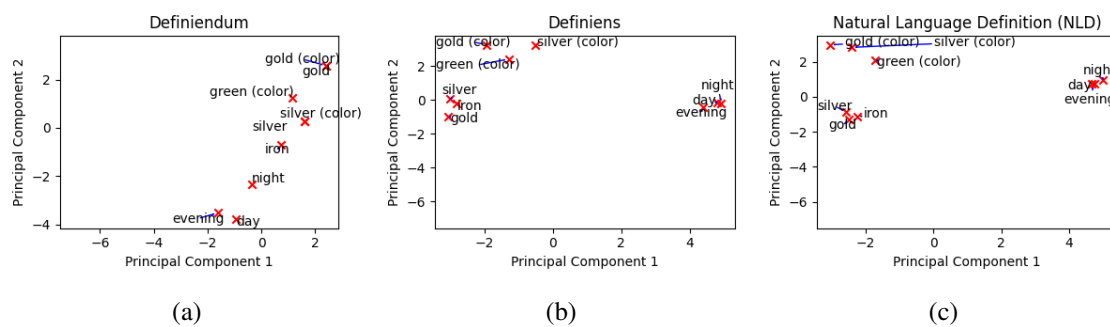


Figure 4.10 – PCA distribution of domain entities based on the embedding of: (a) Definiendum; (b) Definiens; (c) NLDs using BERT-Base language model.

So far, we described several approaches for representing domain entities textually, including definienda, definientia, example sentences, and NLDs. We also presented two key notions supporting the hypothesis that NLDs offer the most effective representation of domain entities. Beyond these theoretical foundations, additional factors support the choice of NLDs, such as: existing resources can provide NLDs, and domain experts can curate them during the ontology development process to ensure alignment with domain knowledge; ontology development methodologies (METHONTOLOGY and NeOn) also position NLDs early in the development cycle, typically after the Knowledge Acquisition phase and before Conceptualization; NLDs tend to be relatively short, averaging only a few dozen words, which reduces sequence length for language models, lowering computational costs during text processing. Additionally, NLDs aim to explain a domain entity's meaning clearly and in a specific context, minimizing polysemy issues. However, a major limitation of NLDs arises from their dependence on clarity and precision. Next, we explore how the choice of resources for extracting NLDs influences their structure and

effectiveness.

Similar to definiens, the effectiveness of NLDs depends significantly on the clarity and precision of the text used to describe the domain entity. Several factors influence how NLDs are constructed, including vocabulary selection, the level of detail and length, the intended audience, and the degree of domain-specific expertise. To account for these variations, this work follows the distinction outlined by Seppälä et al. (2016) (presented in Section 2.1.3), categorizing NLDs based on the type of resources from which they are derived. Specifically, we use the term general NLDs to refer to those extracted from lexicographic, lexico-semantic, and terminological resources, and domain-specific NLDs to describe those derived from ontologies, standards, and normative terminological resources. This classification aligns with the datasets described in Section 4.1 and Section 4.2, where the DLP dataset consists exclusively of general NLDs derived from WordNet and Wikipedia, and the BFO dataset contains only domain-specific NLDs derived from domain ontologies available in the OBO Foundry repository.

Table 4.4 presents several examples of general and domain-specific NLDs from our datasets. A direct comparison highlights the trade-off between intention, richness, and precision. General NLDs typically have the intention to provide the meaning of the domain entity by offering a broader conceptual coverage but are limited in their depth and specificity, while domain-specific NLDs have the intention to disambiguate the meaning of a domain entity regarding their specific context usage, providing a detailed semantic clarity and precision to the defined domain entities. For example, the general definiens contained in the NLD for "Inferior horn of the thyroid cartilage" describes it as "The largest cartilage of the larynx", which is a broad and somewhat imprecise definiens since it is the same definiens as "Superior horn of the thyroid cartilage". In contrast, the domain-specific NLD provides a much richer and more precise definiens: "The inferior horn of thyroid cartilage is short and thick, directed downward, with a slight inclination forward and medialward, and presents, on the medial side of its tip, a small oval articular facet for articulation with the side of the cricoid cartilage". This definiens includes detailed structural properties and spatial relations, essential for formal anatomical ontologies where exact positional and functional attributes must be defined. Similarly, the definiens contained in the general NLD for "Andesite" simply states, "A dark grey volcanic rock", which offers a basic surface-level description suitable for non-specialist audiences. However, it lacks essential compositional details necessary for a geological ontology, for example. In contrast, the domain-specific NLD expands significantly by defining "Andesite"

as "An extrusive igneous, volcanic rock, of intermediate composition, with aphanitic to porphyritic texture". This version clarifies the rock's formation process, texture, and mineralogical properties, aligning better with a structured classification of rock types in the field of geology.

Table 4.4 – Examples of General NLDs and Domain-Specific NLDs.

Definiendum	General NLD from DLP dataset	Domain-specific NLD from BFO dataset
Inferior horn of thyroid cartilage	Inferior horn of thyroid cartilage is the largest cartilage of the larynx	The inferior horn of thyroid cartilage is short and thick. it is directed downward, with a slight inclination forward and medialward, and presents, on the medial side of its tip, a small oval articular facet for articulation with the side of the cricoid cartilage
Superior horn of thyroid cartilage	Superior horn of thyroid cartilage is the largest cartilage of the larynx	The superior horn of thyroid cartilage is long and narrow, directed upward, backward, and medialward, and ends in a conical extremity, which gives attachment to the lateral hyothyroid ligament
Andesite	Andesite is a dark grey volcanic rock	Andesite is an extrusive igneous, volcanic rock, of intermediate composition, with aphanitic to porphyritic texture
Rock	Rock is a material consisting of the aggregate of minerals like those making up the Earth's crust	Rock is a naturally occurring solid aggregate of one or more minerals or mineraloids
Day	Day is the time for Earth to make a complete rotation on its axis	Day is a one-dimensional temporal region which spans the duration an astronomical body completes one rotation on its axis

The examples presented in Table 4.4 effectively illustrate the differences between general and domain-specific NLDs, highlighting distinctions in intention, richness, and precision. However, these examples should not be interpreted as rigid or absolute patterns. While general NLDs often prioritize broader coverage and simplicity, they can, in certain cases, exhibit significant richness and precision comparable to domain-specific NLDs, as in the example of the "rock" domain entity. Similarly, domain-specific NLDs,

despite their focus on specialized contexts and formal clarity, can sometimes be overly constrained by the structure of the top-level ontology from which they are derived. For instance, the general NLD for "Day" describes it as "The time for Earth to make a complete rotation on its axis", a concise yet accurate description. The domain-specific NLD, however, defines it as "A one-dimensional temporal region which spans the duration an astronomical body completes one rotation on its axis", explicitly referencing the BFO top-level concept of "One-Dimensional Temporal Region". While precise, this direct ontological reference risks circularity by embedding the classification within the textual representation, potentially biasing the classification results. Although this bias can occur intentionally or unintentionally in both types of NLDs, it has a minimal impact on the overall extracted datasets.

In summary, the textual representation of domain entities using definienda, definiens, example sentences, and NLDs offers diverse approaches for capturing the meaning of domain entities during the ontology development process. Each method varies in semantic precision, intention, and clarity, with definienda being concise but often ambiguous, definiens providing more detailed descriptions yet varying in quality across resources, and example sentences supplementing context but lacking formal precision. NLDs are a structured approach that links the definiendum and definiens through a formalized linguistic pattern (Aristotelian format), balancing clarity with semantic detail. However, the contrast between general and domain-specific NLDs, as shown in Table 4.4, highlights differences in intention and granularity. While general NLDs often focus on broader conceptual coverage, domain-specific NLDs emphasize context-specific precision but may closely mirror the ontology structure, risking circularity in classification tasks. These distinctions underscore the importance of carefully selecting textual representations that balance expressiveness and formal alignment, ensuring a comprehensive yet flexible approach for classifying domain entities into top-level ontology concepts.

4.5 Training Methodology

This chapter discussed dataset extraction and the relation between NLDs, distributional hypothesis, and top-level ontology concepts. From the dataset perspective, we represented each domain entity by a row containing three columns: the definienda, the definiens, and the labels. Since each knowledge resource, such as WordNet or Wikipedia, only includes one definiens per domain entity and a domain entity can have more than one

definiendum, we can use an approach to augment the dataset (increase the dataset size) by creating rows for each domain entity definiendum. We kept the same definiens and labels for each new domain entity row. This technique is also called the "explode" approach. For example, the "gold" domain entity in Table 4.3 has three different definienda in BabelNet: gold, Au, and atomic number 79. From that, we create a new row in the dataset for these three definienda and keep the same definiens and labels.

Figure 4.11 presents the number of definienda of the domain entities in the DLP dataset from the alignment between OntoWordNet and BabelNet³. Considering that most dataset domain entities have more than one definiendum, the "explode" approach helps increase the number of examples without artificially generating data. For instance, we can expand the dataset from 59,666 to 387,814 examples. However, this process has implications for the performance of a classification model. Considering the example of the "gold" domain entity (from Table 4.3) and that we followed this sequence of decisions, used Pipeline 2 (described in the previous section) with a KNN machine-learning model with $k = 3$, trained this model with two of the three new rows generated from "gold" domain entity, and predict the labels of the last remaining "gold" row. Based on the distributional hypothesis, the three closest examples predicted by KNN also include the two variations of the "gold" domain entity because they share almost the same NLD with a change in the definiendum part. This fact also occurs in Pipeline 1 but is more easily explainable using Pipeline 2. Although this characteristic appears after using the "explode" approach, this may not be a problem, but we should consider it when evaluating classifier performance, especially in evaluations using training and test sets of the same dataset, like in k-fold cross-validation.

In this work, we explored three training methodologies to deal with the consequences of the "explode" approach. Considering that the training and test samples are from the same dataset, the first method ignores this consequence, i.e., we apply the "explode" approach before dividing the dataset into training and test samples. However, suppose that we aim to evaluate the performance of a classification model with completely novel domain entities. In this case, as our second method, we must apply the "explode" approach after dividing the dataset into training and test samples, ensuring that no test domain entity is present in the training samples. As a third method, we considered that the training and test samples are from different knowledge resources, such as the training samples from WordNet and the test samples from Wikipedia, or vice-versa. In this case,

³In the BFO dataset, each domain entity contains only 1 definiendum and definiens.

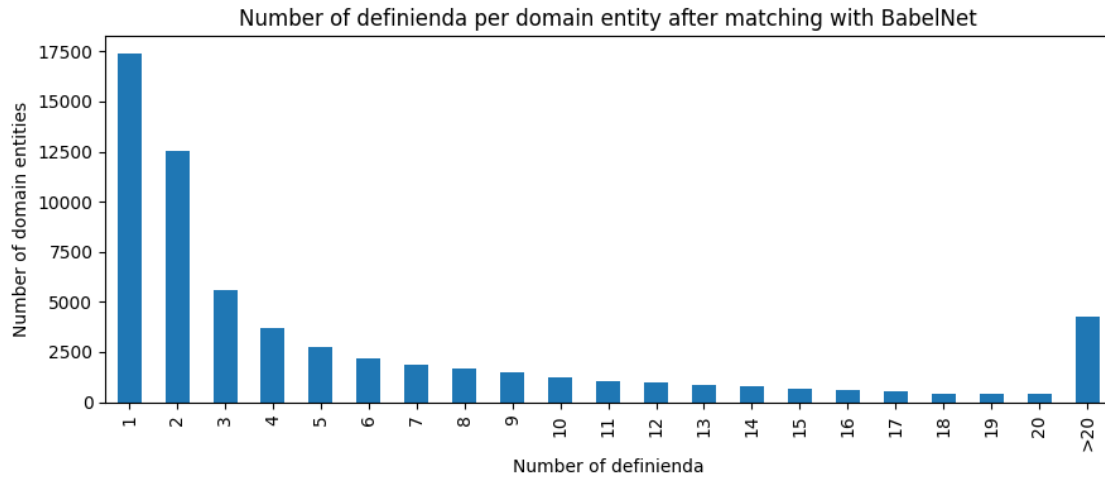


Figure 4.11 – The number of definienda of the domain entities in the dataset.

although we can have the same domain entity in multiple resources, they usually do not have the same definiens. For example, in Wikipedia, "gold" is defined as "a chemical element; it has symbol Au and atomic number 79", an almost entirely different definiens from the one described in Table 4.3. So, the embedding representation of the NLDs of these domain entities is more discrepant than changing the definiendum. Thus, the "explode" approach does not generate such perceptible consequences in this training methodology.

4.6 Classification Pipelines and Evaluation

In this work, we propose a systematic approach for the task of classifying domain entities into top-level ontology concepts, addressing the inherent complexity of this task with a multi-label classification strategy. As detailed in Section 4.3, the multi-label paradigm allows domain entities to be associated with multiple hierarchical concepts, capturing the varying levels of abstraction and specificity within a top-level ontology. This approach more accurately reflects how ontology engineers classify entities during ontology development, where the selection of more generic or more specific top-level concepts is often guided by the context and intended granularity of the ontology. Furthermore, as discussed in Section 4.4, we adopted NLDs as the primary textual representation for domain entities in our classification pipelines. NLDs offer a structured yet expressive way to describe domain entities, balancing intention, richness, and precision more effectively than alternative textual forms such as definienda, definientia, or example sentences. Combining the definiendum, copula, and definiens, their Aristotelian structure provides contextual clarity essential for disambiguation and semantic representation of domain

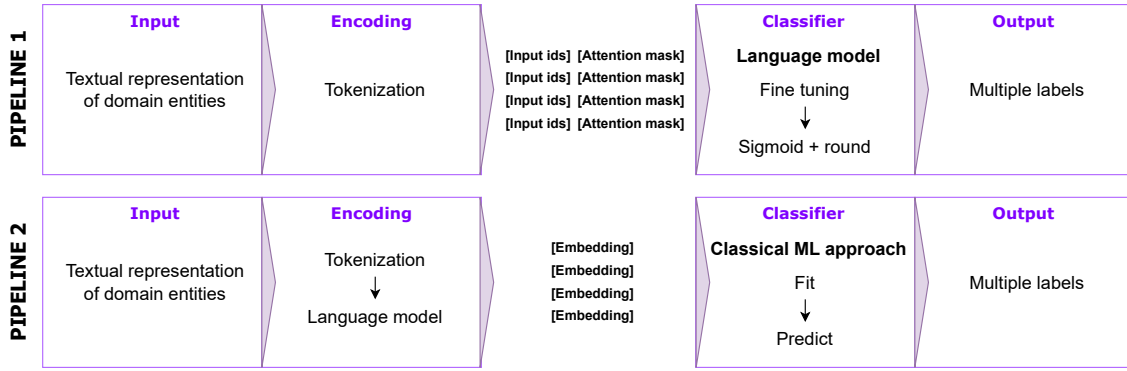


Figure 4.12 – Proposed pipelines for classifying domain entities into top-level ontology concepts using the textual representation of these domain entities.

entities. This choice aligns with the need for high-quality textual inputs in textual classification tasks, as NLDs encode both descriptive features and relational context, which can be leveraged by Machine Learning Models to improve classification accuracy. By integrating a multi-label classification scheme with semantically rich textual representations, we proposed two kinds of classification pipelines which we will discuss next.

The central challenge in classifying domain entities into top-level ontology concepts using NLDs arises from the need to accurately capture the complex semantic information embedded within these definitions. NLDs often convey subtle distinctions between concepts, requiring models capable of understanding not only the direct meaning of words but also their contextual relationships, hierarchical implications, and domain-specific nuances. In this context, we employed Language Models in two distinct classification pipelines (Figure 4.12):

Pipeline 1 is designed to fine-tune a pre-trained Language Model (LM) specifically for classifying domain entities into top-level ontology concepts in a multi-label scenario. Fine-tuning involves adapting the pre-trained LM to the specific task by adjusting its internal weights during the training phase. From that, we allow the model to refine its understanding of the intricate patterns and nuanced meanings present in the textual representation of domain entities, which are often subtle and require deep contextual understanding. However, the fine-tuning process is computationally expensive, especially when dealing with large language models that contain millions or even billions of parameters. The computational cost increases with the number of training epochs—each pass through the dataset—since more epochs generally lead to better model convergence but at the expense of longer training times. Also, the size of the original LM can significantly impact resource requirements. For instance, larger models require more memory and processing power, making it challenging to fine-tune them on standard hardware without

access to specialized computing infrastructure, such as GPUs or TPUs. In addition, fine-tuning an LM can introduce the risk of overfitting, especially in a dataset where the data combines many definienda that name a domain entity with a single definiens. In such cases, the model may become too closely aligned with the specific patterns of the training data, leading to poor generalization to new, unseen data.

In contrast, **Pipeline 2** bypasses the fine-tuning process entirely, opting instead to leverage pre-trained LMs to generate embeddings—dense vector representations of the input textual representation of domain entities that encapsulate the semantic information of the text. In this context, using these pre-trained embeddings, Pipeline 2 avoids the computationally intensive task of fine-tuning an LM, significantly reducing computational requirements. The core idea behind Pipeline 2 is to decouple the representation learning from the classification task, in which the LM acts as a feature extractor, transforming the unstructured text data into a high-dimensional vector space where we preserve the semantic relatedness between texts, according to the distributional hypothesis. Also, this approach benefits from the fact that pre-trained language models, such as BERT or GPT, have been trained on vast amounts of text data and have already learned general linguistic patterns that are effectively reused for many downstream tasks without further adjustment. Once we generate the embeddings, we feed them into classical machine learning algorithms like K-Nearest Neighbor (KNN), Decision Tree, or Support Vector Machine (SVM) for classification. This step capitalizes on the strengths of traditional Machine Learning methods, which are well-suited for handling embedding data. In addition, classical algorithms like KNN offer intuitive interpretability, as the classification decision is based on the proximity of the new data point to known training examples. Despite these advantages, Pipeline 2 may not achieve the same level of predictive accuracy as fine-tuning the language model directly. Since we generate the embeddings from a pre-trained model without task-specific adjustments, the model may not fully capture the nuances of the language used to represent domain entities.

The desired output of both classification pipelines is multiple top-level ontology concepts in which a given input subsumes. In this context, we represent all the possible top-level concepts in a one-hot encoded vector, i.e., each concept is represented by a binary value, with 1 indicating the presence of the concept and 0 indicating its absence. This vector serves as the target for the Classification step of both pipelines. In this context, for Pipeline 1, we used a sigmoid activation to transform the average pooled output from the language model into probability values. After that, we rounded these probability values to

handle the one-hot encoding representation of the top-level concepts. On the other hand, for Pipeline 2, we do not need any transformation from a mathematical function since traditional machine learning approaches, such as K-Nearest Neighbor and Decision Tree, can handle the predicted decisions directly.

Considering the above description of both Pipeline 1 and Pipeline 2, we need to evaluate certain criteria to decide which one to use:

- **Classification results:** Pipeline 1 generally has an advantage in classification results because fine-tuning allows the model to adapt specifically to the task at hand. Pipeline 2, while efficient, may lack the task-specific accuracy that fine-tuning provides since it depends on the data used to train the Language Model.
- **Computational cost:** Pipeline 1 is computationally expensive due to the fine-tuning process, which involves adjusting millions or billions of parameters. Pipeline 2, in contrast, is more computationally efficient, as it reuses pre-trained embeddings without requiring model adjustments, although classification models like decision trees and k-nearest neighbor struggle with large datasets.
- **Explainability:** Pipeline 2 excels in explainability, thanks to the transparency of classical machine learning models, like KNN or Decision Trees. Pipeline 1, with its reliance on deep learning, often functions as a black box, making it harder to interpret the decision-making process during classification.
- **Overfitting risk:** Pipeline 1 carries a higher risk of overfitting, especially with small or repetitive datasets, since fine-tuning can make the model overly specialized to the training data. Pipeline 2, by using pre-trained embeddings and traditional machine learning classifiers, is generally less prone to overfitting.
- **Flexibility and experimentation:** Pipeline 2 allows for more flexible experimentation since we generate the embeddings once, store them, and reuse them. From that, it is easy to test different classical machine-learning models without the necessity of retraining the language model. Pipeline 1, on the other hand, requires retraining for every new experiment, which can be time-consuming and resource-intensive.

5 EXPERIMENTS

In this section, we present five study cases designed to validate the hypothesis that NLDs encapsulate sufficient semantic information to relate domain entities to their corresponding top-level ontology concepts. Each study case focuses on the same overarching task: classifying domain entities into top-level ontology concepts. However, each study case explores distinct aspects of this classification challenge, thereby providing a comprehensive evaluation of our hypothesis. In Study Case 1, we investigated the effectiveness of various textual representation approaches (e.g., definiendum, definiens, example sentences, NLDs) for representing domain entities, as well as the impact of different data augmentation strategies, including the "explode" approach. Study Case 2 expanded this analysis to multilingual settings by leveraging generic NLDs in different languages. Study Case 3 explored the generalizability of different language models and pipelines in cross-resource scenarios, using generic NLDs from distinct knowledge sources. Study Case 4 examined the performance of pre-trained language models in cross-domain scenarios, testing their ability to classify entities from diverse knowledge domains into top-level concepts using domain-specific NLDs. Finally, Study Case 5 evaluated the cross-ontology performance, analyzing how domain-specific ontologies perform and generalize within the embedding representation of domain-specific NLDs. Together, these study cases provide a comprehensive framework for understanding the role of NLDs and pre-trained language models in the task of classifying domain entities into top-level concepts.

5.1 Experimental Setting

In addressing the task of classifying domain entities into top-level ontology concepts with unbalanced labels, we used a stratified k -fold cross-validation approach with $k = 10$. This k -fold methodology involves partitioning the dataset into k equally sized folds, ensuring each fold maintains the same proportion of class labels observed in the original dataset. After that, we combined $k - 1$ of these folds for each cross-validation iteration to form the training sample upon which we trained the classification pipelines. Also, we used the remaining single fold as the test sample to evaluate the performance of the classification pipelines. This process is systematically repeated k times, with each fold getting the opportunity to be the test sample exactly once. From this, we allow for a more robust assessment of classification pipeline generalization on the input data.

In order to evaluate the results, we used the macro F1-score (Equation 5.1) to ensure a better classification analysis in the unbalanced scenario. We also conducted the study cases and experiments on a machine equipped with an Intel i7-10700 CPU (4.8GHz), 32 GB of RAM, and a GeForce RTX 3060 GPU with 12GB of VRAM.

$$F1_{\text{macro}} = \frac{1}{N} \sum_{i=1}^N 2 \cdot \frac{\text{precision}_i \cdot \text{recall}_i}{\text{precision}_i + \text{recall}_i} \quad (5.1)$$

where $F1_{\text{macro}}$ is the macro F1-score; N is the number of labels; i is the i th class in N ; precision_i is the precision for the i th class, defined as the number of true positive predictions divided by the total number of positive predictions (true positives plus false positives); recall_i is the recall for the i th class, defined as the number of true positive predictions divided by the total number of actual positives (true positives plus false negatives).

5.2 Study Case 1: Textual Representation of Domain Entities

In this study case, we performed experiments regarding the effectiveness of different textual representations of domain entities to classify domain entities into top-level ontology concepts (as described in Section 4.4). In these experiments, we compared the two pipelines proposed in Section 4.6 using train and test samples from the DLP dataset (described in Section 4.1). For Pipeline 1, we employed fine-tuning the BERT-Base language model for our specific task trained over 4 epochs. On the other hand, for Pipeline 2, we also used the BERT-Base language model to generate the embedding representation of the input text and fed a K-Nearest Neighbors (KNN) classifier to predict the output labels. Also, we compared the performance of the "explode" approach for training and test samples from the same knowledge resource. In this case, we evaluate the textual representations provided by the WordNet knowledge resource since only WordNet contains both NLDs and example sentences.

Table 5.1 presents a comparative view of dataset sizes of each textual representation approach across our preprocessing states of the dataset, considering the original dataset size from the alignment between OntoWordNet and BabelNet, the size after merging the labels of the domain entities that belong to multiple top-level concepts and removing the ones that have disjoint labels, and after the application of the "explode" approach to increase the dataset size. For the NLD and definiendum datasets, the original size is

Table 5.1 – The number of examples in each text representation dataset.

Dataset	Original size	After preprocessing	After "explode"
NLD	65,018	59,666	387,814
Definiendum	65,018	59,666	387,814
Definiens	65,018	59,666	59,666
Example sentence	8,417	7,329	7,329
Definiendum + Exemple sentence	8,417	7,329	43,101

65,018, which was reduced to 59,666 after merging domain entities that belong to multiple top-level concepts and removing each one that presents disjoint labels (as discussed in Section 4.3). After using the "explode" approach for data augmentation, the size expands dramatically to 387,814. In contrast, the definiens dataset remains unchanged at 59,666 examples because each domain entity has only one definiens in WordNet. The example sentence dataset starts at a much smaller original size of 8,417 and decreases to 7,329 after preprocessing, with no increase after the "explode" step. Combining each definiendum of a domain entity with its respective example sentence, we increased the dataset size to 43,101 examples after the "explode" process.

Figure 5.1 presents the macro F1-score results achieved using the NLDs as input of Pipelines 1 and 2. This figure also presents the specific results of using the "explode" approach before and after the k-fold split. Notably, when we used "explode" before the k-fold split, the results were more expressive than when we used "explode" after, with more than 90% across different levels of the DOLCE-Lite-Plus (DLP) top-level ontology, i.e., NLDs showed robust results from more general top-level concepts to more specific ones. In this experiment, Pipeline 1 showed better overall results regarding Pipeline 2, specifically in the experiment using "explode" after the k-fold split.

Figure 5.2 describes the macro F1-score results achieved by combining the definienda

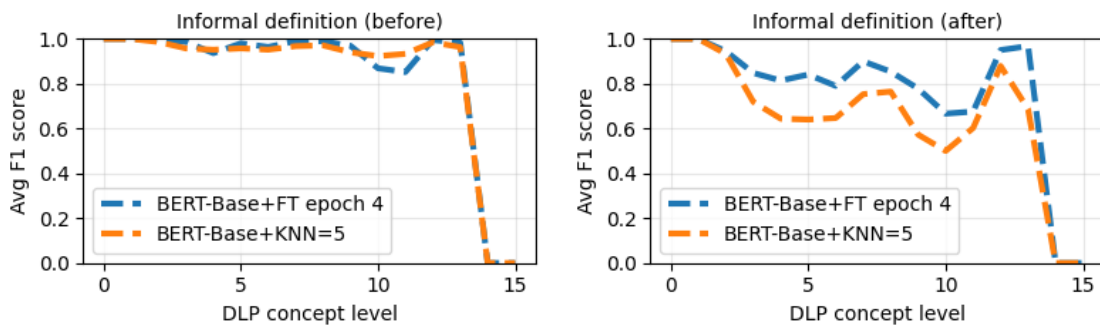


Figure 5.1 – The average macro F1-score, for each DLP level, obtained by employing NLDs as input for the pipelines. The figures depict the pipeline results utilizing the "explode" approach before (on the left) and after (on the right) the k-fold split.

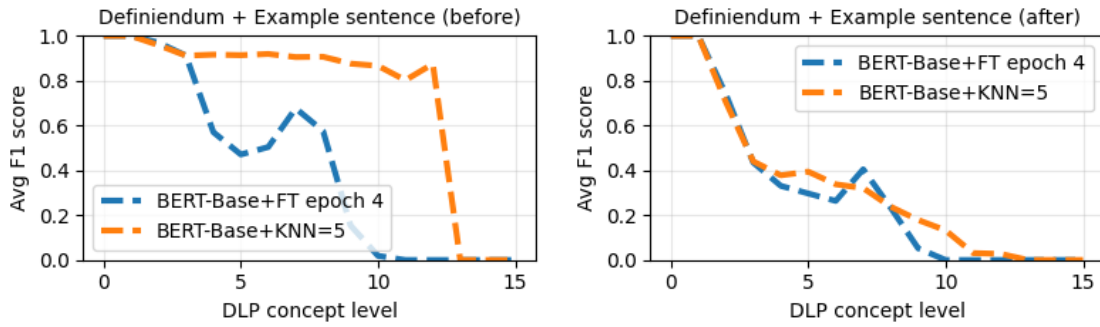


Figure 5.2 – The average macro F1-score, for each DLP level, obtained by employing a combination of definienda and example sentences as input for the pipelines. The figures depict the pipeline results utilizing the "explode" approach before (on the left) and after (on the right) the k-fold split.

and their respective example sentences as input for the evaluated classification pipelines. This figure shows that Pipeline 2 has better results than Pipeline 1 across most DLP levels, using the "explode" approach before the k-fold split. Also, Pipeline 1 has a more significant performance degradation after the 3rd level of DLP. In the same line, using the "explode" approach after the k-fold split, both pipelines showed poor performances over most DLP levels. Despite the performance degradation of Pipeline 1 in both "explode" scenarios, Pipeline 2, with embeddings from the BERT-Base language model and a KNN classifier, presented a close performance compared to Pipeline 2 of the experiment presented in Figure 5.1.

Figure 5.3 shows the experiments using only the definiens (on the left) and only the example sentences (on the right) of the domain entities as input of the evaluated pipelines. In these experiments, we do not need to apply the "explode" approach since WordNet has exactly one definiens and zero or one example sentence for each domain entity. For using only the example sentences, the results present a performance that decreases progressively as the DLP levels increase for both evaluated pipelines. These results indicate a potential challenge in capturing the full semantic scope of a domain entity, which is required for its classification into top-level ontology concepts, using only the example sentences as input. On the other hand, there is a notable variation in the average macro F1 when using only the definiens as input for the pipelines. In this case, the average macro F1 scores are substantially higher than those in the other experiment, achieving more than 80% across several DLP levels. In this context, the results suggest that using WordNet as the knowledge resource, the definiens are a better text representation approach for domain entities than example sentences. However, both approaches are still worse than using their systematic combination with their respective definienda, like in the experiments presented

in Figure 5.1 and Figure 5.2.

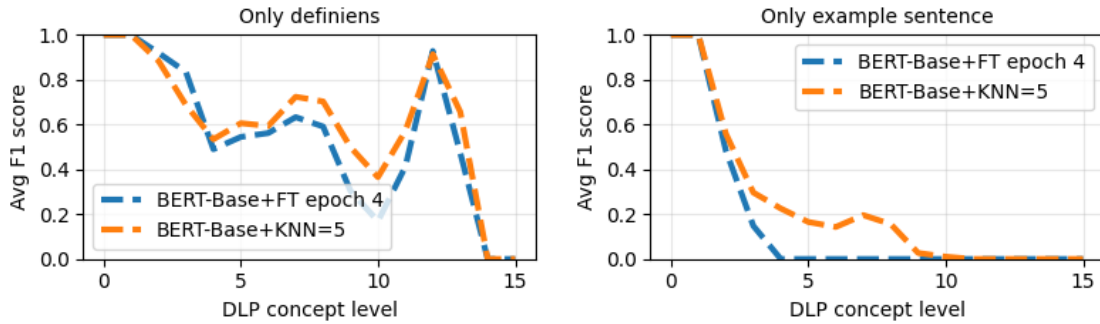


Figure 5.3 – The average macro F1-score, for each DLP level, obtained by employing only the definiens (on the left) and only the example sentences (on the right) as input for the pipelines.

Figure 5.4 presents the experiments using only the definienda as input of the evaluated pipelines. In these experiments, although several definienda are polysemic, i.e., they use the same terms but have different meanings, we do not conduct any duplicate removal in order to analyze the impact of the polysemy in classifying domain entities into top-level ontology concepts. In this context, the results showed similar behavior to those presented using other text representation approaches for domain entities, with performance decreasing as the DLP concept level increases. Also, in both "explode" approaches, the evaluated pipelines showed almost close average macro F1 scores, suggesting this kind of text representation tends to be less affected by the nuances of having the definienda that represent the same domain entity in the training and test samples that we used to evaluate the pipelines. However, even though using only the definienda to represent domain entities presented interesting results, they are unreliable due to the polysemy problem discussed in Section 4.4.

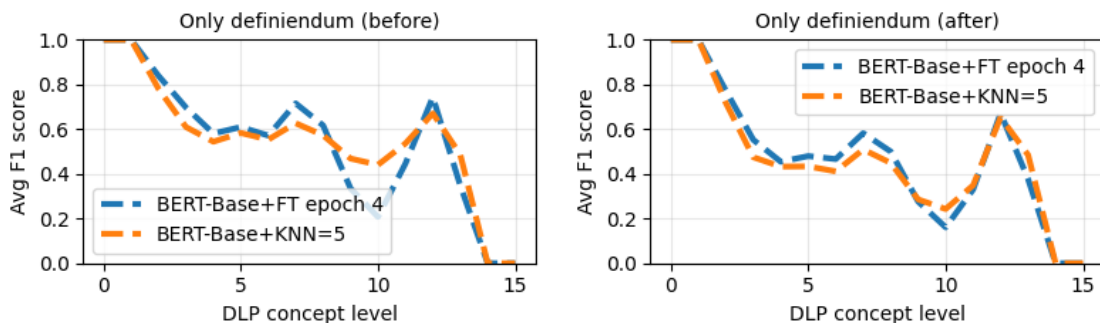


Figure 5.4 – The average macro F1-score, for each DLP level, obtained by using only the definiendum as input for the pipelines. The figures depict the pipeline results utilizing the "explode" approach before (on the left) and after (on the right) the k-fold split.

Figure 5.5 presents the comparison of the performance of different textual representation approaches — Definiendum, Definiens, and NLD — using the BFO dataset

for classifying domain entities using a BERT-Base+KNN pipeline. In this experiment, we mixture the data from different ontologies and domains as those presented in Table 4.2 in the training and test samples across the 10 folds evaluated. The results in the figure show the average macro average F1-scores across all these test folds in the six BFO concept levels, revealing distinct behaviors among the textual representation methods. In this context, the (domain-specific) NLD approach consistently outperformed the others, maintaining robust macro F1 scores across all levels, particularly at lower levels (more specific top-level concepts). Overall, the performance of all textual representation approaches slightly decreased as the depth increased, demonstrating that the BERT-Base+KNN pipeline struggles with specific top-level concepts. Using definiens as a textual representation approach, the results showed comparable performance at higher levels but suffered sharper declines as the concepts became more specific, reflecting its limitations in capturing the breadth of semantic meaning needed for deeper levels. Similarly, the definiendum approach demonstrated adequate performance at the upper levels but experienced significant drops beyond the third concept level.

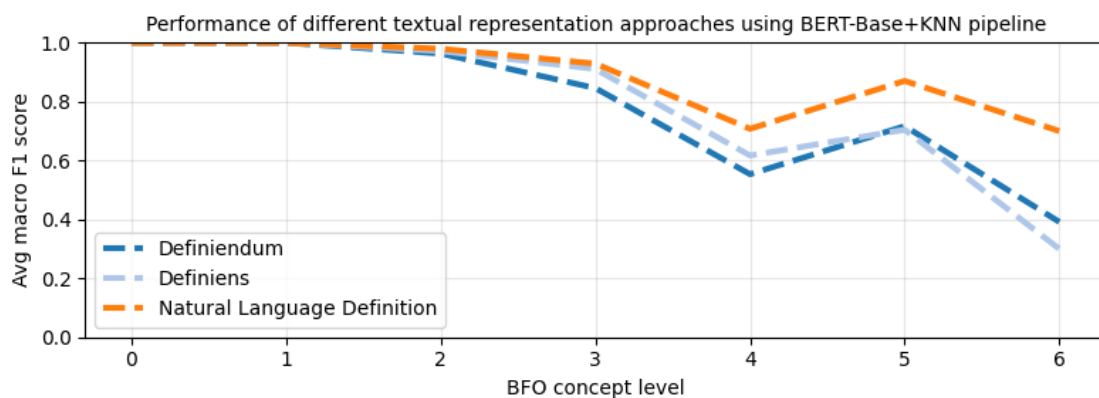


Figure 5.5 – Average macro F1-score results for different textual representation approaches (definiendum, definiens, and NLDs) using the BFO dataset and BERT-Base+KNN classification pipeline.

In this study case, we evaluated the performance of different textual representation approaches of domain entities to classify domain entities into top-level ontology concepts. Although several approaches achieved promising results, such as using only the definiens or the definiendum or the combination of definiendum and example sentences, the NLDs proved to be the best way to represent domain entities in this task for both DLP and BFO datasets. Among the advantages is the possibility of extracting larger datasets, not suffering from the problem of polysemy, and presenting a better average macro F1-score for both "explode" approaches for data augmentation. In addition, Pipeline 2, which adopted

the contextual embedding from the BERT-Base language model and used a KNN classifier to classify them into top-level ontology concepts, achieved the average better stability and performance across all the experiments conducted in this study case. Also, both general NLDs from the DLP dataset and domain-specific NLDs from the BFO dataset achieved outstanding F1 scores, with more than 80% across most of the top-level ontology levels. Thus, the results suggest that using NLDs as textual representation for domain entities and Pipeline 2 are the better options for classifying domain entities into top-level ontology concepts. From this, we will employ this combination for the following study cases conducted in this work.

5.3 Study Case 2: Multi-Language Experiments

We follow our investigation after the results of the previous study case that suggest that NLDs are the best way to represent domain entities and that Pipeline 2 achieved better stability and performance of the macro F1-score in classifying domain entities into top-level ontology concepts. From that, in this case, we evaluate our hypothesis across NLDs from different languages obtained from Wikipedia knowledge resources. Table 5.2 presents the number of examples in each language dataset across the same preprocessing and data augmentation approaches carried out in the previous study case for NLDs. In detail, the English dataset presents 2 or 3 times more examples than most other language datasets, which can affect classification performance. Also, we evaluated Pipeline 2 using the "explode" approach before and after the k-fold split. In addition, the training and test samples are in the same language for every experiment.

Figure 5.6 presents the average macro F1-score results across all DLP levels for the experiment using datasets from different languages. Based on the results, we achieved a higher macro F1-score across all evaluated languages using the "exploding" approach before the k-fold split, with more than 90% of the average macro F1-score. Spanish and English lead with better scores, followed closely by Persian, and while Japanese and Russian trail at the lower end, the scores are still high. However, using "explode" after the k-fold split, all the macro F1-scores decreased, with most languages achieving less than 40% of the average macro F1-score. In this case, the English dataset stands out as the highest scorer, followed by French, Spanish, and Portuguese. This result suggests there are challenges in embedding quality, issues related to the language's complexity, or the dataset's size matters for this task using the "explode" approach after the k-fold split.

Table 5.2 – The number of examples in each different language dataset.

Dataset	Original size	After preprocessing	After "explode"
English	65,018	59,666	387,814
Spanish	25,742	23,209	192,360
German	25,438	23,209	149,446
Persian	20,361	18,596	141,817
French	26,173	23,967	141,730
Russian	23,724	21,634	137,830
Arabic	21,243	19,446	129,790
Chinese	20,776	18,937	125,147
Swedish	23,290	21,339	120,950
Japanese	21,229	19,876	115,496
Dutch	22,561	20,642	108,541
Portuguese	21,826	20,001	106,785
Italian	22,279	20,332	103,778

Overall, the scores for other languages besides English are relatively close, reflecting the consistent performance of Pipeline 2.

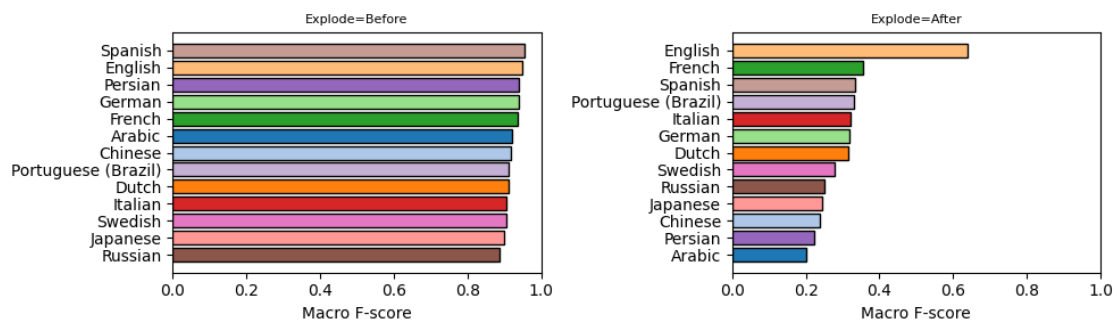


Figure 5.6 – Macro F1-score results for each language dataset using Pipeline 2 and the "explode" approach before (left) and after (right) the k-fold split, respectively.

5.4 Study Case 3: Cross-Resource Experiments

The previous two study cases have demonstrated that NLDs are the best way to represent domain entities for classifying them into top-level ontology concepts and show that our hypothesis, which says "NLDs represent semantic information that allows domain entities to be related to their top-level ontology concepts," is also valid for languages other than English (with some considerations in using the "explode" approach after the k-fold split). However, we used only the BERT-Base language model in the pipelines explored in these study cases. Also, we presented that the results are drastically affected by the "explode" approach. In this context, if we use two datasets with NLDs from different

Table 5.3 – The number of examples in each resource dataset.

Dataset	Original size	After preprocessing	After "explode"
WordNet 3.0	65,018	59,666	387,814
Wikipedia	33,547	30,603	371,766

resources, one as the training sample and the other as the test sample, we can avoid the consequences regarding the "explode" approach, as discussed in Section 4.5. Additionally, we can also evaluate the performance of the pipelines presented in Section 4.6 in a cross-resource scenario.

In this study case, we evaluated several language models for both pipelines described in Section 4.6. For Pipeline 1, we employed the BERT-Base and GPT2, fine-tuned using 10 epochs. In Pipeline 2, we utilized the BERT-Base, BERT-Large, ROBERTA-Base, ALBERT-Base, GPT2, T5, BART, Gemma2B, Mistral7B, and Llama7B, where 2B and 7B means the number of parameters of the respective language model. Also, for Pipeline 2, we only used the KNN classifier because it is the one that best fits our hypothesis, as presented in the previous study cases. In addition, Table 5.3 presents the number of examples of each evaluated dataset after performing preprocessing and "explode" steps. From that, we used the NLDs from WordNet to train the pipelines and the NLDs from Wikipedia to evaluate the pipelines.

Figure 5.7 presents the average macro F1-score of each evaluated pipeline across all DLP depths. Our best pipeline combined the Mistral7B large language model with a KNN classifier, achieving 86.6% of the macro F1-score. Following this kind of pipeline with a KNN classifier, other large language models, such as Llama7B and Gemma2B, also achieved results close to 80% of the average macro F1-score. Interestingly, although BERT-Large has more parameters than BERT-Base, they presented similar results. The pipelines with fine-tuned versions of BERT-Based and GPT2 achieved around 75% of the average macro F1-score, showing distinct results compared with their same version with a KNN classifier. In this case, the pipeline with GPT2 and KNN achieved the worst results in our experiments. Other pipelines with KNN and language models, such as ROBERTA, ALBERT, BART, and T5, showed results ranging from 60% to 75% of the average macro F1 scores.

Figure 5.8 presents in detail the average macro F1-score of each evaluated pipeline for each DLP depth. As in the other study cases, the scores appear to decrease as the level of DLP increases. Also, all pipelines generally maintain their performances until level 10, which can be associated with the 10th level having the more significant number of

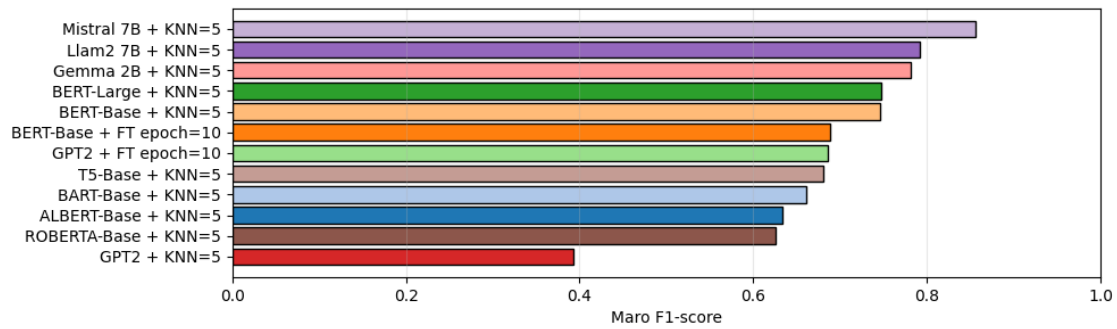


Figure 5.7 – The macro F1-score results for each evaluated pipeline across all DLP levels.

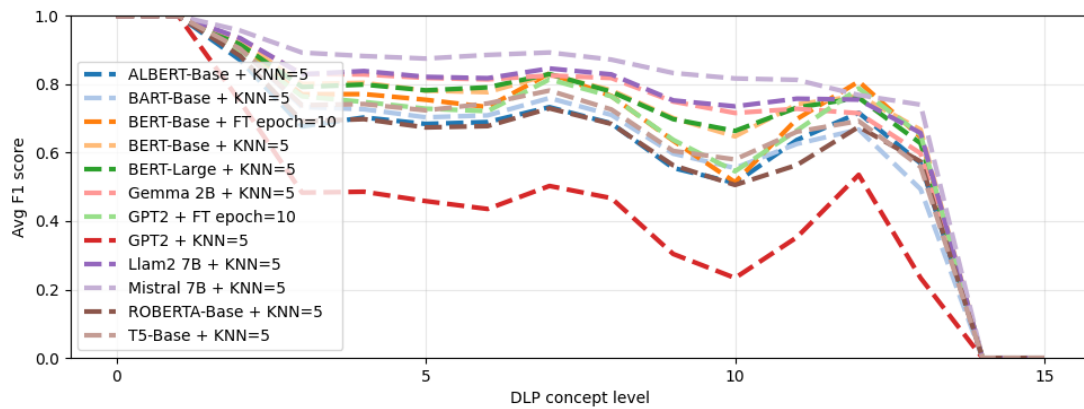


Figure 5.8 – The macro F1-score results for each evaluated pipeline across each DLP level.

top-level concepts regarding all DLP levels. Also, we can justify the better performance of the pipeline with the Mistral7B large language model and the KNN classifier with their stability regarding a better classification performance from the more generic to the more specific top-level ontology concepts.

This case study presented a rich evaluation of our hypothesis that considers the NLDs to represent semantic information that allows domain entities to be related to top-level ontology concepts. From that, we employed various language models and state-of-the-art large language models in the proposed classification pipelines. Also, the results suggest that we can avoid the computational cost of fine-tuning by employing the embedding from a pre-trained language model and using classical machine learning approaches, like KNN. In this context, we showed that using a KNN classifier with contextual embedding derived from NLDs yields better results than fine-tuning methods. This outcome supports the notion that the distributional properties of NLDs encapsulate the top-level ontology concepts associated with domain entities. Furthermore, the enhancement of this process by state-of-the-art models such as Mistral, Llama, and Gemma indicates that the classification of domain entities into top-level ontology concepts via NLDs and contextual embedding benefits from advancements in language model improvements.

5.5 Study Case 4: Cross-Domain Experiments

In this study case, we evaluated the ability of different pre-trained language models — BERT-Base, Bio-BERT, and Llama 3.2 1B — to classify domain entities into top-level ontology concepts across multiple knowledge domains in the BFO dataset. The classification was performed using the KNN pipeline (Pipeline 2) and domain-specific NLDs as textual representations of the domain entities. The BFO dataset spans 12 diverse domains, providing a comprehensive and representative testing framework. This evaluation aimed to assess how effectively these models can extract and represent semantic information embedded in domain-specific NLDs and generalize their knowledge across domains.

For each evaluated domain, all the domain-specific data were used exclusively in the test sample, ensuring no overlap with the training data. From that, the results, as shown in Figure 5.9, reveal that BERT-Base consistently achieved competitive F1 scores across most domains with the other language models, demonstrating its robustness as a general-purpose model. Bio-BERT, which specializes in biomedical text, achieves the best F1 score for most of the analyzed domains, including Anatomy and Development, Diet, Metabolomics and Nutrition, Health, Information, and Investigations domains. Llama 3.2 1B, despite its larger model size, showed weaker performance compared to the other models, except in the Microbiology and Organisms domains, indicating potential misalignment with the task requirements. Among the domains, Microbiology and Information Technology achieved the highest scores across all evaluated language models in the KNN pipeline, with BERT-Base slightly outperforming the other models in the Information Technology domain and Llama 3.2 1B outperforming in the Microbiology domain. Overall, the results suggest a poor performance in a cross-domain evaluation scenario since most of the results have not achieved more than 40% in the average F1 score. Figure 5.10 also confirms these poor results, showing a sharp decline in performance as the BFO concept level deepens. Across all domains, the average F1 score sharply drops after the second concept level where the subdivisions in the BFO top-level ontology began, indicating the increasing difficulty of accurately classifying entities into more specific hierarchical concepts. This further highlights the limitations of language models in generalizing effectively across diverse domains and deeper ontology hierarchies in cross-domain tasks, i.e., the language models struggle to represent domain-specific NLDs across different domains in classifying domain entities into top-level ontology concepts.

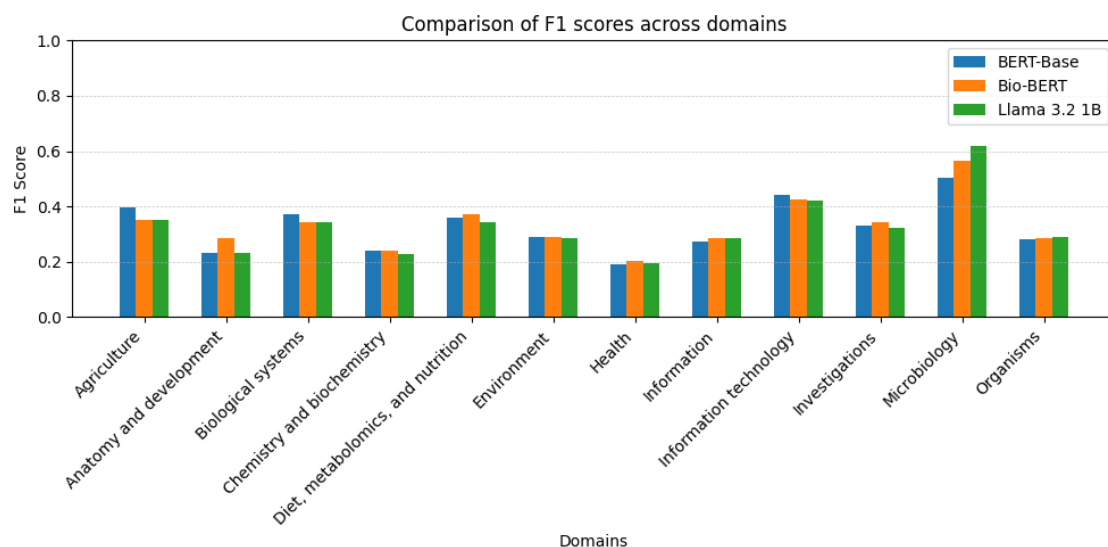


Figure 5.9 – The macro F1-score results in our cross-domain evaluation using different Language Models in the KNN classification pipeline.

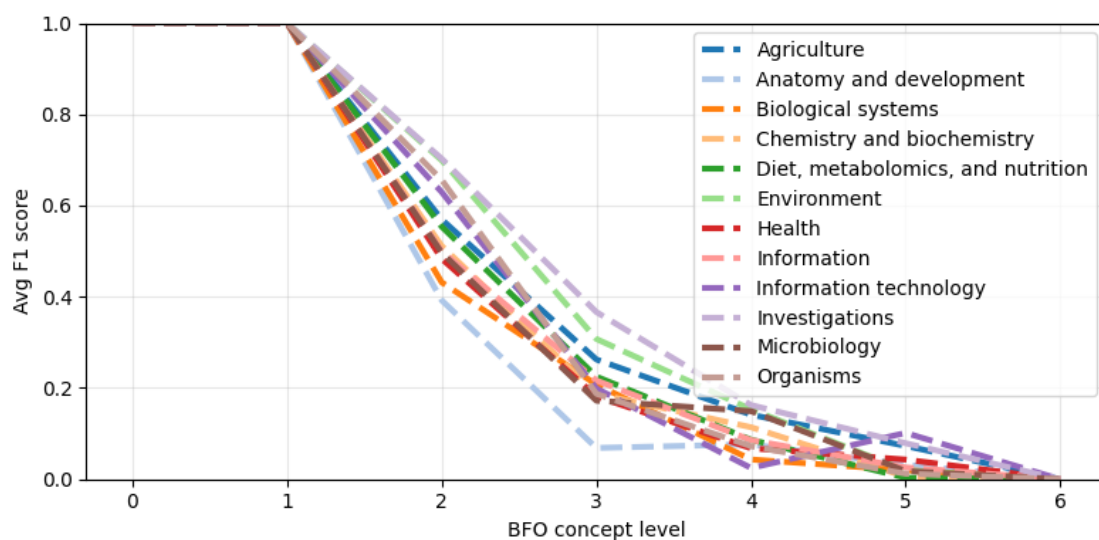


Figure 5.10 – The macro F1-score results for each evaluated knowledge domain across each BFO level using the BERT-Base+KNN classification pipeline.

When comparing Figure 2.1a and Figure 5.10, we can notice that the language models struggled to generalize across domains and performed poorly on more specific BFO concept levels. From that, the results indicate that language models are highly sensitive to domain-specific NLDs in training data. This dependency underscores a limitation of these models: while they can learn effectively with sufficient domain representation in the training data, their ability to generalize to unseen domain data is limited. Figure 5.11 supports this analysis, where we randomly and incrementally move a percentage of domain-specific test data into the training data across the 12 evaluated domains. As the percentage of domain data incorporated into the training set increases, there is a consistent

(with expressive fluctuations) in the F1 scores across all domains. This trend confirms a strong dependence of BERT-Base+KNN classification pipeline performance on the availability of domain-specific training data. Also, the fluctuations in the F1 scores further reveal that the classification performance is also dependent on the domain ontology data randomly selected to move to the training data.

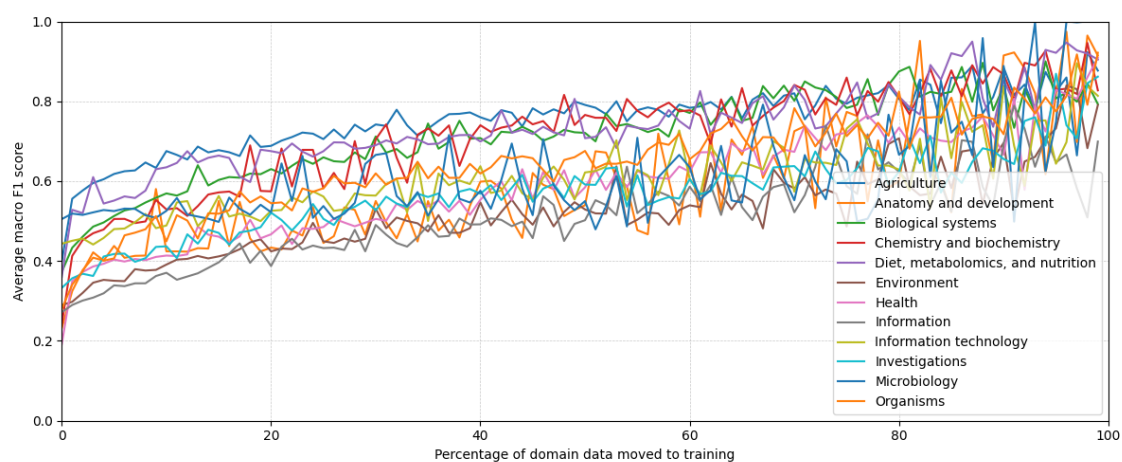


Figure 5.11 – The macro F1-score results by moving a percentage of data from the test sample to the training sample for each domain using the BERT-Base+KNN classification pipeline.

The findings in this study case provide insight into the challenges language models face with domain data, i.e., domain-specific NLDs. While pre-trained models like BERT-Base, Bio-BERT, and Llama 3.2 1B are robust for tasks aligned with their pretraining objectives, their performance in classifying domain entities into top-level concepts in a cross-domain scenario drops significantly when faced with domain-specific nuances not represented in their training set. The sharp dependence on domain data also indicates that these models, despite their large-scale pre-training, are not inherently capable of understanding the deeper relationships within specialized knowledge domains unless explicitly exposed to sufficient examples. Also, the findings highlight the importance of augmenting training datasets with domain-specific data to enhance performance in this classification task.

5.6 Study Case 5: Cross-Ontology Experiments

In this study case, we focused on evaluating cross-ontology performance in classifying domain entities into top-level ontology concepts using all the 110 domain ontologies described in Table 4.2 organized by knowledge domain. Also, we employed the BERT-Based+KNN pipeline using the domain-specific NLDs as the textual representation

approaches for the domain entities.

For each domain ontology, we evaluated its performance by using all the NLDs from the ontology exclusively in the test sample, meaning that no NLD data from the ontology was included in the training sample. In this setup, high precision indicates that the classifications made by the BERT-Base+KNN pipeline were highly accurate, with few false positives, meaning the predicted top-level concepts for the NLDs matched the actual ones. High recall, on the other hand, reflects the pipeline’s ability to classify most of the relevant NLDs correctly, with few false negatives, indicating broad coverage. Conversely, low precision suggests frequent misclassifications, where the predicted top-level concepts for the NLDs differed from the actual ones, leading to many false positives. Low recall indicates that the pipeline failed to classify a substantial number of relevant NLDs, resulting in significant false negatives and limited coverage. Figures 5.12 and 5.13 illustrate these metrics by plotting the average recall (x-axis) and average precision (y-axis) for each domain ontology in this evaluation scenario. Below, we detailed the results for each domain:

- **Agriculture domain:** the PSO ontology achieved perfect precision but low recall, indicating highly specific classifications with limited coverage. Conversely, AGRO displayed better recall but lower precision, reflecting more comprehensive but less accurate classifications.
- **Anatomy and development domain:** ontologies like ZFA, FBBT, and WBBT achieved both high precision and recall, highlighting their strong alignment with the training data. In contrast, PLANA, CLO, and COLAO showed poor performance, with frequent misclassifications (false positives) and missed entities (false negatives).
- **Biological systems domain:** NBO demonstrated near-perfect precision and recall, indicating strong alignment of its NLDs with the training sample. GO, OHPI, GENO, and INO showed balanced yet slightly lower performance due to minor mismatches in represented classes.
- **Chemistry and Biochemistry Domain:** the GNO ontology achieved perfect precision and recall, while PR, MOP, MOD, and CHEMINF showed balanced but moderate performance. Ontologies like CHEBI, PROCO, and TXPO performed poorly, reflecting significant classification errors and misalignment.
- **Diet, Metabolomics, and Nutrition Domain:** the ONE ontology achieved high

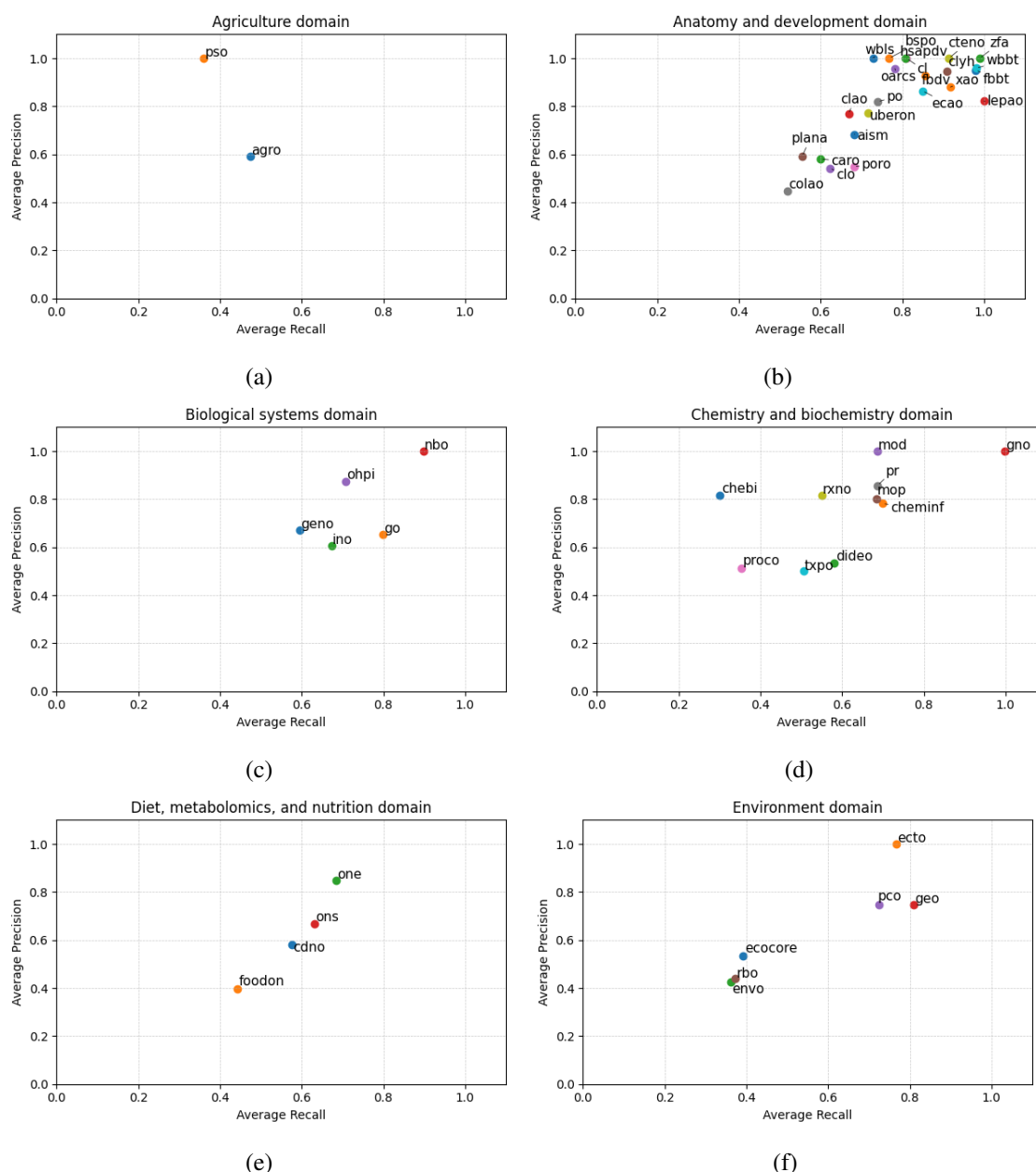


Figure 5.12 – Average precision and recall of ontologies in the agriculture, anatomy and development, biological systems, chemistry and biochemistry, diet, metabolic and nutrition, and environment domains.

precision and recall, while ONS and CDNO performed moderately with some misclassification. FOODON exhibited low precision and recall, indicating poor alignment with the training data.

- **Environment Domain:** the ECTO ontology achieved perfect precision and high recall, while GEO and PCO showed good, balanced performance. ENVO, ECO-CORE, and RBO performed poorly, highlighting significant misalignment.
- **Health Domain:** MONDO, DOID, MPIO, and OVAE ontologies achieved high precision and recall, while MAXO and TRANS displayed moderate alignment.

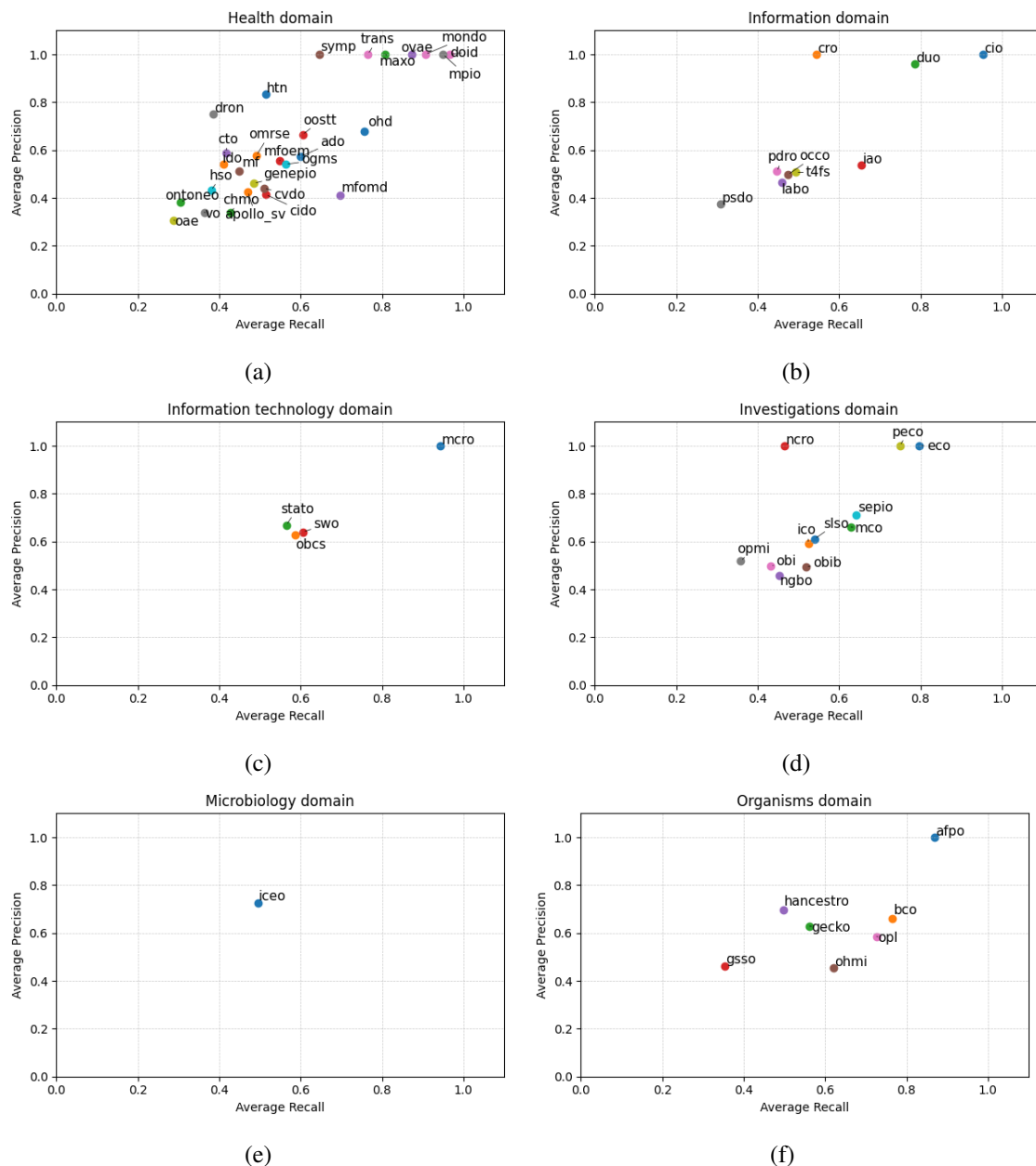


Figure 5.13 – Average precision and recall of ontologies in the health, information, information technology, investigations, microbiology, and organisms domains.

Poorly performing ontologies, such as OAE and APOLLO_SV, exhibited frequent classification errors and limited coverage.

- **Information domain:** the CIO ontology achieved perfect precision and high recall, accurately representing and classifying domain entities. DUO showed balanced performance, while PSDO performed poorly, reflecting frequent misclassifications and significant gaps in coverage.
- **Information technology domain:** the MCRO ontology achieved near-perfect precision and recall, demonstrating strong alignment with the training data. Ontologies like STATO, SWO, and OBCS ontologies showed moderate but lower performance

due to misclassification and missed entities.

- **Investigations domain:** the NCRO, PECO, and ECO ontologies achieved perfect precision, indicating strong classification. Other ontologies demonstrated balanced but lower performance, reflecting limited alignment with the training sample.
- **Microbiology domain:** ICEO, the sole evaluated ontology, showed a slightly balanced but overall poor performance, with limited alignment to the top-level concepts in the training data.
- **Organisms domain:** the AFPO ontology achieved the highest precision and recall, demonstrating strong classification performance. OPL and BCO showed moderate alignment, while GSSO, OHMI, and HANCESTRO ontologies performed poorly, reflecting significant misclassification issues.

The above results reveal considerable variability in performance across domains and ontologies and show how the NLDs of an ontology fit with the NLDs of other ontologies regarding classifying in the same top-level concepts. High-performing ontologies consistently align their NLD representations with top-level concepts in the training data, achieving both accurate and comprehensive classifications. Conversely, poor-performing ontologies often suffer from misalignment, leading to frequent misclassifications (false positives) and missed entities (false negatives). From that, we developed another experiment in order to evaluate the performance of each ontology by moving a portion of its data to the training sample (as in the previous study case). Figure 5.14 illustrates the impact of moving a percentage of training data from specific ontologies (ChEBI, DOID, GO, and UBERON) to the training sample using the BERT-Base+KNN pipeline and NLDs to represent domain entities textually. ChEBI and GO exhibit sharp initial improvements in average macro F1-score as domain data is added to the training set, highlighting the pipeline’s reliance on domain-specific data for effective classification, with ChEBI going from under 40% to above 80% and GO from under 60% to above 80% with few numbers of domain entities in the training sample. In contrast, DOID consistently achieves near-perfect F1 scores across all scenarios, suggesting its NLDs are inherently well-aligned with the top-level concepts and generalize effectively. UBERON starts with a comparatively lower F1-score, indicating weaker initial generalization, but shows steady improvement as more domain data is incorporated into the training set.

The results of this study case provide evidence that some ontologies can share, but many ontologies occupy a specific and distinct region within the embedding space generated by the BERT-Base+KNN pipeline (or other Language Model). This phenomenon

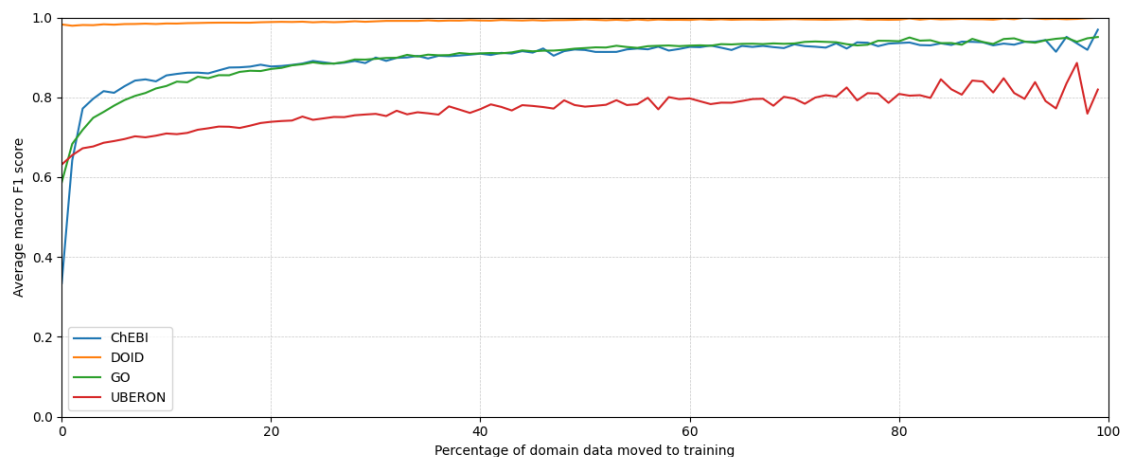


Figure 5.14 – The macro F1-score results by moving a percentage of data from the test sample to the training sample for each ontology using the BERT-Base+KNN classification pipeline.

reflects the fundamental properties of how language models, such as BERT, encode semantic representations. The clustering of domain-specific ontologies in the embedding space arises due to the Distributional Hypothesis (described in Section 2.2.1), which posits that words and phrases used in similar contexts tend to have similar representations. For ontologies, this hypothesis translates into domain-specific NLDs forming dense clusters that capture the semantic nuances and specific structures of each ontology. This alignment, while advantageous for tasks requiring within-domain classification, limits generalization across ontologies, as the embedding regions of one ontology may diverge significantly from those of another. This domain-specific clustering explains why ontologies such as DOID exhibit near-perfect performance even without extensive domain-specific training data—they inherently align well with the embedding space of the training data. Conversely, ontologies like GO, ChEBI, and UBERON, while less aligned initially, show significant improvements in performance with the inclusion of small amounts of domain-specific data in the training data. This suggests that their NLDs, once exposed to the model, quickly establish a consistent and separable region in the embedding space, enabling better classification. The reliance on domain-specific regions in the embedding space also points to a broader limitation of pre-trained Language Models in cross-ontology scenarios. While models like BERT and its variants excel in capturing general semantic relationships, they struggle to bridge the semantic gaps between domain-specific NLDs from unrelated ontologies. This is not simply a result of the training stage but rather the difference across domain-specific NLDs of different ontologies and how embedding representations are constructed. Language models are designed to map inputs into high-dimensional spaces where proximity reflects similarity. When the NLDs of a specific

ontology differ from the other ontologies, their embeddings naturally occupy distinct regions, making it difficult for generalization to occur without explicit domain alignment using a KNN classifier. This finding has critical implications for classifying domain entities into top-level concepts, suggesting that the key to improving performance lies in leveraging the locality of the embedding space rather than attempting to force global generalization when using domain-specific NLDs.

5.7 Comparison with Related Works

Classifying domain entities into top-level concepts has been an active area of research, particularly with the advent of transformer-based language models, like BERT, and their ability to represent complex semantic relationships inside text. However, the choice of textual representation for domain entities, classification methods, and evaluation datasets greatly influences the performance of these approaches. From that, this section aims to compare the performance of our proposed pipelines with related works (described in Section 3.2), highlighting the distinctions in methodology and outcomes. We specifically evaluate our methods against two prominent studies. The first study, conducted by Jullien, Valentino and Freitas (2022), employed a fine-tuned RoBERTa-Large model with example sentences from the Brown Corpus for ontology classification. We compare their results to our RoBERTa-Large+KNN pipeline, which utilizes NLDs and incorporates training strategies such as embedding generation before and after data splitting (described in Section 4.5). This comparison allows us to investigate how the use of NLDs and different data split strategies impact classification performance. The second study, conducted by Sousa et al. (2023), evaluated a fine-tuned BERT model on the classification of domain entities using Definiendum, Definiens, and NLDs as textual representations. We compare these results with our BERT-Base+KNN pipeline, focusing on the differences in performance across two datasets (Sousa23-5c and Sousa23-6c) and their respective textual representations. This comparison also provides insight into the strengths and limitations of fine-tuning versus leveraging pre-trained embeddings with a KNN classifier.

Table 5.4 presents the comparison of our approach against Jullien, Valentino and Freitas (2022) study. The comparison between our RoBERTa-Large+KNN approaches and the RoBERTa-Large fine-tuning method by Jullien, Valentino and Freitas (2022), reveals significant performance differences across five DOLCE-Lite-Plus (DLP) top-level

Table 5.4 – Comparison between the results of Jullien, Valentino and Freitas (2022) work with example sentences from the Brown Corpus and our approach using different kinds of domain entity textual representation and the RoBERTa-Large language model.

Approach	Socially constructed person	Cognitive event	Information Object	Geographical Object	Biological Object
Brown Corpus Example Sentences - RoBERTa-Large+Fine-Tuning (Jullien; Valentino; Freitas, 2022)	0.946	0.915	0.863	0.873	0.857
WIKI NLD - RoBERTa-Large+KNN (After) (Ours)	0.952	0.734	0.865	0.951	0.975
WIKI NLD - RoBERTa-Large+KNN (Before) (Ours)	0.998	0.991	0.995	0.995	0.999
WN NLD - RoBERTa-Large+KNN (After) (Ours)	0.975	0.842	0.919	0.979	0.989
WN NLD - RoBERTa-Large+KNN (Before) (Ours)	0.997	0.978	0.988	0.997	0.998
WN Term+Example Sentence - RoBERTa-Large+KNN (After) (Ours)	0.790	0.790	0.643	0.595	0.761
WN Term+Example Sentence - RoBERTa-Large+KNN (Before) (Ours)	0.976	0.959	0.949	0.962	0.971

ontology concepts: Socially Constructed Person, Cognitive Event, Information Object, Geographical Object, and Biological Object. Jullien’s fine-tuned RoBERTa-Large model achieved consistent but moderate F1 scores, ranging from 85.7% to 94.6% across the concepts. In contrast, our configurations, particularly WIKI NLD - KNN (Before) and WN NLD - KNN (Before), demonstrated superior performance, exceeding 99% F1 score in four out of five concepts. Notably, Cognitive Event reached 99.1%, and Biological Object scored 99.9% F1 score. These results highlight the benefits of using NLDs over exam-

Table 5.5 – Comparison between the results of Sousa et al. (2023) work and our approach using different kinds of domain entity textual representation and the BERT-Base language model.

Approach	Dataset	Definiendum	Definiens	NLD
BERT+FineTuning (Sousa et al., 2023)	Sousa23-5c	0.71	0.88	0.88
BERT-Base+KNN (Ours)	Sousa23-5c	0.82	0.90	0.92
BERT+FineTuning (Sousa et al., 2023)	Sousa23-6c	0.69	0.87	0.87
BERT-Base+KNN (Ours)	Sousa23-6c	0.70	0.87	0.89

ple sentences, as NLDs provide structured and semantically richer information, aligning more effectively with the task of classifying domain entities into top-level concepts. Also, the choice of data split strategy played a critical role in classification performance. The KNN (Before) setting, where we combine the different definiendum with the definiens of a domain entity to build its NLDs before data splitting, consistently outperformed the KNN (After) configuration. This suggests that having different variants of NLDs of the same domain entity in the training and test samples improves the classification results. Conversely, the WN Term+Example Sentence setup produced lower F1 scores, particularly for Cognitive Event and Information Object, where scores dropped to 64.3% and 59.5%, respectively, under the "After" condition. This demonstrates that example sentences alone, even when combined with domain terms, lack the semantic depth necessary for accurate classification. Overall, The NLDs from Wikipedia and WordNet showed consistently higher performance and stability across all categories, underlining the advantage of using general NLDs over limited-context example sentences. These findings underscore the importance of textual representation, and data split strategy in optimizing classification pipelines for classifying domain entities into top-level concepts using transformer-based language models like RoBERTa-Large.

Table 5.5 presents the comparison of our approach against Sousa et al. (2023) work. The comparison between our BERT-Base+KNN approach and the BERT+FineTuning approach proposed by Sousa et al. (2023) highlights distinct performance trends across the Sousa23-5c and Sousa23-6c datasets for the classification of Definiendum, Definiens, and NLDs into DOLCE-Lite-Plus top-level concepts. Also, we employed all the evaluated approaches using an explode approach "before" the k-fold split, as in Sousa et al. (2023) original work. On the Sousa23-5c dataset, our KNN-based method consistently outperformed the fine-tuned BERT approach across all textual representa-

tion approaches of the domain entities. For the Definiendum, the KNN method achieved an F1 score of 82%, surpassing the fine-tuned model's score of 71%. Similarly, for the Definiens, the KNN approach scored 90%, slightly exceeding the fine-tuned model's 88%, and for the NLD classification, the KNN method achieved 92%, outperforming the fine-tuned model's 88%. These results suggest that the KNN-based approach, leveraging the contextual embeddings of the BERT-Base model, more effectively captured the semantic context used to align the domain entities to top-level ontology concepts. On the Sousa23-6c dataset, the results were more closely aligned between the two approaches. For the Definiendum, the KNN-based approach achieved a score of 70%, slightly higher than the fine-tuned model's 69%, demonstrating minimal differences in this scenario. Both methods performed equally well for the Definiens, scoring 87%. However, the KNN approach again demonstrated a better F1 score in the NLD classification, achieving 0.89 compared to the fine-tuned model's 87%. Overall, the differences in performance across datasets and textual representation approaches reinforce the effectiveness of our BERT-Base+KNN classification pipeline (Pipeline 2) for classifying domain entities into top-level concepts, particularly when using NLDs as the textual representation for domain entities.

6 CONCLUSION

This study addresses the challenge of classifying domain entities into their respective top-level ontology concepts. This task is pivotal for improving data integration and interoperability, enabling structured knowledge representation, and enhancing the development of domain ontologies. The proposed approach is a step forward in ontology development automation and helps to reduce the manual effort and deep expertise in the target domain required for ontology engineering while maintaining accuracy and efficiency when classifying new domain entities into top-level ontology concepts. Our hypothesis postulated that NLDs encapsulate the semantic information necessary for associating domain entities with specific top-level ontology concepts. As part of this study, we constructed multi-label, multi-language, and multi-resource datasets to evaluate the proposed approach rigorously. We extracted these datasets from BabelNet and the domain ontologies that are available in the OBO Foundry repository. The datasets included mappings to DOLCE-Lite-Plus and Basic Formal Ontology (BFO) top-level ontologies and incorporated various textual representation approaches of domain entities, such as generic and domain-specific NLDs, *definienda*, *definientia*, and example sentences. In order to validate this hypothesis, we employed state-of-the-art language models and developed two classification pipelines. One focused on fine-tuning pre-trained language models for the classification task, while the other utilized classical machine learning algorithms to classify embeddings generated by these models. The experiments evaluated the approach's effectiveness across various textual representation techniques, languages, resources, knowledge domains, and domain ontologies. The results consistently demonstrated that using NLDs as the textual representation of domain entities provided the best classification performance. In addition, combining language models with a K-Nearest Neighbor (KNN) classifier in a single pipeline delivered the best overall results across all conducted study cases. This success is attributed to the main notions behind our hypothesis about similarity in ontologies and the distributional relatedness in the embedding representation from language models. In addition, our findings reveal that language models represent generic and domain-specific NLDs differently. Generic NLDs from broad resources like Wikipedia and WordNet provide concise descriptions suitable for wide semantic coverage. In contrast, domain-specific NLDs from specialized ontologies in the OBO Foundry offer richer, more nuanced contextual details. In this context, our classification pipelines performed better with generic NLDs than with domain-specific NLDs

because language models create distinct clusters in the embedding space to represent the same top-level concepts using domain-specific NLDs. These clusters vary for each domain ontology, influenced by the Distributional Hypothesis of language models, which adapts representations based on the specific distributions and context of the domain data. The key contributions of this research include:

- Development of an automated approach to classify domain entities into top-level ontology concepts using NLDs.
- Validation of NLDs as a superior textual representation for capturing semantic nuances necessary for classifying domain entities into top-level ontology concepts.
- Insights into the differences between generic and domain-specific NLDs and their influence on language model embeddings.
- Creation of multi-label, multi-language, and multi-resource datasets from BabelNet and OBO Foundry ontologies, ensuring diverse and representative benchmarks.
- Design and evaluation of two classification pipelines, incorporating fine-tuning and classical machine learning methods.
- Analysis of the role of language models in capturing the semantic complexity of domain-specific knowledge for ontology classification.

Despite the above contributions, this work has some limitations. Our approach focuses exclusively on classifying domain entities into top-level ontology concepts and does not address the extraction of other elements necessary for ontology development, such as relationships between entities or domain-specific axioms. Additionally, the reliance on high-quality NLDs and the challenges posed by underrepresented languages and domains remain areas for improvement. Future work will aim to address these limitations by extending the approach to incorporate the extraction of relationships and axioms using the top-level information of domain entities from this study, contributing to the broader goal of developing semi-automated tools for ontology creation. Further efforts will also focus on enhancing the robustness of the approach for low-resource languages and less-defined domains, making it more broadly applicable across diverse contexts.

6.1 Publications

Main publications related to this research:

- **(Qualis A1) Lopes, A., Carbonera, J. L., Schmidt, D., & Abel, M. (2022).** Predicting the top-level ontological concepts of domain entities using word embeddings, informal definitions, and deep learning. *Expert Systems with Applications*, 203, 117291.

The main contribution of the article is the development of a deep learning approach that automatically classifies domain entities into top-level ontological concepts using word embeddings and informal definitions. The proposed architecture combines a feed-forward neural network and a bi-directional recurrent neural network with long short-term memory units (BiLSTM). This method shows improved performance, outperforming baseline approaches by 6% in the micro F1-score. Additionally, the work introduces a novel dataset based on the OntoWordNet ontology for training and evaluating the classification model

- **(Qualis A1) Lopes, A., Carbonera, J., Schmidt, D., Garcia, L., Rodrigues, F., & Abel, M. (2023).** Using terms and informal definitions to classify domain entities into top-level ontology concepts: An approach based on language models. *Knowledge-Based Systems*, 265, 110385.

The main contribution of the article is the proposal of a novel approach that classifies domain entities into top-level ontology concepts using a combination of terms representing the entities and their informal definitions (NLDs). The approach leverages transformer-based language models, achieving state-of-the-art performance, with 94% micro F1-score on the Dolce-Lite-Plus dataset. Additionally, the paper introduces two new datasets extracted from the OntoWordNet ontology aligned with Dolce-Lite and Dolce-Lite-Plus ontologies.

- **(Qualis A1) Lopes, A., Carbonera, J., Rodrigues, F., Garcia, L., & Abel, M. (2024).** How to classify domain entities into top-level ontology concepts using large language models. *Applied Ontology*, (Preprint), 1-29.

The main contribution of the article is the development of an automated approach to classify domain entities into top-level ontology concepts using informal definitions (NLDs). The study leverages large language models (LLMs) and evaluates the effectiveness of the approach across different languages, knowledge resources, and textual representation methods. The research highlights the potential of NLDs to capture semantic information, showing that a K-Nearest Neighbor (KNN) classifier using embeddings from a language model, such as Mistral, provides the best results in this classification task.

Additional publications:

- **(Qualis B3)** Garcia, L. F., Rodrigues, F. H., **Lopes, A.**, Kuchle, R. D. S. A., Perrin, M., & Abel, M. (2020). What Geologists Talk About: Towards a Frequency-Based Ontological Analysis of Petroleum Domain Terms. In ONTOBRAS (pp. 190-203).
- **(Qualis B3)** **Lopes, A.**, Carbonera, J., & Abel, M. (2022). Learning Domain Ontologies Based On Top-Level Ontology Concepts Using Language Models And Informal Definitions. WTDO. ONTOBRAS.
- **(Qualis A3)** Rodrigues, F. H., **Lopes, A.**, dos Santos, N. O., Garcia, L. F., Carbonera, J. L., & Abel, M. (2023, October). On the Use of ChatGPT for Classifying Domain Terms According to Upper Ontologies. In International Conference on Conceptual Modeling (pp. 249-258). Cham: Springer Nature Switzerland.
- **(Qualis A3)** **Lopes, A.**, Carbonera, J., & Abel, M. (2023, November). Disjointness axioms between top-level ontology concepts as a heuristic for word similarity evaluation. In 2023 IEEE 35th International Conference on Tools with Artificial Intelligence (ICTAI) (pp. 538-545). IEEE.
- **(Qualis A3)** **Lopes, A.**, Carbonera, J., Santos, N., Rodrigues, F., Garcia, L., & Abel, M. Cross-Domain Classification of Domain Entities into Top-Level Ontology Concepts Using BERT: A Study Case on the BFO Domain Ontologies. In International Conference on Enterprise Information Systems (pp. 141-148).
- **(Qualis B3)** **Lopes, A.**, Carbonera, J., Rodrigues, F., Garcia, L., & Abel, M. (2024). Unveiling influential factors in classifying domain entities into top-level ontology concepts: an analysis using GO and ChEBI ontologies. In ONTOBRAS.

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APPENDIX A — RESUMO EXPANDIDO

A.1 Introdução

Nas últimas décadas, duas áreas importantes da Inteligência Artificial (IA) voltaram a convergir: ontologias e Processamento de Linguagem Natural (PLN). As ontologias, oriundas da filosofia e conectadas a áreas como metafísica, epistemologia e filosofia da linguagem, são amplamente utilizadas na modelagem e representação de conhecimento. Elas fornecem uma estrutura formal e padronizada para organizar informações complexas, favorecendo a integração e a interoperabilidade entre sistemas. Paralelamente, os avanços em PLN, especialmente com o surgimento de Modelos de Linguagem de Grande Escala (LLMs), como BERT, GPT e Mistral, revolucionaram a forma como representamos e interpretamos textos. Essa convergência impulsionou o renascimento da aprendizagem de ontologias a partir de texto, abrindo novas possibilidades para a geração automática ou semi-automática de ontologias com base em dados textuais. No entanto, apesar dos avanços, o campo ainda apresenta diversos desafios e questões em aberto, especialmente no que se refere à classificação de entidades de domínio em conceitos de ontologias de topo.

Dentre os diversos usos das ontologias, destacam-se a representação de conhecimento e a gestão de dados, uma vez que as ontologias aumentam a clareza semântica ao definir precisamente entidades e suas relações. Isso reduz ambiguidade e garante uma compreensão comum de domínios complexos. Para promover tais benefícios durante o desenvolvimento de ontologias de domínio, metodologias como NeOn, SABiO e SABiOx recomendam o uso de ontologias de topo – estruturas ontológicas abstratas e amplamente aplicáveis que organizam conceitos fundamentais, como objetos, eventos e qualidades. Contudo, identificar a qual conceito de uma ontologia de topo uma entidade de domínio pertence continua sendo uma tarefa altamente especializada, feita manualmente e que demanda conhecimento aprofundado tanto no domínio de aplicação quanto na própria engenharia de ontologias. Essa limitação torna o processo de construção e manutenção de ontologias mais lento e oneroso.

Nesse cenário, esta tese aborda o problema da classificação de entidades de domínio em conceitos de ontologias de topo por meio do uso de Definições em Linguagem Natural (NLDs). NLDs são descrições textuais que expressam o significado, características e contexto de uma entidade, geralmente formuladas com base em estruturas canônicas,

como a definição aristotélica "X é um Y que Z". Partimos da hipótese de que essas NLDs encapsulam informações semânticas suficientes para permitir a associação entre entidades de domínio e conceitos de ontologias de topo. Para isso, apoiamos-nos em dois fundamentos: a noção de similaridade semântica em taxonomias ontológicas – segundo a qual entidades específicas que compartilham um conceito mais geral possuem atributos em comum – e a Hipótese Distribucional, que sustenta os modelos de linguagem contemporâneos e estabelece que palavras em contextos semelhantes possuem significados semelhantes. Dessa forma, assumimos que NLDs de entidades ontologicamente similares apresentarão padrões linguísticos semelhantes, permitindo sua representação vetorial e posterior classificação automática.

Com o objetivo de validar essa hipótese, propusemos a extração de conjuntos de dados multi-rótulo, multilíngues, multidomínio e multi-recurso, baseados em BabelNet e em ontologias do repositório OBO Foundry. Esses conjuntos foram alinhados às ontologias de topo DOLCE-Lite-Plus (DLP) e Basic Formal Ontology (BFO), garantindo uma diversidade representativa de entidades e suas respectivas classificações. Também desenvolvemos dois pipelines de classificação: o primeiro baseado em ajuste fino de modelos de linguagem para a tarefa de classificação e o segundo baseado na geração de embeddings via modelos de linguagem, seguidos da aplicação de algoritmos clássicos de Aprendizado de Máquina, como K-Nearest Neighbors (KNN), Árvores de Decisão e Máquinas de Vetores de Suporte (SVM). A proposta foi testada em cinco estudos de caso que exploram diferentes representações textuais, idiomas, fontes de conhecimento e domínios, com resultados expressivos, especialmente no uso de NLDs como entrada para os classificadores.

A.2 Fundamentação Teórica

A.2.1 Ontologias

A partir da intersecção entre filosofia, ciência da computação e linguística, a ontologia é tratada não apenas como a investigação do ser, mas como um artefato computacional para modelar o conhecimento de maneira formal, explícita e compartilhada. Dentre os diferentes níveis de generalidade das ontologias — de topo, núcleo, domínio e aplicação —, este estudo concentra-se na relação entre entidades de domínio e ontologias de topo. As ontologias de topo, como DOLCE-Lite-Plus e BFO, fornecem categorias

fundamentais que organizam o conhecimento em estruturas taxonômicas amplas e consistentes, promovendo interoperabilidade entre sistemas e domínios distintos. Apesar de divergirem em seus compromissos ontológicos — DOLCE priorizando aspectos cognitivos e linguísticos, e BFO assumindo uma postura realista —, ambas compartilham distinções estruturais.

A.2.2 Definições em Ontologias

No processo de desenvolvimento ontológico, as definições em linguagem natural (NLDs) assumem papel central. Elas não apenas descrevem entidades, mas comunicam sua essência e contexto de forma textual, acessível e estruturalmente padronizada. Ao seguir uma forma aristotélica — composta por definiendum, definiens e cópula —, as NLDs viabilizam a representação semântica rica e estruturada das entidades de domínio. Ainda que elaboradas em fases iniciais do processo de modelagem, antes da formalização ontológica, essas definições encapsulam informações semânticas suficientes para subsidiar sua associação com conceitos de topo. Este trabalho parte da hipótese de que NLDs, mesmo isoladas de uma estrutura formal, carregam evidências semânticas que podem ser exploradas por modelos de linguagem e técnicas de aprendizado supervisionado.

A.2.3 Metodologias para o Desenvolvimento de Ontologias

A tarefa de classificar automaticamente entidades de domínio em conceitos de topo enfrenta limitações nas abordagens tradicionais de engenharia de ontologias, que dependem fortemente de análise manual e conhecimento especializado. Ainda que metodologias como METHONTOLOGY, NeOn, SABiO e LOT sistematizem o processo de construção ontológica, a etapa de vinculação semântica entre entidades e conceitos superiores permanece desassistida por ferramentas automáticas. Neste contexto, propõe-se uma abordagem que antecipa o uso de NLDs como fonte de representação semântica a ser explorada por pipelines de classificação automática, antes mesmo da etapa de conceitualização presentes em metodologias de desenvolvimento de ontologias.

A.2.4 Similaridade Semântica

A estrutura hierárquica das ontologias oferece suporte teórico para a noção de similaridade semântica, a qual considera que entidades mais próximas na taxonomia compartilham características comuns. Medidas de similaridade baseadas em ancestralidade comum, profundidade taxonômica ou propriedades herdadas fornecem meios objetivos para avaliar o grau de semelhança entre entidades. A similaridade, por sua vez, distingue-se da noção mais ampla de relação semântica, que abarca também conexões não hierárquicas. Neste trabalho, adota-se a noção estrita de similaridade, pois ela reflete com mais precisão a estrutura lógica das ontologias de topo, cuja organização pressupõe hierarquias bem definidas. Assim, assume-se que NLDs semanticamente próximas sinalizam entidades ontologicamente similares, tornando possível classificá-las a partir de representações vetoriais extraídas de seus textos.

A.2.5 Representação Vetorial de Texto

A representação vetorial de texto é ancorada na Hipótese Distribucional, que estabelece que palavras que ocorrem em contextos semelhantes tendem a ter significados semelhantes. Essa hipótese sustenta o desenvolvimento de técnicas de vetorização textual que evoluem de abordagens simples, como codificação one-hot e Bag-of-Words, para modelos de aprendizado de embeddings como Word2Vec e GloVe, culminando em representações contextuais com modelos baseados em transformadores, como BERT, RoBERTa, T5 e Mixtral. Esses modelos não apenas codificam palavras, mas aprendem estruturas semânticas complexas com base na distribuição linguística em grandes corpora. Ainda que sua natureza relacional capture associações amplas, argumenta-se que sua capacidade de representar semântica em alta dimensionalidade torna-os instrumentos promissores para mapear NLDs em conceitos de topo, desde que combinados com algoritmos apropriados de classificação.

A.3 Abordagem Proposta

A abordagem proposta neste trabalho visa automatizar a classificação de entidades de domínio em conceitos de ontologias de topo, utilizando Definições em Linguagem Nat-

ural (NLDs) como representação textual e explorando modelos de linguagem de última geração em conjunto com técnicas de aprendizado supervisionado. Para viabilizar essa abordagem, são construídos dois conjuntos de dados distintos: um mapeado à ontologia DOLCE-Lite-Plus (DLP) e outro à Basic Formal Ontology (BFO), contemplando diferentes fontes, domínios e idiomas. No caso do conjunto DLP, realiza-se o alinhamento entre a ontologia OntoWordNet e a rede semântica BabelNet, permitindo a extração de entidades de domínio com definições textuais associadas a conceitos de topo. Este processo envolve o mapeamento semântico entre synsets do WordNet e definições provenientes de recursos como Wikipedia e Wiktionary, resultando em uma base com 118 conceitos de topo e exemplos textuais em até 291 idiomas.

Em contrapartida, o conjunto de dados vinculado à BFO é construído de forma mais manual, com a análise e extração de definições diretamente de 110 ontologias disponíveis no repositório OBO Foundry, abrangendo 12 diferentes domínios do conhecimento. A seleção considera ontologias com estrutura formal compatível e que contenham NLDs aderentes à forma aristotélica, possibilitando seu uso como entrada textual. Ao contrário do conjunto DLP, que depende de mapeamentos automáticos, o conjunto BFO prioriza precisão semântica e especialização, incorporando NLDs mais técnicas e específicas. Em ambos os casos, é realizada uma análise de cobertura e profundidade taxonômica, assegurando diversidade conceitual e representatividade ontológica nos dados.

Originalmente construídos em formato multi-classe, os dois conjuntos de dados são convertidos para o formato multi-rótulo, reconhecendo que uma mesma entidade pode ser coerentemente associada a mais de um conceito de topo. Esta reformulação amplia a complexidade da tarefa, aproximando-a de cenários reais em que classificações ambíguas ou multifacetadas são comuns. Adicionalmente, são definidas estratégias de representação textual das entidades de domínio, incluindo definiendum (termo), definiens (explicação), sentenças de exemplo e a própria NLD completa. A análise comparativa entre essas formas de representação permite isolar os efeitos de cada estrutura textual sobre a performance da classificação, sendo as NLDs completas aquelas que, a priori, encapsulam maior densidade semântica.

Com os dados representados textualmente, propõe-se uma metodologia de treinamento supervisionado composta por duas frentes principais. A primeira consiste no ajuste fino (fine-tuning) de modelos de linguagem pré-treinados, especializados diretamente para a tarefa de classificação multi-rótulo com base no texto. A segunda abordagem utiliza os modelos de linguagem apenas como codificadores semânticos, extraindo embeddings

vetoriais das NLDs que, por sua vez, são classificados por algoritmos tradicionais de aprendizado de máquina, como K-Nearest Neighbors (KNN), Support Vector Machines (SVM) e Árvores de Decisão. Para ambas as abordagens, são definidos procedimentos de avaliação que incluem validação cruzada (k-fold), divisão estratificada dos dados e técnicas de expansão do conjunto de dados (data augmentation), como a substituição do termo definido dentro da NLD para gerar múltiplas instâncias semanticamente equivalentes.

Os pipelines de classificação são estruturados de forma modular e flexível, permitindo o intercâmbio entre diferentes modelos de linguagem e classificadores. Esta arquitetura possibilita testar múltiplas combinações entre representações textuais, métodos de codificação e algoritmos de classificação. O desempenho é mensurado por métricas robustas, com destaque para o macro F1-score, especialmente adequado para cenários multi-rótulo com classes desbalanceadas. A proposta contempla ainda a execução de experimentos em múltiplos idiomas e domínios, bem como avaliações cruzadas entre fontes (cross-resource), domínios (cross-domain) e ontologias (cross-ontology), a fim de testar a capacidade de generalização dos pipelines propostos.

Ao alinhar NLDs, representações vetoriais e técnicas de classificação supervisionada, a abordagem apresentada nesta seção busca preencher uma lacuna no processo de engenharia de ontologias, oferecendo uma solução automática e escalável para a tarefa de associação entre entidades de domínio e conceitos de ontologias de topo. Essa solução se destaca por sua aplicabilidade em contextos multilinguísticos, multidomínio e com múltiplos recursos, reforçando o potencial de reutilização semântica das NLDs e a versatilidade dos modelos de linguagem modernos.

A.4 Experimentos e Resultados

Foram conduzidos cinco estudos de caso distintos, cada um direcionado à análise de uma variável específica: forma de representação textual, idioma, recurso de origem, domínio de conhecimento e ontologia utilizada. Os experimentos são realizados sobre os conjuntos de dados previamente extraídos, considerando tanto DOLCE-Lite-Plus (DLP) quanto Basic Formal Ontology (BFO), em cenários multi-rótulo e multilíngue. As métricas de avaliação utilizadas incluem o macro F1-score, dada sua adequação a conjuntos com múltiplas classes e distribuições desbalanceadas, além de precisão e revocação médias para análises complementares.

O primeiro estudo de caso investiga o impacto de diferentes formas de represen-

tação textual das entidades de domínio na tarefa de classificação. São comparadas representações baseadas apenas no definiendum, no definiens, em sentenças de exemplo e nas Definições em Linguagem Natural (NLDs) completas. Os resultados evidenciam que as NLDs superam sistematicamente as demais representações, alcançando macro F1-scores superiores a 90% nos níveis superiores da DLP. Essa superioridade se mantém mesmo quando se utiliza apenas a parte explicativa da definição (definiens), mas com desempenho inferior ao da NLD integral. Conclui-se que a NLD, por conter maior densidade semântica e contextual, é a forma mais eficaz de representação para esta tarefa.

O segundo estudo de caso avalia a robustez da abordagem em cenários multilíngues. São selecionados subconjuntos de dados em diferentes idiomas, incluindo inglês, espanhol, japonês e russo, com variação no tamanho e na qualidade das amostras. Os pipelines apresentam alta performance nos idiomas com maior cobertura, como inglês e espanhol, atingindo novamente F1-scores acima de 90%. Já em idiomas com menor representatividade, como japonês e russo, observa-se uma leve redução na performance, embora os resultados ainda se mantenham satisfatórios. Esses achados demonstram que a abordagem é generalizável a múltiplos idiomas, desde que haja dados em quantidade e qualidade adequadas.

No terceiro estudo de caso, examina-se a generalização entre diferentes fontes de conhecimento (cross-resource). Para isso, treina-se o modelo utilizando dados provenientes de uma fonte (por exemplo, Wikipedia) e testa-se com dados de outra (por exemplo, WordNet). O pipeline baseado em embeddings gerados pelo modelo Mistral7B, combinado com o classificador KNN, destaca-se ao alcançar um macro F1-score de 86,6%, superando pipelines baseados em fine-tuning direto. Conclui-se que pipelines baseados em embeddings são mais resilientes à variação entre recursos, preservando a semântica essencial das NLDs mesmo quando estas provêm de fontes distintas.

O quarto estudo de caso aborda a generalização entre diferentes domínios de conhecimento (cross-domain), com foco nos dados extraídos do repositório OBO Foundry. A performance dos pipelines varia significativamente entre os domínios, com resultados médios frequentemente abaixo de 40% de macro F1-score nos testes iniciais. Entretanto, ao incluir percentuais crescentes de dados do domínio-alvo no conjunto de treinamento, observa-se melhora consistente, ultrapassando 60% de F1 em diversos casos. Esses resultados demonstram que o modelo é sensível ao vocabulário e estrutura conceitual específicos de cada domínio, e que a inclusão estratégica de dados específicos durante o treinamento melhora substancialmente a performance.

No quinto e último estudo de caso, realiza-se uma avaliação cruzada entre diferentes ontologias (cross-ontology), também com base nas ontologias do OBO Foundry. Observa-se grande variação na performance entre as ontologias avaliadas. Ontologias como DOID (Disease Ontology) e NCRO (Non-Coding RNA Ontology) atingem macro F1-scores próximos de 99%, enquanto outras apresentam desempenhos mais modestos. A análise por ontologia mostra que a qualidade e a consistência das NLDs disponíveis em cada fonte impactam diretamente os resultados. Além disso, evidencia-se que NLDs específicas de domínio oferecem estrutura semântica mais clara e alinhada aos conceitos de topo, favorecendo a classificação correta.

Além disso, realizamos uma comparação com trabalhos relacionados, demonstrando que a abordagem proposta apresenta desempenho superior ou competitivo em relação a métodos anteriores, mesmo quando utiliza apenas definições textuais simples. Os resultados confirmam a hipótese central da tese: NLDs encapsulam informações semânticas suficientes para permitir a classificação precisa de entidades de domínio em conceitos de ontologias de topo. A adoção de modelos de linguagem modernos, combinados com técnicas adequadas de representação e classificação, potencializa o uso dessas definições na engenharia de ontologias, especialmente em contextos automatizados, multilinguísticos e multidomínio.

A.5 Conclusão

Este trabalho confirma a hipótese de que definições em linguagem natural (NLDs) encapsulam informações semânticas suficientes para permitir a classificação automatizada de entidades de domínio em conceitos de ontologias de topo, utilizando modelos de linguagem e aprendizado supervisionado. A abordagem proposta, composta por pipelines de ajuste fino e geração de embeddings combinados com classificadores tradicionais, demonstra desempenho expressivo em múltiplos contextos — superando 90% de macro F1-score em diversos cenários — e mantém robustez mesmo em situações multilíngues, entre diferentes fontes, domínios e ontologias. Os resultados evidenciam que NLDs são a forma mais eficaz de representação textual para essa tarefa, validando seu uso tanto em definições gerais quanto específicas. Apesar disso, limitações como a qualidade das NLDs e os desafios em domínios sub-representados permanecem, sugerindo oportunidades futuras de aprimoramento. No conjunto, a proposta contribui com uma solução escalável, reutilizável e efetiva para apoiar a engenharia de ontologias, reduzindo o esforço manual

e promovendo a automação semântica com base em linguagem natural.